



HOLOBIONT | URBANISM

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Holobiont Urbanism

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Ho.lo.bi.ont

adjective

A holobiont is a biomolecular network composed of the host plus its associated microbes, e.g., humans, animals, or plants. As such, their collective genomes form a “hologenome.” Etymology: *Holos*, Greek, meaning whole or in total. Biont, a discrete unit of living matter.

Ur.ban.ism

noun

Refers to both the material aspects of urban living and the cultural aspects of city life.

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ABSTRACT

Holobiont Urbanism is a research endeavor that sets out to study, map, and visualize the microbiome of New York City, inviting participants to reimagine the city they live in as more than a vast metropolis, but rather as a complex and adaptive biological superstructure. Quantitative and artistic methods are used to produce data visualizations that are the basis for a data art installation created by using thermal imaging to capture a live video stream of streets in New York, web-based 3D technologies to artistically render the video, and a design aesthetic crafted from a scientific framework. The project seeks to distort the participant's perception of the known reality so as to see the city through the lens of the microbial world. Once aware of the *companion species* that live among us, on us, and in us, participants may begin to review what it means to be human and their relationship to their bodies and to their urban environments.

ACKNOWLEDGEMENTS

I would like to thank Kevin Slavin for giving me the opportunity to work with the MIT Media Lab and the incredibly talented group of people he brought together for this project. I will forever be grateful to Kevin for taking a chance on a person he hardly knew. I think it was a *gut* feeling, which in the context of this project just made sense.

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Chris Woebken
Dr. Elizabeth Hénaff
Dr. Chris Mason

With special appreciation to the Mori Building Company of Japan.

Thesis Advisors:

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PREFACE

In October 2015 I interviewed Kevin Slavin as part of my thesis *community of practice* research. At that time my thesis project was focusing on biodesign and bio-fabrication, and I had heard that Kevin was doing research around urban bees. This information came to me during a previous interview with Oliver Medvedik, Ph.D., co-founder of GenSpace and Director of Science at Terreform ONE. During my conversations with Dr. Medvedik, he mentioned that I should speak with Kevin. Separately, my professor at the time, Ed Keller, The New School Director of the Center for Transformative Media and professor in Design Strategies, mentioned the name “Kevin Slavin” to me.

It was through this building of my *community of practice* that I first met Kevin on a rainy day in Brooklyn. After a few more meetings, Kevin invited me to work on the project, called at the time *Biological Immaterials* (a nod to the *Immaterials* exhibition by Timo Arnal). By November 2015, I was officially on the team. Thus, the following is the culmination of six month’s of research into the microbiome of New York City.

This paper is meant to serve as an impetus and process document—a lab notebook of sorts. However, it is not meant to document every aspect of the six month research endeavor. For a complete source of documentation, refer to *Additional Documentation* in the *Appendix* section.

INTRODUCTION



FIGURE 1 | “A DARK SKY OVER DEATH VALLEY” PHOTO CREDIT: DAN DURISCOE¹

DESIGN ACROSS SCALES²

When I look at the night’s sky littered with stars and distant galaxies, I am filled with a sense of awe imagining how immense the universe is. Yet, as a former astrophysicist, I also have a keen understanding that the immensity is really just *a lot of little stuff*. The universe is made up of many, many tiny particles (about 10^{78} – 10^{82} atoms in the known, observable universe³) that erupted into being nearly 14 Billion years ago as a result of the Big Bang. Everything there is *out there* in the cosmos and *down here* on Earth was all born out of the same cosmic *soup*.

The relationship between the *macro* and the *micro*, the idea of scale, from the farthest reaches of our known universe (macro) to the smallest components that make up the building blocks of matter (micro), has been the driving theme in my work and studies for over two decades. Most fascinating to me, however, is how these seemingly opposed forces are in fact mirrors of each other. Like a fractal—an endless repetition of

¹ Nemiroff, Robert, and Bonnell, Jerry. “Astronomy Picture of the day.” <http://apod.nasa.gov/apod/ap070508.html> (July 13, 2008).

² Design Across Scales is the name of a course taught by Neri Oxman at the MIT Media Lab

³ Villanueva, John Carl. “How Many Atoms are There in the Universe?” Universe Today. <http://www.universetoday.com/36302/atoms-in-the-universe/> (April, 2016).

patterns that exist at every scale—the micro and macro scale are deeply intertwined. For example, string theory argues that, at the smallest level, the known elements are one-dimensional, vibrating strings⁴. Yet, the mathematics that holds this theory in place works only if the universe operates in a 10-dimensional reality. This has interesting implications, like the possibility that there is not just one universe, but an infinite number—which physicists call the multi-verse. In the realm of biology, scientists are now seeing that systems are far more interconnected than ever before understood. For example, the human microbiome, the bacteria that live in the human gut and were once thought to be an autonomous entity, may dictate such fundamental human traits as behavior. A human being is now understood to be a *bimolecular network*—the “collective genomic content of a host and its microbiome.”⁵ At both the large and small ends of the scale, the “fabric of the cosmos”⁶ is far more interconnected than we ever before realized.

Holobiont Urbanism is the embodiment of designing across scales as it seeks to position the microbial environment (micro scale) within the urban environment (macro scale) and elucidate the interconnectedness between the two. While not explicitly dealing with the architecture of the city, woven through this project is the looming metropolis of New York—a core protagonist—which begs us to question the implications of the microbial world on the built environment.

⁴ Greene, Brian. “Making Sense of String Theory”. Ted Talk. Feb 2005.
https://www.ted.com/talks/brian_greene_on_string_theory?language=en

⁵ Bordenstein, Seth R. “Host Biology in Light of the Microbiome: Ten Principles of Holobionts and Hologenomes.” PLOS|Biology.

⁶ Phrase borrowed from Brian Greene’s book *The Fabric of the Cosmos: Space, Time, and the Texture of Reality*.

MICRO-MACRO METHODOLOGIES

There is a symmetrical poetry in the data collection methodology of astronomers searching the sky for cosmic signal (macro) while biologists are searching Earth for new life forms (micro) – both in search for the origins of life. Astronomers use massive telescopes (some taller than the statue of liberty) to survey the sky and advanced computational methods to parse out the signal from noise. At the same time, biologists survey the environment collecting metagenomic samples to analyze microbial life on this planet and use large sequencing machines to scan through massive amounts of DNA code.

DESIGN AND SCIENCE

Interdisciplinary work is when people from different disciplines work together. But antidisciplinary is something very different; it's about working in spaces that simply do not fit into any existing academic discipline—a specific field of study with its own particular words, frameworks, and methods.⁷ —Joi Ito

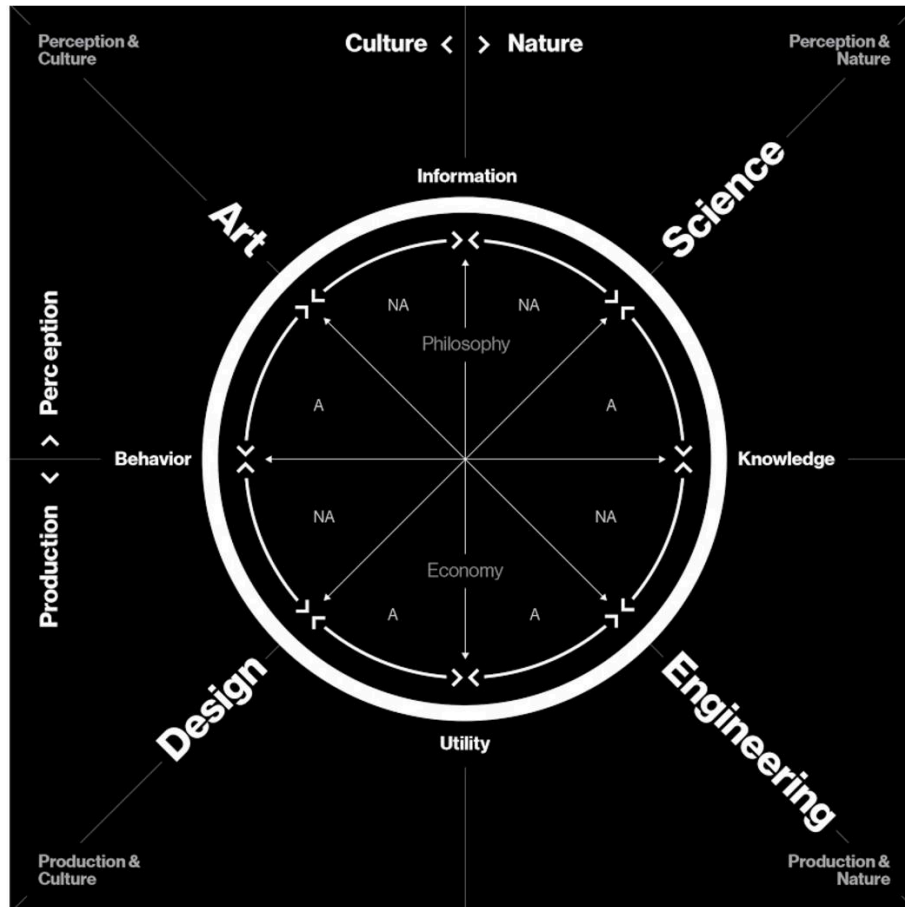


FIGURE 2 | NERI OXMAN, JOURNAL OF DESIGN AND SCIENCE, JODS.MITPRESS.MIT.EDU

Holobiont Urbanism is a project that in its very essence, described in Neri Oxman's *Krebs Cycle of Creativity*, sits at the intersection of design, engineering, science, and art. The project positions itself in the first and fourth quadrants of Oxman's diagram as it seeks to bleed

⁷ Oxman, Neri. "Krebs Cycle of Creativity." *Journal of Design and Science*. <http://jods.mitpress.mit.edu/pub/designandscience> (April, 2016).

across the lines of perception and culture and perception and nature in order to challenge participants to reevaluate their understanding of themselves and the world that surrounds us all.

Bringing together a team of engineers, scientists, designers, artists, and technologists, *Holobiont Urbanism* is redefining a new academic discipline that does not fit into a traditional framework. This work is, as Ito writes, “antidisciplinary,” in that through the work, analysis and prototyping, we have developed our own “words, frameworks and methods” to navigate this new space we are defining. The project is not *biodesign* nor is it *bio art*, yet at the same time it goes beyond a traditional metagenomic scientific project. *Holobiont Urbanism* is a mélange of disciplines that together embody the future of design—a practice that connects design and science.

FROM STAR TO PLANET TO CRYSTAL TO MICROBE TO STAR

“From Star to Planet to Crystal to Microbe to Star” is a phrase taken from a lecture by Ed Keller in his Parsons course, “Post Planetary Design.” In this lecture Professor Keller draws the connection between the sun (*star*), which provides energy to the Earth (*planet*), which undergoes millions of years of geological transformation and stratification by microorganisms that literally change the geological structure of the planet. As an example of the connection, he points first to the Hamersley Iron Province in Western Australia which is characterized by a 2,500 million year old group of Archaean rocks formed by chemical sedimentation of minerals in an ancient marine environment; then to the *crystal* such as Stromatolite formations which “are layered bio-chemical accretionary structures formed in shallow water by the trapping, binding and cementation of sedimentary grains by biofilms (microbial mats) of microorganisms, especially cyanobacteria”; then to the *microbe* one of Earth’s earliest life forms; and lastly, back to the *star* as man searches the cosmos for an understanding of his own existence (e.g. Nostalgia for the Light).

PROJECT BRIEF

There is a vast negative space around us, and we think it's empty, the same way we imagined life before germ theory. It's not empty. It's more alive than anything we build. This is a project in shifting what Foucault called "the liminal horizon" so that you understand the whole world differently when you look away from what we've shown you. —Kevin Slavin

Over half of the world's population lives in urban areas and, according to the U.N., by 2050 nearly 70%⁸ of the world will live in cities, and with the rise in world population, more mega-cities (cities with more than 10 million in population), like Tokyo's 13 million, will form. As more of us move to urban areas, understanding our cities, at both the macro and micro scale, will become ever more important.

Scientists are only just beginning to understand what the microbiome is and, more importantly, its implications for humans. Yet what is clear is that cities, where more and more humans live, are not just accumulations of concrete and massive buildings; they are *complex, adaptive biological systems*, and the human inhabitants of those cities are utterly dependent upon the microbial world for survival. Can visualizing the parallel and unseen world of New York City help the inhabitants understand their connection to it? Through this visualization can we enable the inhabitants to *feel* a connection to the microbial world?

The goal of this work is to answer these questions and to recreate that sense of awe and delight (such as that one might have when looking at the night's sky) for the participants of this project as they begin to see the invisible microbial world all around them.

⁸ "World's Population Increasingly Urban with More Than Half Living in Urban Areas." UN. <http://www.un.org/en/development/desa/news/population/world-urbanization-prospects-2014.html> (March, 2016).

Holobiont Urbanism is a research endeavor conceived of by Kevin Slavin, head of the MIT Media Lab's Playful Systems Group, with the generous support of the Mori Building Corporation of Japan. This international collaboration between geneticists, computational biologists, engineers, architects, artists, designers, and technologists seeks to study and ultimately visualize the microbiome of New York City.

The MIT Media Lab has embarked upon a revolutionary technique of scalable metagenomic data collection by using bees as the "collection" agents or "sensors." The team has partnered with urban, rooftop apiarists and built custom beehives that allow for the detritus of the bees to collect at the bottom of the hives. Using these bees as citizen scientists, the team is then able to collect the debris, extract the DNA, and run metagenomic sequence analyses which provide data about the microbes present in the environments where the bees roam, an approximate 1.5 mile radius from their hive. Four hives have been placed in different neighborhoods of Brooklyn (one in Fort Greene, two in Crown Heights) and Queens (Astoria).

The DNA is sequenced using shotgun, next-generation sequencing technology, and the data is analyzed using MetaPhlAn v2.0, "... a computational tool for profiling the composition of microbial communities from metagenomic shotgun sequencing data."⁹ MetaPhlAn is one of the most advanced algorithms used in the field.

Mapping this data is, in part, about developing the microbial "fingerprint" of these New York neighborhoods. Yet it is also about developing evocative visualizations to help bring this microbial world to life to help the citizens of this metropolis called New York understand that we are not just humans, we are *holobionts*.

⁹ "MetaPhlAn v2.0 Tutorial." Atlassian Bitbucket.
<https://bitbucket.org/biobakery/biobakery/wiki/metaphlan2> (March, 2016).

TEAM ROLES

The breakdown of the team's roles and responsibilities are as follows:

- >> Kevin Slavin
 - _MIT Media Lab
 - _Principal Investigator
- >> Miguel Perez
 - _MIT Media Lab
 - _Project Lead
 - _Hive Engineer
- >> Regina Flores Mir
 - _Parsons School of Design
 - _Design
 - _World Development
 - _Data Visualization
- >> Chris Woebken
 - _Extrapolation Factory
 - _Design
 - _Microbe Development
- >> Devora Najjar
 - _The Cooper Union
 - _Science
 - _Sample Collection
 - _DNA Extraction
 - _DNA Sequencing
- >> Dr. Elizabeth Hénaff
 - _Weill Cornell Medical College
 - _Science
 - _Computational Biology
 - _DNA Sequencing
- >> Dr. Chris Mason
 - _Weill Cornell Medical College
 - _Science
 - _Science Advisor

A BRIEF NOTE ON BEES AS CITIZEN SCIENTISTS

While there appears to be ongoing research using sensors to model bee behavior (i.e., strapping sensors onto bees), namely research into Colony Collapse Disorder (CCD), to the best of my knowledge (at time of writing) Holobiont Urbanism is the first research endeavor in the world to use bees as sensors, i.e., as citizen scientists. Specifically, we are working in cooperation with these bees to collect environmental samples for genomic sequencing, otherwise known as metagenomics. Our specially designed beehive trays are not intrusive to the hive, and since we have no interest in even collecting the honey from the hive, we see these bees as true partners and collaborators in the quest for urban sampling. Mapping the New York City microbiome using traditional methods like in PathoMap is time consuming and resource-intensive (a team of scientists literally goes out to the street to swap surfaces) which is what makes the Holobiont Urbanism project unique and innovative.

BACKGROUND

THE MICROBIOME

What was the human perception of our world before the Hubble Space images were published? What was our understanding of ourselves before the Human Genome Project was complete? Science, in its very essence, is about understanding the structure and behavior of the natural world. *Voyages of discovery* and *feats of imagination* such as these large-scale scientific missions are ultimately a way for humans to understand ourselves, to understand our world, and ultimately, to understand our place in the cosmos.

The Human Microbiome Project, a large-scale project to map the human microbiome, was launched in 2008 to provide scientists an understanding of the intricacies of the human body and our connection to our environment. Like the Hubble space images, this was the first time in history that humans could visualize the complex systems that make up the human body and the ways that these systems can affect human health.

The microbiome is the vast sea of microbes all around our world as well as inside of our human bodies. The Human Microbiome refers to the microbes that live inside a human being. "... [T]he healthy human microbiome is a balanced ecosystem. Microbes perform essential functions such as digesting food and synthesizing vitamins."¹⁰ Yet, are the microbes inside our body more *us* than we realize? We now know that bacterial cells outnumber human cells by 10:1¹¹. This means that only about 10% of the human body is *human*. Recent research has shown that the microbiome may have a much greater impact on what makes us human. For example, "[s]tudies have also linked the microbiome to human mood and behavior, as well as gut health, human development, and metabolic

¹⁰ "Explore Your Microbiome." μBiome. <http://ubiome.com/> (April, 2016).

¹¹ Ibid.

disorders.”¹² This past winter the American Museum of Natural History in New York opened a major exhibition called *The Secret World Inside You* that highlighted potential links between the human microbiome and health disorders such as obesity, depression, anxiety, autism, and memory loss.

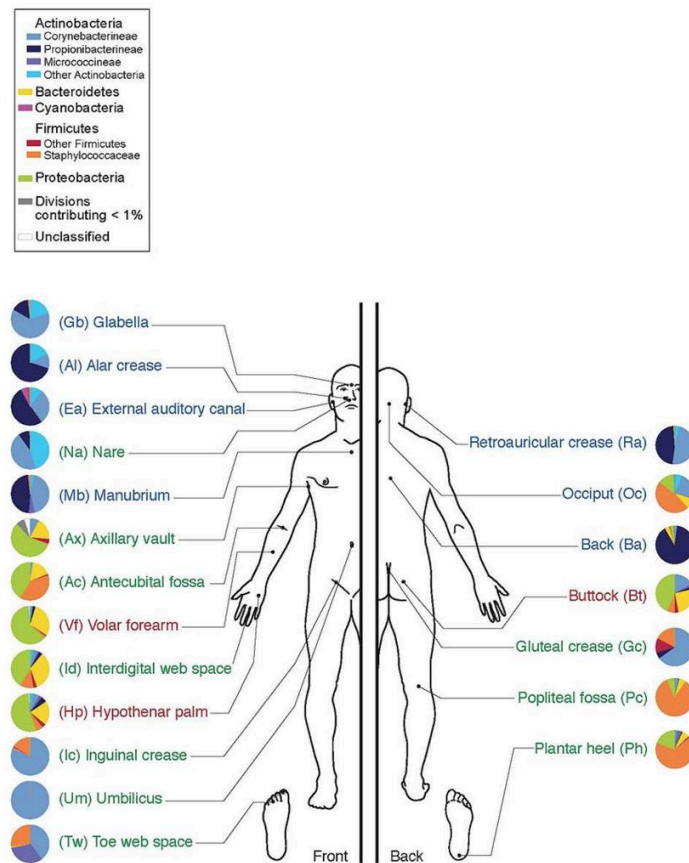


FIGURE 3 | THE HUMAN MICROBIOME PROJECT, HMPDACC.ORG

In January 2016 (only five months before this paper was published), a *Nature* paper called *A New View Of The Tree Of Life* from scientists at UC Berkeley revealed that life on Earth is much more vast and complex than we had ever before understood, and most importantly, human beings represent only a fraction of Earth’s life. Microbes, the tiniest living

¹² Ibid.

creatures, are more plentiful and more abundant than any other creature on Earth.

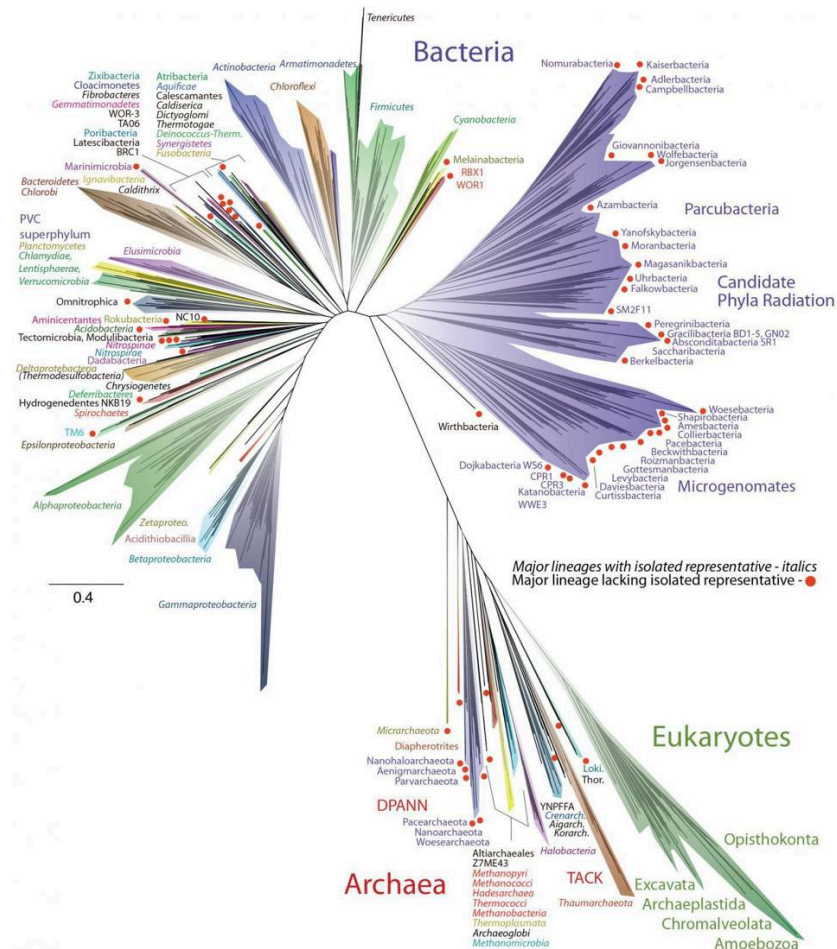


FIGURE 4 | JILL BANFIELD, UC BERKELEY AND LAURA HUG, UNIVERSITY OF WATERLOO.

“The results [of this new tree] reveal the dominance of bacterial diversification and underline the importance of organisms lacking isolated representatives, with substantial evolution concentrated in a major radiation of such organisms.”¹³ The paper also highlights that many of these life lineages are still unknown and further analysis is needed to understand our evolutionary history.

As methods for genomic sampling improve, scientists will only continue to understand the diversity of life. Yet what is overwhelmingly clear is

¹³ Hug, Laura. “A New View of the Tree of Life.” *Nature Microbiology*. <http://www.nature.com/articles/nmicrobiol201648> (April, 2016): doi: 10.1038/nmicrobiol.2016.48

that because of this microbial diversity, humans, plants, and animals are much more complex than we ever understood. Another paper, *Host Biology in Light of the Microbiome: Ten Principles of Holobionts and Hologenomes*, published in August 2015 (only eight months before this paper was published) highlights that

Complex multicellular eukaryotes are not and have never been autonomous organisms, but rather are biological units organized from numerous microbial symbionts and their genomes.¹⁴

Eukaryotes, humans, plants and animals, are *holobionts*— “biomolecular networks composed of the host plus its associated microbes.” This vast microbial world called the microbiome, the microbial life living on us, in us, and around us in the environment, is far more influential than we realize.

¹⁴Bordenstein, Seth R. “Host Biology in Light of the Microbiome: Ten Principles of Holobionts and Hologenomes.” PLOS|Biology.
<http://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.1002226> (April, 2016).

COMPANION SPECIES — DONNA HARAWAY & KEVIN SLAVIN

In her 2003 pamphlet, “The Companion Species Manifesto,” Donna Haraway writes about the interconnected relationship between people and their dogs, suggesting that the two species have co-evolved together. Haraway examines evolutionary biology by investigating the species that grow with us and concludes that by inviting these canine (and other) species into our lives, humans have inadvertently affected our own human evolution. Although the “Companion Species Manifesto” never explicitly points to microbial evolution as a “companion species”, Kevin Slavin, in his comical 2013 talk at Eyeo Festival points out the connection between toxoplasmosis, cats, and human behavior.

ARCHITECTURE: DESIGNING FOR THE MICROBIOME

How might humans design our cities and buildings in partnership the microbial environment to optimize human health or a more balanced relationship with our ecosystem? Although *Holobiont Urbanism* does not specifically deal with microbial design solutions for the urban landscape, it is a natural extension of the work we are doing and has played a role in the research.

Dr. Jessica Green is one of the world's preeminent researchers on the microbiome and the built environment. One of her core research areas deal with HVAC (heating, ventilation, and air conditioning) systems inside modern buildings and how we might better design our buildings with the microbiome in mind. Her work also involves mapping the microbial environments in different rooms in a building, i.e., bathrooms vs. office spaces. In her Ted Talk, *We're Covered in Germs. Let's Design for That*, Dr. Green says:

Mechanical engineers design air handling units to make sure that people are comfortable, that the air flow and temperature is just right. They do this using principles of physics and chemistry, but they could also be using biology. If you look at the microbes in one of the air handling units in this building, you'll see that they're all very similar to one another. And if you compare this to the microbes in a different air handling unit, you'll see that they're fundamentally different. The rooms in this building are like islands in an archipelago, and what that means is that mechanical engineers are like eco-engineers, and they have the ability to structure biomes in this building the way that they want to.¹⁵

¹⁵ Green, Jessica. "We're Covered in Germs. Let's Design for That." TED. https://www.ted.com/talks/jessica_green_good_germs_make_healthy_buildings (April, 2016).

In Dr. Green's lab they also study *microbial clouds*—the cloud-like aura, consisting of millions of microorganisms (fungi, bacteria, and viruses), that surrounds humans.

Dispersal of microbes between humans and the built environment can occur through direct contact with surfaces or through airborne release; the latter mechanism remains poorly understood. Humans emit upwards of 10^6 biological particles per hour, and have long been known to transmit pathogens to other individuals and to indoor surfaces.¹⁶

Their research, published in September 2015, *Humans Differ In Their Personal Microbial Cloud*, revealed that an individual's microbial cloud is like a “unique cocktail”, like a finger print, and that when studying the air in a room

most occupants could be clearly detected by their airborne bacterial emissions, as well as their contribution to settled particles, within 1.5–4 h. Bacterial clouds from the occupants were statistically distinct, allowing the identification of some individual occupants. Our results confirm that an occupied space is microbially distinct from an unoccupied one, and demonstrate for the first time that individuals release their own personalized microbial cloud.¹⁷

I interviewed one of Green's post-doctoral students, Dr. Roxana Hickey, a microbial ecologist, about some of the research they are doing in the Biology and the Built Environment Center at the University of Oregon. The goal of the interview was to guide some of the design thinking in order to understand how best to visualize the urban microbial world. The

¹⁶ Meadow, James F., et al. “Humans Differ in Their Personal Microbial Cloud.” PeerJ. <https://peerj.com/articles/1258/> (April, 2016).

¹⁷ Ibid.

transcripts of that interview can be found in the *Unpublished Source Material* section of this document.

HUMAN MICROBIAL CLOUD: [Speculative Design]

In the future, is it possible that we will have sensors (like temperature or humidity sensors) that will be able to detect our air-borne microbial cloud? And if so, what are the implications for data-privacy? How might homes of the future be designed for optimal microbial balance? Will forensic science radically change if we can use microbial air samples to determine who has been in a room or not? How might air travel change when microbial optimization can be achieved? What social discriminations arise when dealing with the microbial environment?

URBAN METAGENOMIC SEQUENCING

While the Green Lab in Oregon is investigating the microbiome of buildings and humans, the Mason Lab in New York is researching the microbiome of urban transportation systems. In 2015, the Mason Lab, located at Weill Cornell Medical College, took a team of scientists to the streets to swab surfaces in the New York City subway system. The project called *PathoMap* (refer to Mason, et al.¹⁸) had a singular goal—to study the microbiome and metagenome of the underground built environment of NYC. This subway microbial landscape mapping project in NYC was the first time such an endeavor has been undertaken. The success of PathoMap turned into a larger project to map the world's subway systems.

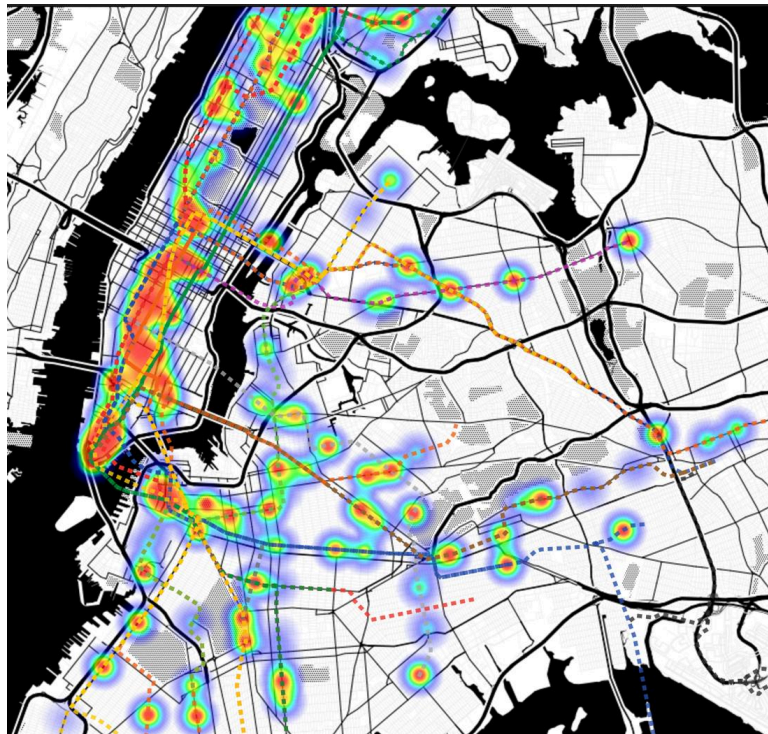


FIGURE 5 | PATHO MAP, PATHOMAP.ORG

¹⁸ Afshinnekoo, Ebrahim, et al. "Geospatial Resolution of Human and Bacterial Diversity with City-Scale Metagenomics." *Cell Systems* 1, no. 1 (July 29, 2015): 72–87. doi:10.1016/j.cels.2015.01.001.

MetaSUB: Metagenomics and Metadesign of Subways and Urban Biomes was created with an international collaboration to “build a molecular portrait of cities.”

Subway surfaces define the daily commute for billions of people each year, and yet there is almost nothing known about the impact of surface type, season, commuter type, or subway design on their commute. We aim to bring a molecular view of the cities to improve their design, use, and impact on health.¹⁹



FIGURE 6 | METASUB, DR. CHRIS MASON SWABBING A TURN-STYLE IN NYC SUBWAY

Both MetaSub and PathoMap are the first of their kind of large-scale projects to research and map the microbiome of the urban environment. I interviewed Dr. Hénaff, one of the lead researchers²⁰ in both projects, as part of my primary research. The transcript can be found in the *Unpublished Source Material* section.

¹⁹ “Building a Molecular Portrait of Cities.” MetaSUB. Weill Cornell Medical College. <http://www.metasub.org/> (April, 2016).

²⁰ Dr. Chris Mason and Dr. Elizabeth Hénaff, both lead researchers in PathoMap and MetaSub, work on the team for the *Holobiont Urbanism* project.

What is most fascinating to me about urban metagenomics is how new this field is. During my year-long research effort for my thesis, paper after paper was being published on the topic of the human and urban microbiome. Even in April, as I was writing this document, an article was published called *Cities Have Individual Microbe Signatures*.²¹ Yet, no other research group (that we know of) is using bees. This project is radical in that way, and moreover, it begins to hint at an idea that I have written about at length in previous work—that we must work in *cooperation* with nature. We are just beginning to understand the implications of the microbial world's effect on our bodies and our environments. As more research is completed, humans will further understand just how interconnected we are to each other and to our world. Urban metagenomics will foster this understanding.

²¹ “Cities Have Individual Microbial Signatures.” Science Daily.
<https://www.sciencedaily.com/releases/2016/04/160419144724.htm> (April, 2016).

NASA TWIN STUDY: MICROBIOME

In NASA's "Twins Study: Understanding the Impact of Space Travel on Humans," one of the core areas of research is in the gut microbiome. NASA used metagenomic sequencing of the microbiome in the GI tract of the twin astronauts to understand the diet differences. NASA also takes periodic samples at the International Space Station to understand how the microbiome lives in extreme environments and the ways this can impact human health. Fellow collaborator, Dr. Chris Mason, also works on this project.

PROJECT DESIGN

DESIGN QUESTIONS

The design research and thinking for this work was initially focused around five central questions that fall broadly into three domains of inquiry:

>> **Biodesign:**

1. What is the microbiological quality of New York City?
2. How does the microbiome of a city affect the people living there?

>> **Data Visualization:**

3. What might location-based microbial data tell us about the ecology of specific neighborhoods?

>> **Urban Interaction Design:**

4. If humans could perceive the microbial world, what would it look like?
5. How can we visualize the effects of the built environment on the microbial world? What does that distortion look like? Clouds? Shadows?

After months of refinement and through the process of *making*, the research has refocused on these central questions:

1. How can we change the participant's perception to reveal that New York City is a biomolecular network?
2. How do we place the participant into the *microbe's* point of view?
3. How can we turn a microscope onto the city to create a top-down view of the city through the lens of a microscope?

4. How can we reveal the invisible microcosm that lives in the negative space of our built environment, which is more alive than anything we humans could ever build?

DESIGN VALUES

The core design values are as follows:

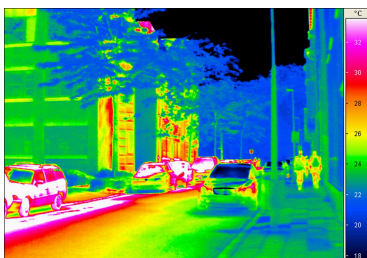
1. Environment—Transform the city into a world that is vastly different from what humans see.
2. Perception—Change the participant's perception, i.e., place participant as microbe.

An additional design value was to create a piece that was procedural or generative. While my thesis focuses on data and hives from New York, the MIT group is planning on eventually incorporating data from Sydney, Melbourne, Venice, and Tokyo. Thus, I wanted the piece to be easily replicated and accessed by users no matter what city he/she was trying to model.

DESIGN AESTHETIC

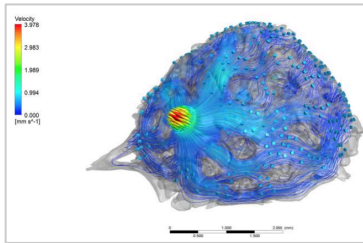
The design aesthetic was crafted from a scientific framework in which each element was identified as a deliberate manifestation of concepts to embody in the final piece. During a design workshop, the team settled on this framework. Here are the final design aesthetic choices:

>>Thermal Imaging



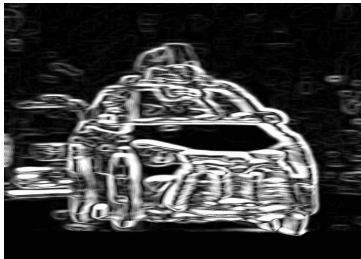
Each species of microbe thrives in its own optimal temperature range. Do microbes perceive the world through temperature? We decided to use thermal video as a video-input and map the microbes to temperature.

>>Flow Fields



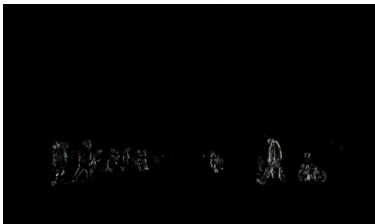
Using the paper, *Computational Fluid Dynamics of Sponge Aquiferous Systems*,²² as a model for the behavior of microbes in the microclimates that exist on other animals, we decided to model the movement using flow fields.

>>Elimination of Color



Microbes do not perceive color like a human being. Thus, we remove color from the video renderings to move closer to the microbial perception of the environment.

>>Depth of Field



Microbes do not perceive depth as humans do; thus, we eliminate objects in the background and only render the foreground of the images.

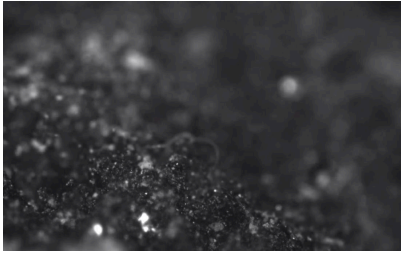
>>3D Distortions



We use 3D distortion effects by mapping the video to a 3D mesh to allow for z-plane depth. By using post-processing effects we can also reimagine reality using our artistic interpretation.

²²Hammel, J.U. et al. *Computational fluid dynamics of sponge aquiferous systems*. December 2012. Source: photon-science.desy.de/annual_report/files/2013/20132819.pdf

>>*Optical Effects*



We apply microscopy effects using post-processing to mimic the optical effects observed while using microscopes in the lab.

MICROBIAL MAPPING

DATA ANALYSIS METHODOLOGY

Understanding the Biology in the Data

In its raw form the data was very difficult to understand. The raw sequencing data came from Dr. Elizabeth Hénaff, PhD in Computational Biology at the Mason Lab at Weil Cornell Medical College, and was collated by Devora Najjar, a scientist from The Cooper Union Microbiology Lab. To help me understand the data, I first met with both the scientists from the respective labs to interview them and ask questions about how to read the data and what, from their perspective, could be understood from this data set.

The data was broken out by the four beehives and named according to the neighborhood they were placed in: Astoria, Crown Heights-Langstroth, Crown Heights-Top Bar, and Fort Green. A number of samples were taken from each hive: honey, beeswax, propolis (a sticky substance bees produce inside the hive), a single bee, etc. Each of those samples was then purified in a lab, and the DNA was extracted. The DNA sample was then sent to another lab to be sequenced. The sequencing process could show which DNA was from microbes and which DNA was from bees. Thus, the data set showed the percentage of microbial life present in each sample that was not attributed to bee DNA. The data was then broken out by taxonomical classification. Below is an example that shows a typical identification from the dataset.

FIGURE 7 | EXAMPLE OF AN ID FROM THE TAXONOMICAL DATA.

```
k__Bacteria|
p__Actinobacteria|
c__Actinobacteria|
o__Actinomycetales|
f__Propionibacteriaceae|
g__Propionibacterium|
s__Propionibacterium_acnes|
t__Propionibacterium_acnes_unclassified
```

In order to understand this taxonomy, I had to research basic biology concepts such as the Phylogenetic Tree (otherwise known as the Biological Tree of Life), which is a diagram biologists use to map all life on Earth, and the Biological Taxonomic Ranking, which is the hierarchy biologists use to classify life. I was able to then understand that the data is identified first by type of life form (Kingdom): bacteria, virus, or eukaryota. Then that life form is labeled according to its phylum, class, order, family, genus, species (and sub-species when available).

Data Wrangling

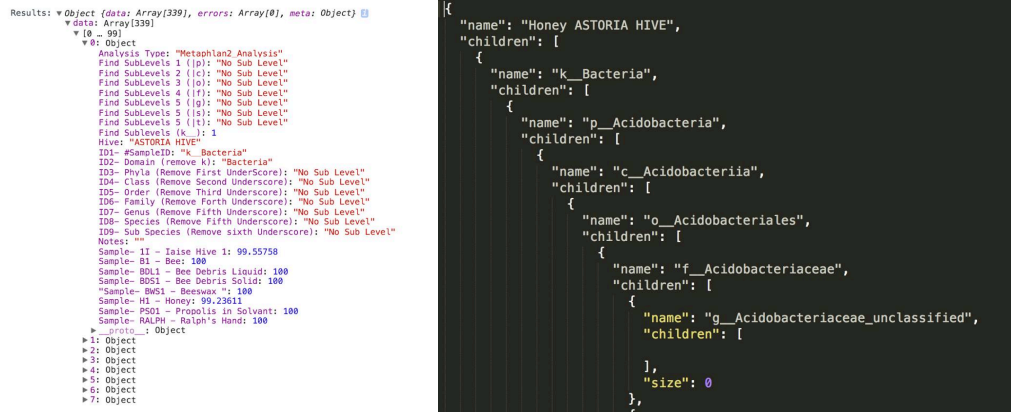
As the data came to me in CSV format, I first began to manipulate it using Excel. I used logic to parse strings to get a sense of how the data was formatted and to really just *play* around in the spreadsheet. After some testing and simple manipulations, I felt like I was ready to move onto a more formal analysis.

Data Parsing—JavaScript

I first used a parsing algorithm in JavaScript to turn the data set into a simple JSON²³ format. I quickly realized that this format was not robust at all because inherent in the data set is the taxonomical classification which is hierarchal, and in order to build a data visualization that accounted for this hierarchy, I had to format my JSON file accordingly. Toward that end, I wrote a recursive algorithm in JavaScript. Figure 8 shows the difference between the hierarchal and non-hierarchal JSON files. I then made data visualizations using the d3.js library (d3.layout.cluster).

²³ JSON stands for JavaScript Object Notation and is a standard format used in programming, in particular, on the web, in order to organize data in a legible way for both machines and humans to read.

FIGURE 8 | (LEFT) POORLY FORMATTED JSON. (RIGHT) HIERARCHAL FORMATTED JSON.



Data Parsing—Python

Although the d3.js library is used industry-wide, for my thesis I decided to push myself to develop my own software. I began making my own data visualizations in openFrameworks, and to do this I needed a slightly more descriptive JSON format. The Python language enabled me to achieve such a format, and since I have wanted to learn Python for some time, I took this opportunity to embrace the language. I used Python to write a new algorithm to format the data into a very specific JSON file.

DATA ART AND VIZUALIZATION

I created four maps of the metagenomic taxonomical data using Radial Reingold–Tilford Tree from the d3.js library. The maps are broken out by hives, then by samples, and then by taxonomical classification. Looking at these maps as a set, it is clear that by mapping this data, we can begin to see the microbial differences and similarities of the three neighborhoods (and four hives).

FIGURE 9 | INFOGRAPHIC OF TAXONOMICAL MAPS FOR ALL FOUR HIVES BY NEIGHBORHOOD AND BY SAMPLE.

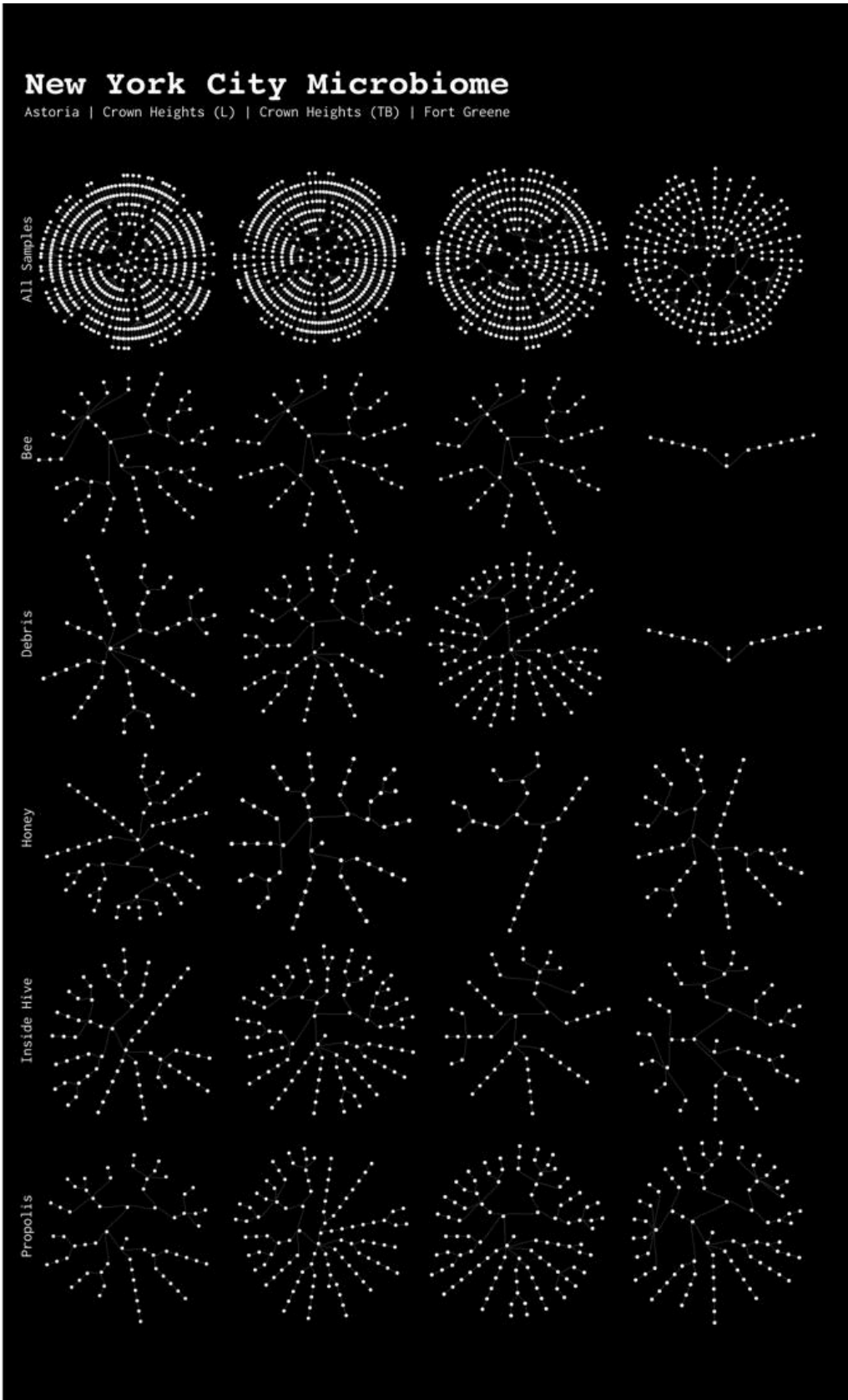
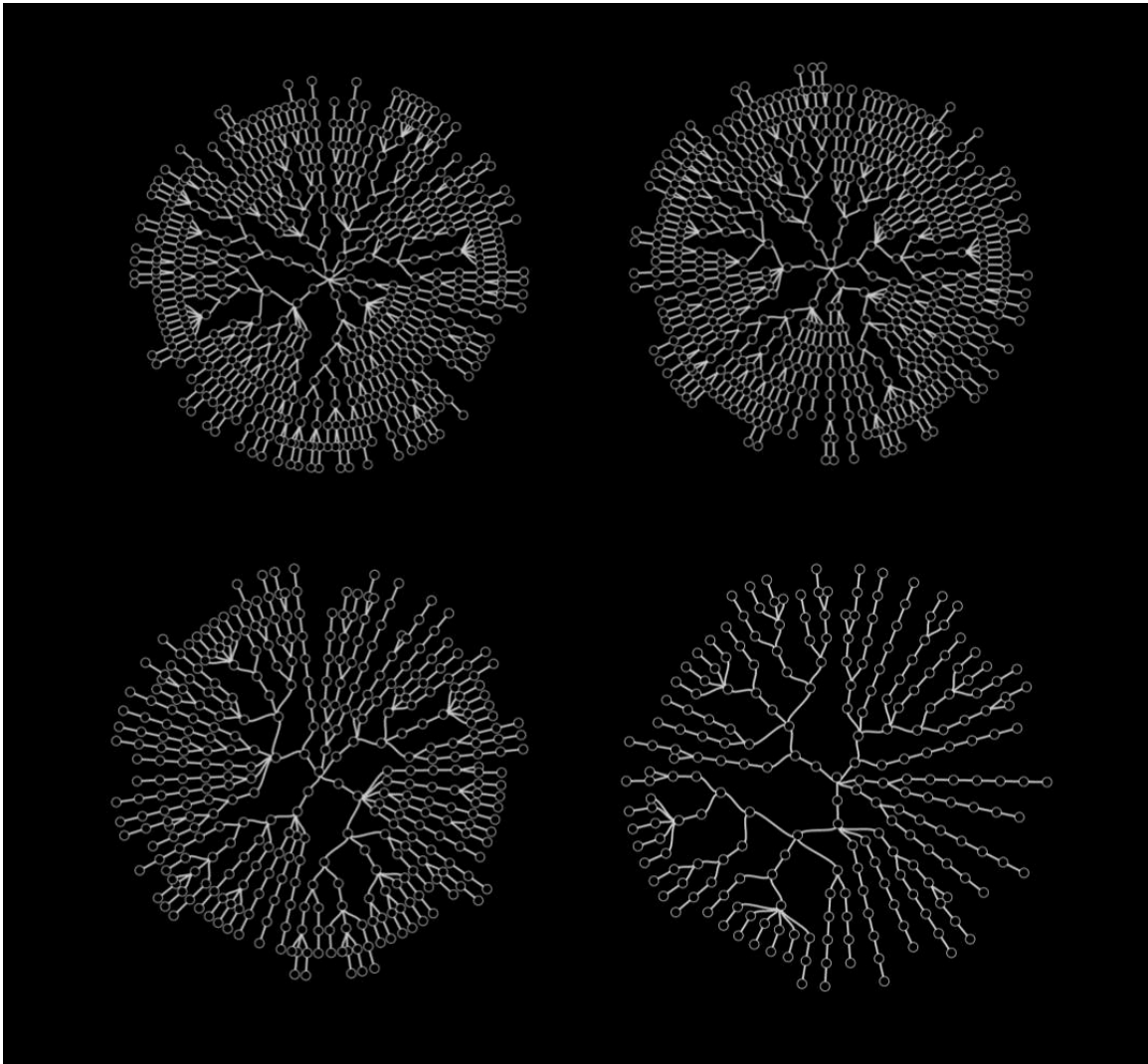
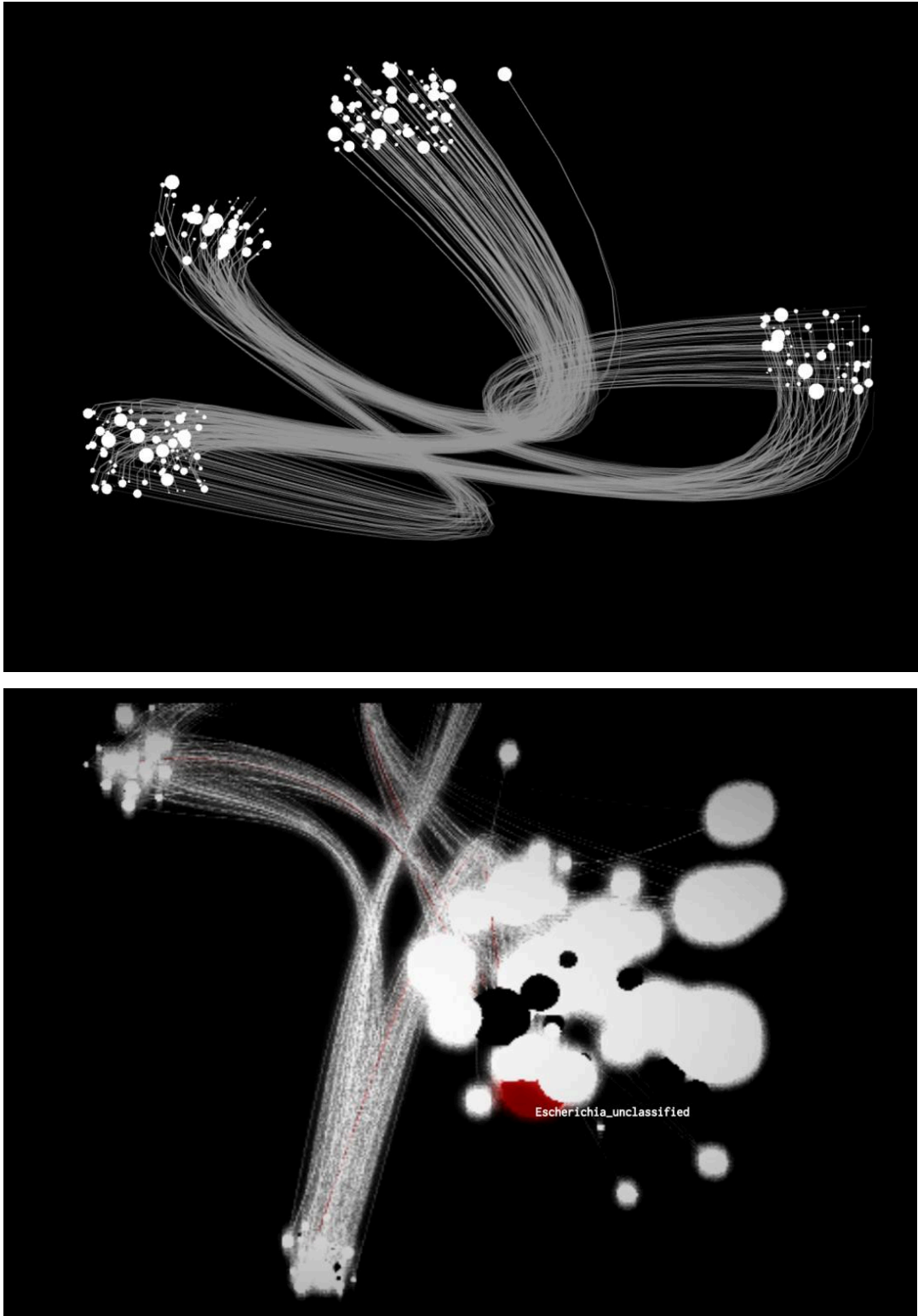


FIGURE 10 | TAXONOMICAL MAPS FOR THE FOUR HIVES. TOP (LEFT) ASTORIA, (RIGHT) CROWN HEIGHTS LANGSROTH. BOTTOM (LEFT) CROWN HEIGHTS TOP BAR, (RIGHT) FORT GREENE.



The second mapping I made was a correlation map that shows, by species, the commonalities of the species in the hives. Each data point represents a microbe. The size of the dot is proportional to the percent abundance of that microbe in the neighborhood. A path is drawn from each microbe in one hive to its twins (if they exist) in other hives. Thus, the paths begin to represent the *flow* of microbes between neighborhoods. By coloring the dots that do not have paths to other neighborhoods, I highlight the unique microbes that define what makes one neighborhood different from another.

FIGURE 11 | (TOP) FIRST PROTOTYPE OF CORRELATION MAP. (BOTTOM) A MORE DEVELOPED VERSION SHOWING UNIQUE VALUES IN BLACK AND UI ELEMENTS IN RED.



Functional Analysis

A second data set called the Functional Analysis was a part of the early data exploration. In total, I spent about a month looking at the data and thinking about how best to interpret it. The Functional Analysis differs from the Taxonomical Analysis in that, rather than showing which microbes are present, the data shows what the microbes do, i.e. what the biological function of the microbe is. Here is an example of an ID from the functional data: "ANAEROFRUCAT-PWY: homolactic fermentation". To understand each data point, a background in microbiology is required, and it became incredibly difficult for me to analyze this data set. Thus, we decided to overlay this data set by assigning meta-tags to the function of the microbe. The tags used were Toxicity, Plant, Water, Human, Animal and Unknown. Use of these meta-tags provided a more top-down understanding of what these biological processes were. Working closely with Dr. Henaff, I looked at only subset of the data; the process of assigning the meta-tags was very time consuming as Dr. Henaff had to manually research each function in the data and determine which tag to apply. From our preliminary analysis, it appears that the vast majority of the functions were unknown, but a significant portion could be attributed to plant and animal activities. While we ultimately abandoned this effort as it proved to be overly time consuming, a complete analysis could be done in the future to understand more wholly the microbial makeup of New York City.

DESIGN PROCESS

FIRST PROTOYPES: FAILED TECHNICAL EXPERIMENTS

Data analysis was just one part of the design effort. The major work was about creating an evocative visualization which will serve as the capstone of the work. As a creative coder, my medium is code and software. I wanted to work with web-based platforms to be able to make the work sharable, distributable, and procedural (i.e., able to go to any city, not just New York, but anywhere urban metagenomics might occur). This was in fact an original design value for the project.

The initial concept for this project was focused around the idea of the *dark city*. Conceived of by Kevin Slavin, it is a reference to the *Unfinished Swan*.²⁴ The idea was to create a dark, black screen. As the user navigates through the screen, the city (buildings, cars, people) are only visible as the microbes interact with those parts that are within their environment.

Tangram Map

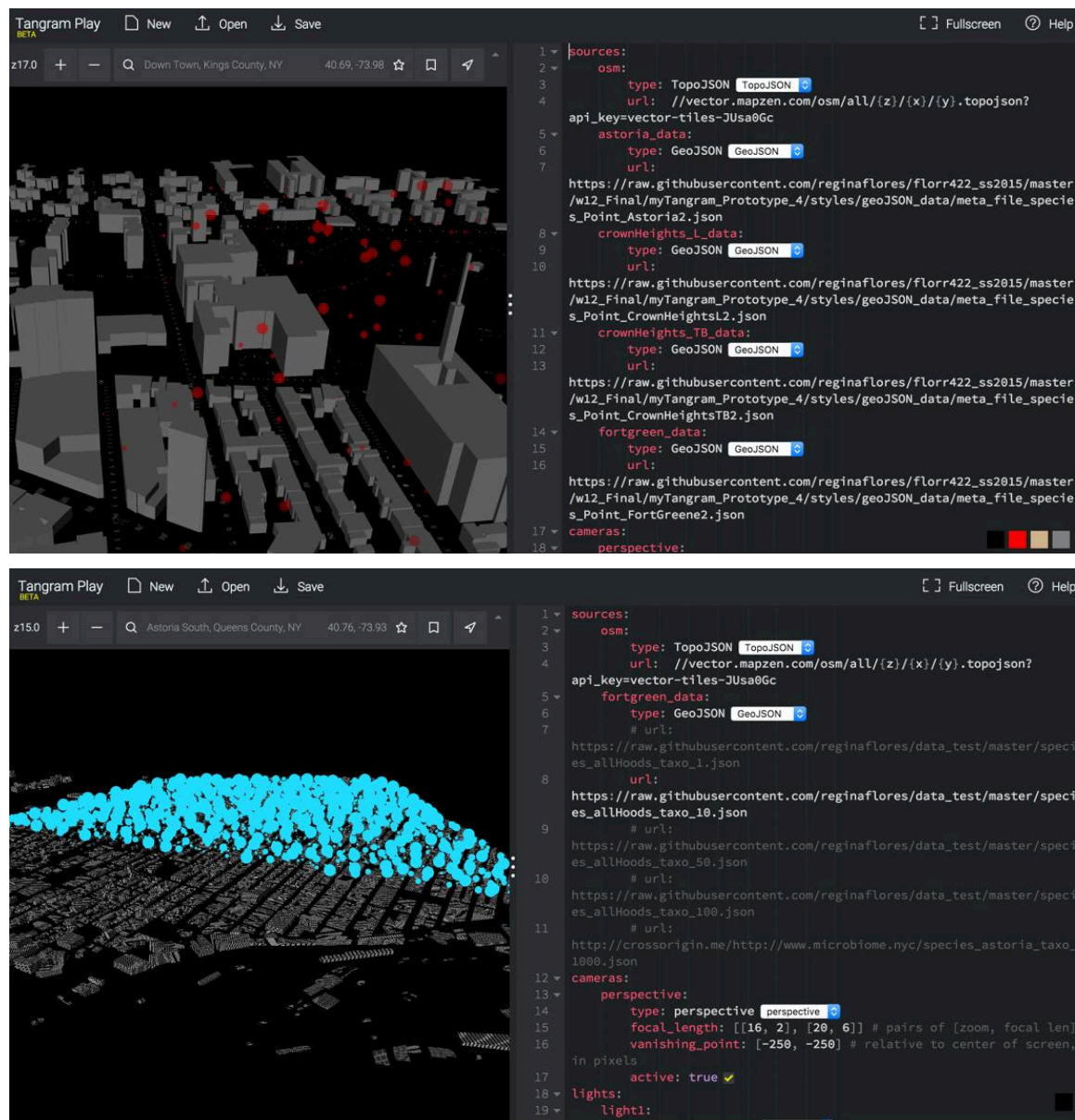
The initial idea for the design work stemmed from creating a volumetric representation of the city. Some of my early work in this project was done in the Tangram library using WebGL shaders and the Open Street Map data. I spent about three months attempting to create a cityscape in Tangram using the taxonomical data to visualize microbes interacting with the built environment.

This proved challenging because Tangram uses a totally new programming paradigm built on shader code in YAML files using geoJSON formatted data. However, the real issue was that Tangram was never built to incorporate physics libraries. I spent several weeks trying to add in

²⁴ "The Unfinished Swan." Giant Sparrow. <http://www.giantsparrow.com/games/swan/> (April, 2016).

occlusion and repulsion (physics) into the Tangram platform so that the microbes would be able to interact with buildings in the cityscape. This proved truly impossible given the time frame and choice of tools. Tangram is a great tool for mapping but ultimately did not have the technological capabilities I required.

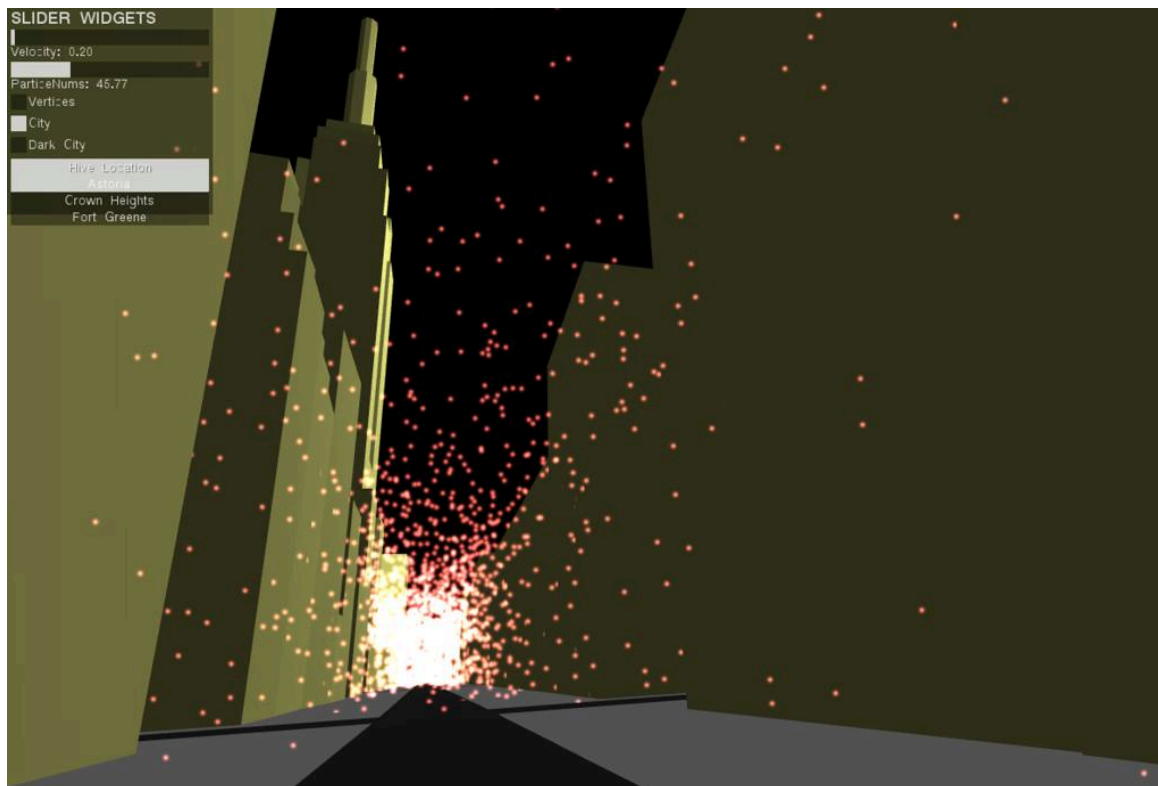
FIGURE 12 | SKETCHES OF EXPERIMENTS MAPPING TAXONOMICAL DATA IN TANGRAM USING GEOJSON DATA. NOTE: TANGRAMPLAY IS THE WEB-BASED TEXT EDITOR FOR THE SOFTWARE. (TOP) ZOOMED IN VERSION OF THE DATA INTERACTING WITH STREET-LEVEL BUILDINGS (BOTTOM) ZOOMED OUT VIEW OF WHOLE DATA SET APPLIED TO THE MAP.

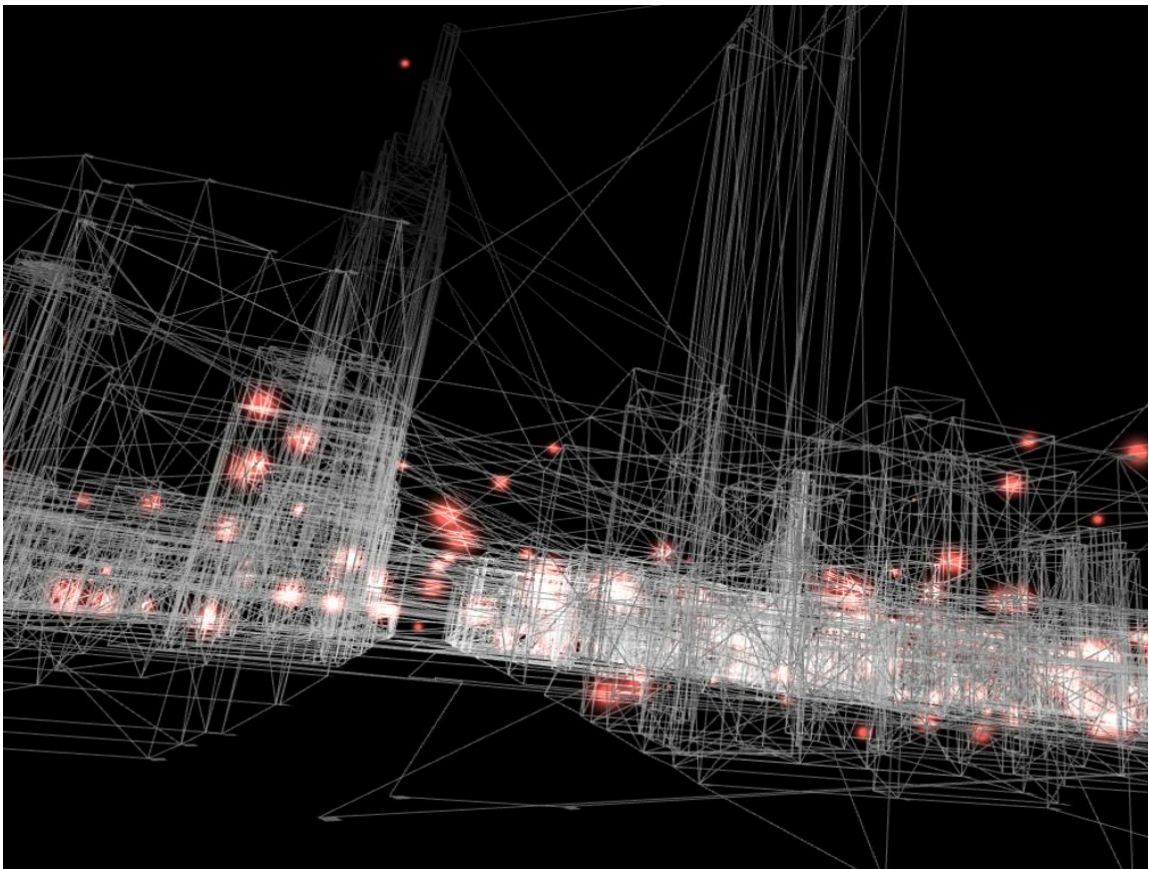
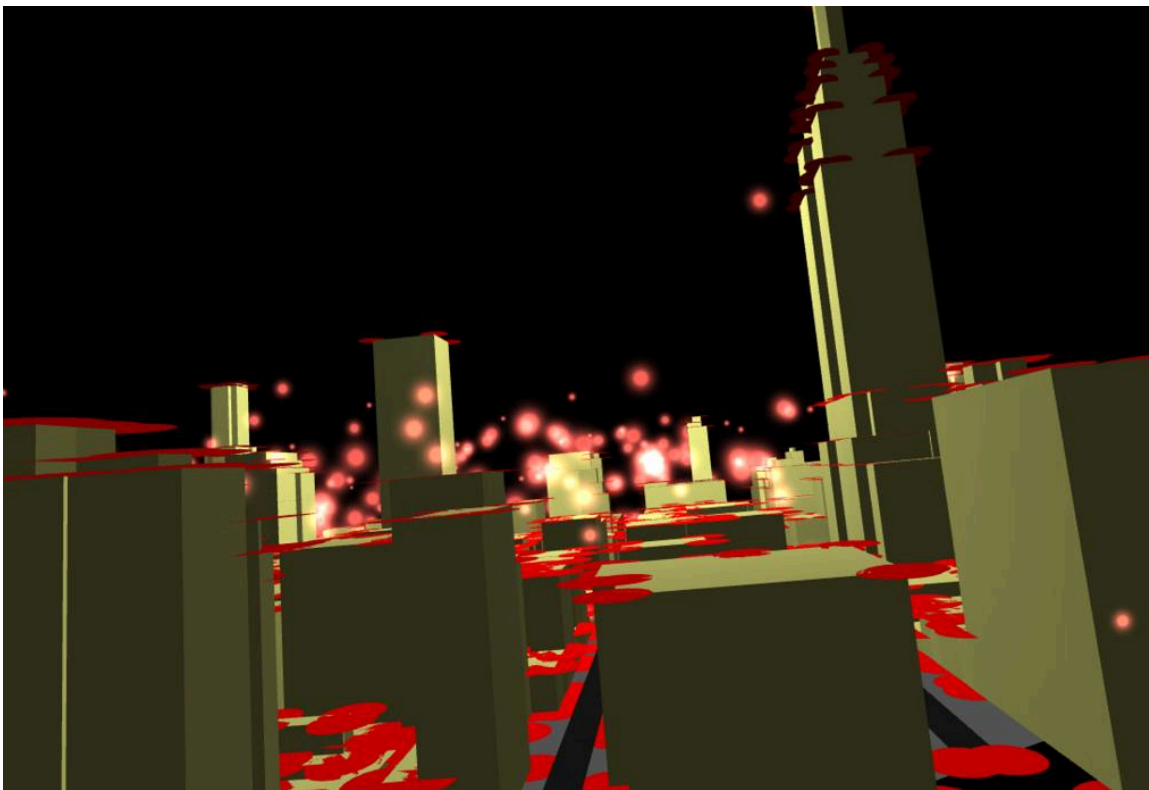


OpenFrameworks: ofxVectorTile Tangram Add-on

I also attempted to use the Tangram add-on in openFrameworks since OF does have physics capabilities. However, this also became quite a feat in attempting to re-engineer these physics engines. Trying to determine the location of the “walls” of buildings was not at all straightforward.

FIGURE 13 | SKETCHES OF OPENFRAMEWORKS EXPERIMENTS USING OFXVECTORTILE. (TOP) DATA RENDERED AS PARTICLE SYSTEM AT THE STREET LEVEL. (MIDDLE) RENDERING RED CIRCLES AT THE EDGES OF BUILDINGS IN AN ATTEMPT TO LOCATE THE WALLS TO BUILD IN COLLISION. (BOTTOM) RENDERING THE VERTICES OF THE CITY IN AN ATTEMPT TO LOCATE WALLS TO BUILD IN COLLISION.



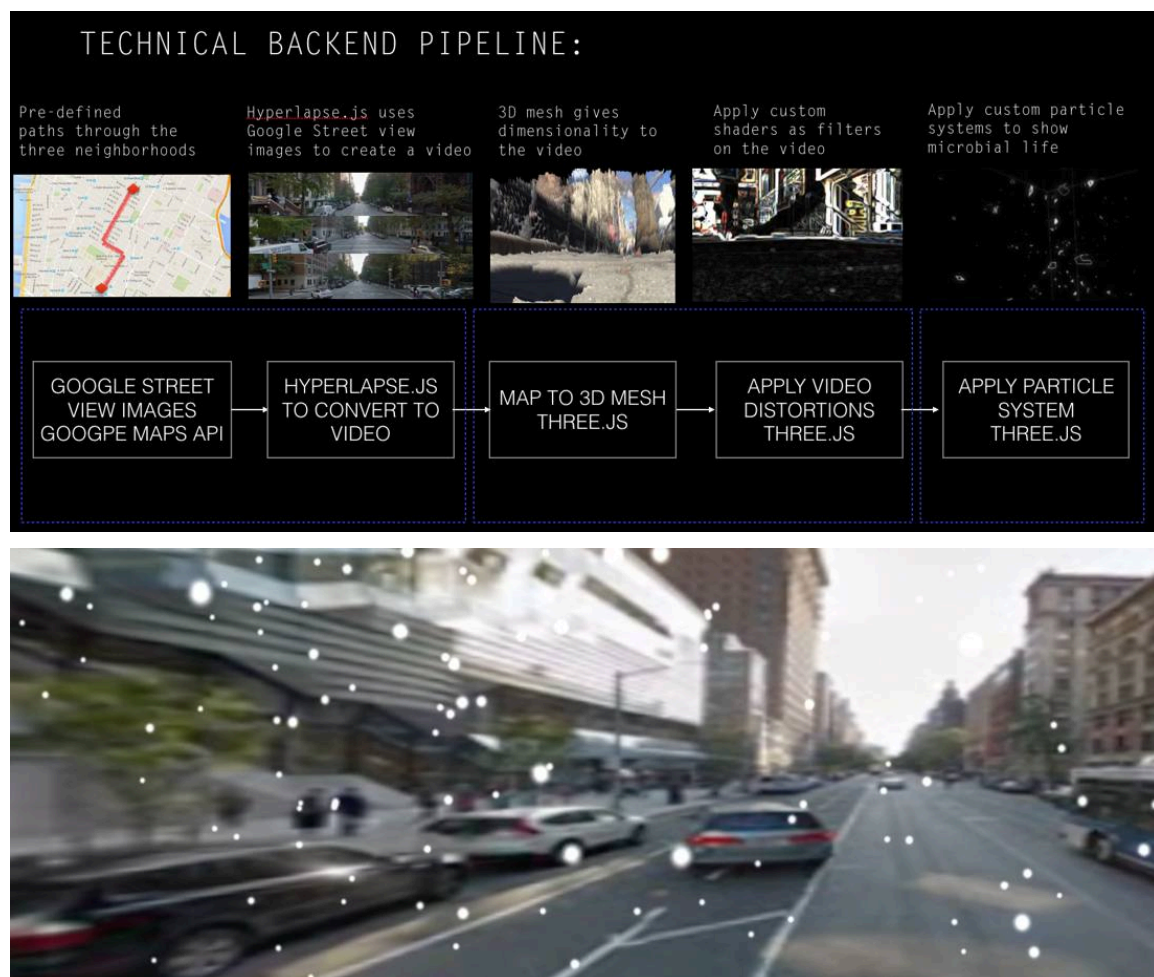


After this experiment failed as well, I went back to the drawing board to generate new concepts for the visualization. I ended up using video elements as a new direction.

Google Maps Hyperlapse

Using video, but continuing with the “map” theme, I experimented with using Google Maps and Hyperlapse.js to create a web-based tool that could generate a video from a user-generated route through a city. The code works, and I was able to successfully integrate the code into a three.js project. However, conceptually I veered away from the idea of using Google Maps as source video.

FIGURE 14 | (TOP) SCHEMATIC OF THE GOOGLE MAPS HYPERLAPSE SOFTWARE. (BOTTOM) SCREEN-SHOT OF THE HYPERLAPSE VIDEO WITH CRUDE PARTICLE SYSTEM IN THE SCENE.



DISCOVERING THE DESIGN AESTHETIC

In 1934, the biologist Jakob Von Uexküll wrote an essay called *A Stroll Through the Worlds of Animals and Men*²⁵ in which he explored the concept of the *umwelt*. The *umwelt*, a German word meaning *environment* or *surroundings*, refers to the idea that animals and humans live in a “self-centered world” in that their perception of the world is only as extensive as what their biology (eyes, ears, antennae, etc.) allows them to *see/hear/feel*. For example, an organism that has no ears cannot perceive sound in the same way a human can. On the other hand, there are sound frequencies that human ears have not evolved to perceive that dogs, for example, can. Thus, Von Uexküll writes that even though organisms may share the same environment, they have different *umwelten*, different perceptions, of how the environment is.

Early on in the formation of the design aesthetic, Kevin Slavin encouraged me to read this essay as he wanted this Von Uexküllian idea of perception to be central to the design aesthetic. *How can we place the participant into the point of view of the microbe? How can we turn the microscope onto the city to reveal the city through this lens?* These questions informed the early prototypes.

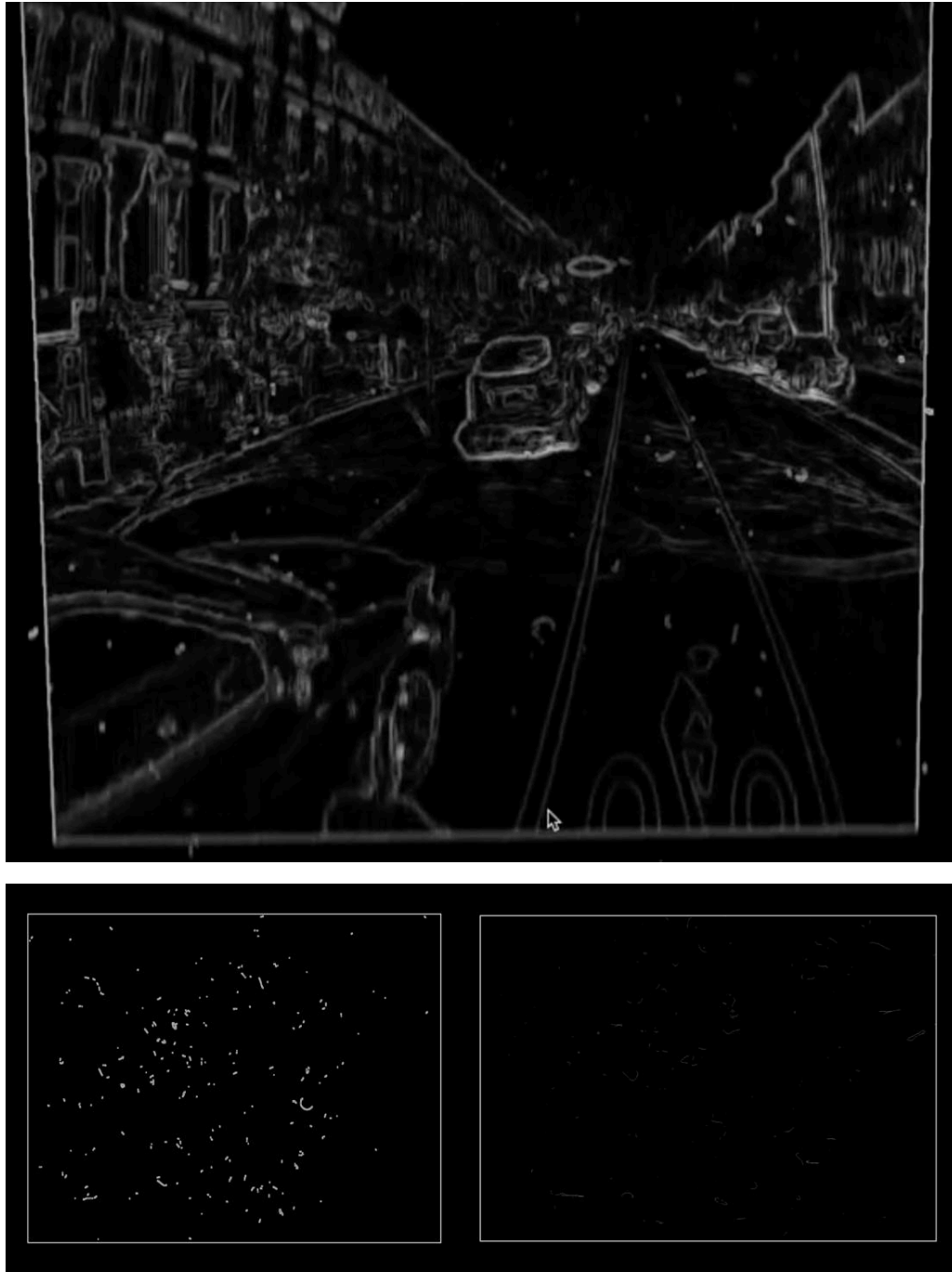
Early Prototypes: Explorations with Video

For the first prototype I used a piece of software I had developed for a previous project. I used a video of a person riding a bike through streets in New York (note the derivation of this thinking: Tangram Map >> Google Street Maps >> POV video of street) and rendered it in a 3D environment in OpenFrameworks and applied edge detection to the video. I then overlaid a particle system into the scene as an example of microbes

²⁵ *A Stroll Through the Worlds of Animals and Men*. Von Uexküll, Jakob. Published 1934. Introduction to the book “Instinct Behavior: The Development of a Modern Concept”. International Universities Press, Inc. New York.

flying through the air. Use of a flat video made the work seem one dimensional—literally the particles looked like a filter on a different plane from the video.

FIGURE 15 | (TOP) FIRST PROTOTYPE OF A FLAT VIDEO OF A NEW YORK CITY STREET WITH EDGE DETECTION AND PARTICLES OVERLAID. (BOTTOM) CLOSE-UP OF THE PARTICLES SYSTEM.



Video Mapping to a 3D Mesh

To give the work dimensionality, I then rendered the video on a mesh in OpenFrameworks and forced the mesh to vary with the video's brightness. This gave the video a wave quality and created a dream-like effect. This was the beginning of exploring how to alter the "known reality" and transform it into something else. I wanted the participant to be able to tell that the video was of a city street (something recognizable) yet be altered enough to cause the participant to question what I was showing.

FIGURE 16 | SECOND PROTOTYPE OF THE VIDEO RENDERED ON A MESH.



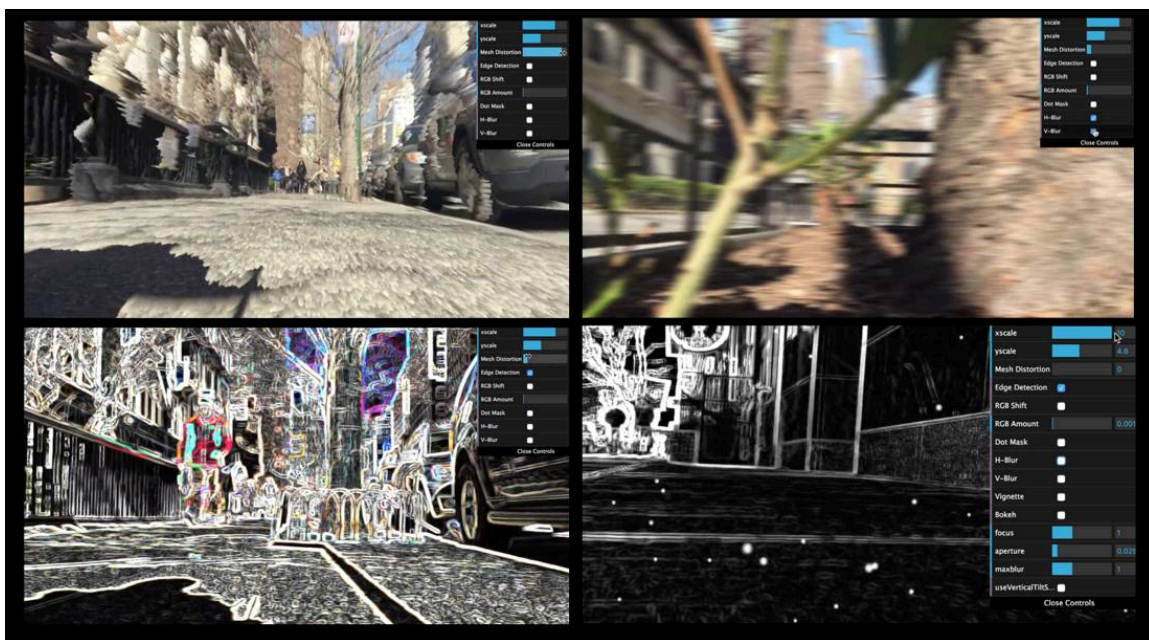
3D WebGL Software Experiments

As I discuss later in the *Exhibition* section, I made a decision early on to use a web-based environment as the distribution platform. Thus, I moved away from OpenFrameworks towards three.js. The next phase of work was to use those early OF sketches as a guide to build-out software in JavaScript. I took test video footage of streets in New York and mapped

it to a 3D mesh in three.js. I then used the post-processing libraries to achieve blurring and edge detection effects.

I also began experimenting writing particles systems in three.js to explore rendering microbes to the screen. Figure 17 shows a series of screenshots that demonstrate this early work.

FIGURE 17 | PRELIMINARY EXPERIMENTS IN BUILDING THE THREE.JS SOFTWARE (TOP LEFT) MAPPING VIDEO TO A 3D MESH (TOP RIGHT AND BOTTOM LEFT) USING POST-PROCESSING EFFECTS SUCH AS BLURRING AND EDGE DETECTION (BOTTOM RIGHT) PRELIMINARY EXPERIMENTS ADDING PARTICLES TO THE SCENE.



Depth of Field: ofxBlur

Returning to the Von Uexküllian idea of “microbe perception” and thinking about the *umwelten*, I wanted to play more with the concept of foreground vs. background. When looking down 5th Avenue, for example, a human being can perceive depth of field. She can see buildings, people, and cars that are blocks away from where she is standing. But the microbe does not have this sense of depth. The microbe can only perceive that which is near—or in other words, the microbe’s depth of field is

much narrower than the human's. Thus, I wanted to explore rendering the video such that only the objects in the foreground could be seen. Anything "far away" would be rendered to blackness. I used OpenFrameworks as a prototyping tool to achieve this using the ofxBlur add-on.

FIGURE 18 | USING OFXBLUR AS VIDEO PROCESSING (LEFT) LOOKING DOWN 5TH AVENUE WITH MINIMAL DISTORTION (MIDDLE AND RIGHT) FULL DISTORTION REMOVES THE BACKGROUND SO THAT ONLY THE FOREGROUND IS VISIBLE.



Layering Effects: Distorting Reality

The videos using ofxBlur were too realistic, too close to known reality. I wanted to distort the reality further, so I used those videos as the inputs into the three.js software I had built. Using blurring effects, the people in the scene appeared as *blobs*. This aesthetic appealed to me. It was still clear that the scene in the video was a street, but the objects were distorted in such a way that reality was beginning to bend. Could this be how microbes perceive human beings, as moving blobs?

I experimented with iterating this process to run the processed videos through various rounds (or treatments) in the three.js software to see the different effects. The more the video was processed, the more "blob-like" the objects appeared.

FIGURE 19 | USING OFXBLUR AS VIDEO INPUT AND THEN APPLYING EDGE DETECTION AND BLURRING IN THREE.JS.

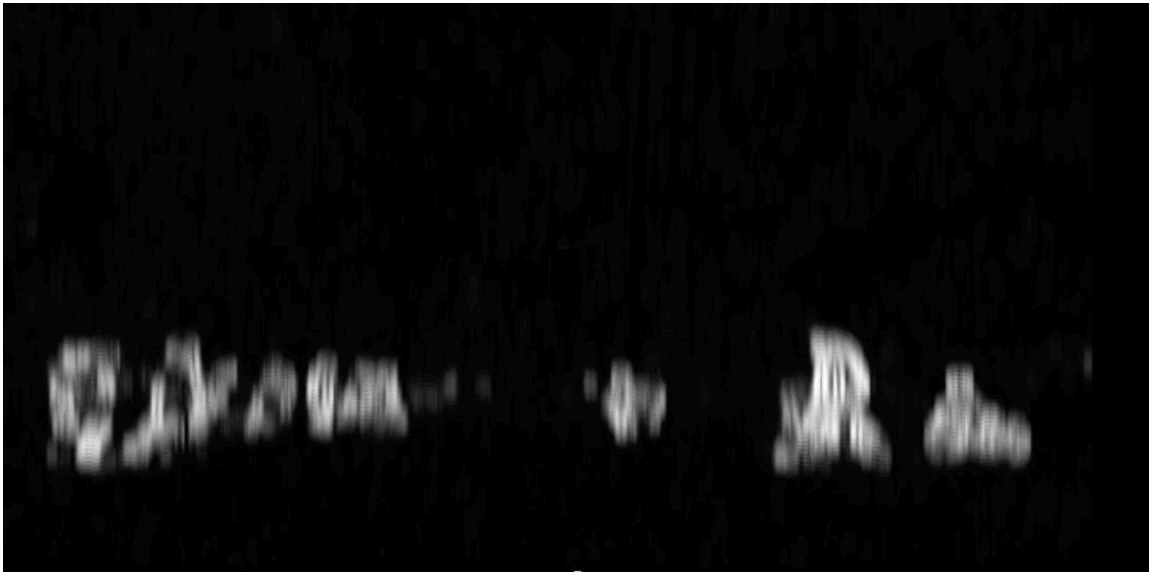
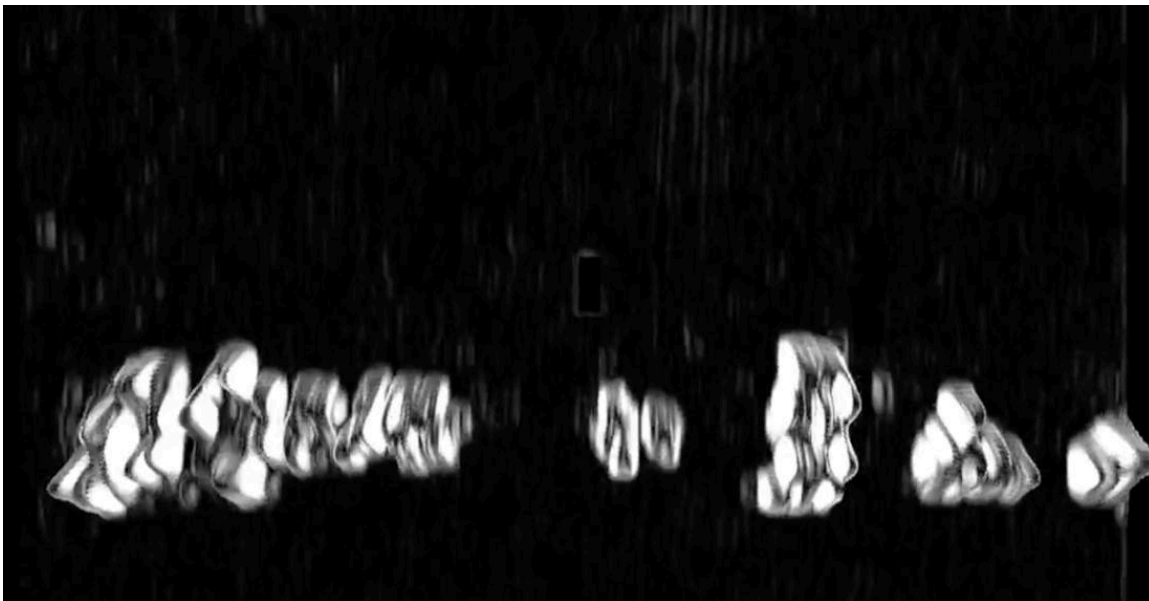


FIGURE 20 | USING OFXBLUR AS VIDEO INPUT AND THEN APPLYING EDGE DETECTION AND BLURRING IN THREE.JS - THEN APPLYING ANOTHER ROUND OF BLURRING AND EDGE DETECTION AS WELL AS DISTORTING THE 3D MESH.



These early prototypes very much informed the overall design aesthetic and were the “inputs” into the “algorithm” that created the overall design aesthetic.

MICROSCOPY EXPERIMENTS

The ability to “peer into the microscopic world” became a driving desire in order to mold the design into the aesthetic framework. Continuing to explore the idea of the microbial world, it became necessary to experiment with microscopy and travel through space into the smallest microscopic scale.

Microscope Flirtations

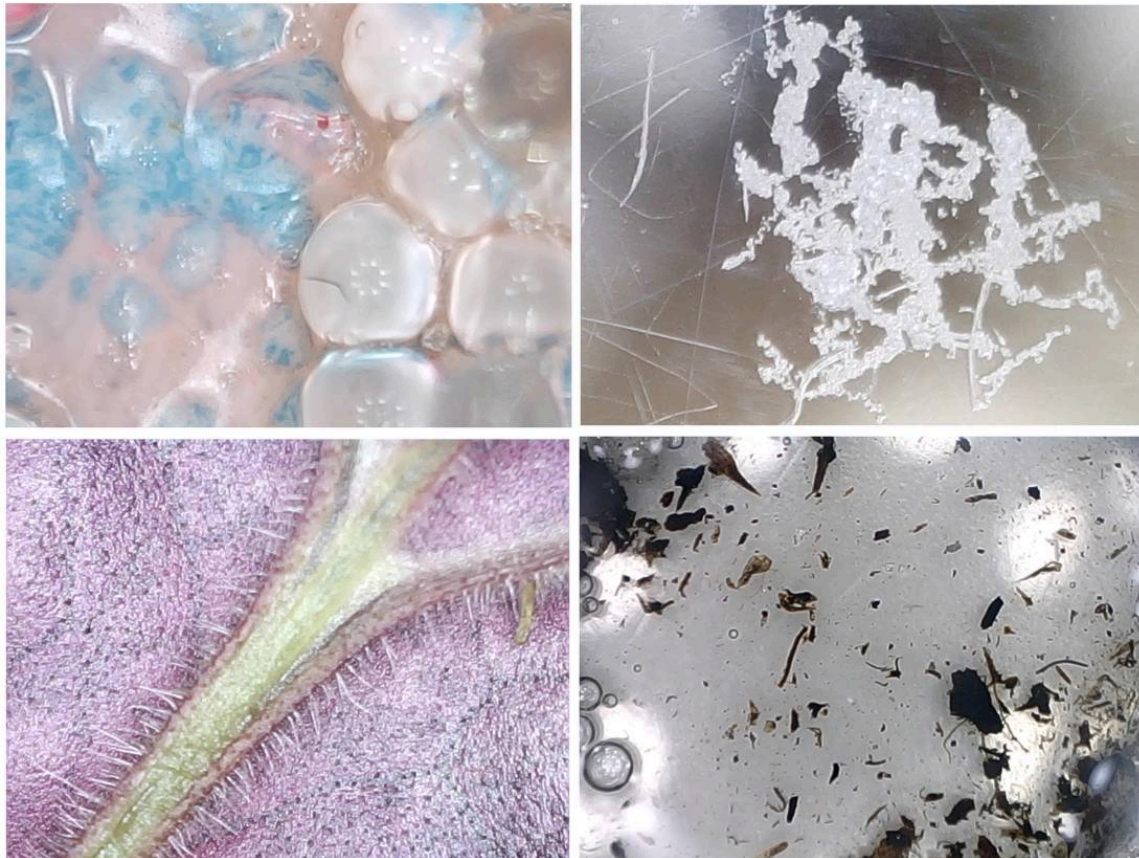
I purchased an inexpensive microscope, AmScope M158C-E Compound Monocular Microscope (40x-1000x), to conduct early experiments. Using the video capture function, I made some rough videos by zooming through layers of magnification.

FIGURE 21 | LAB SET-UP USING A 1000X MICROSCOPE.



Even using such a low-resolution microscope, it was clear that the microscopic world, although rarely seen, offered beautiful insights towards framing a design aesthetic.

FIGURE 22 | FIRST EXPERIMENTS WITH A 1000X MICROSCOPE LOOKING AT RANDOM OBJECTS FOUND ON D12 - (CLOCKWISE) ACRYLIC PLASTIC, SALIVA, PLANT LEAF, DIRT.



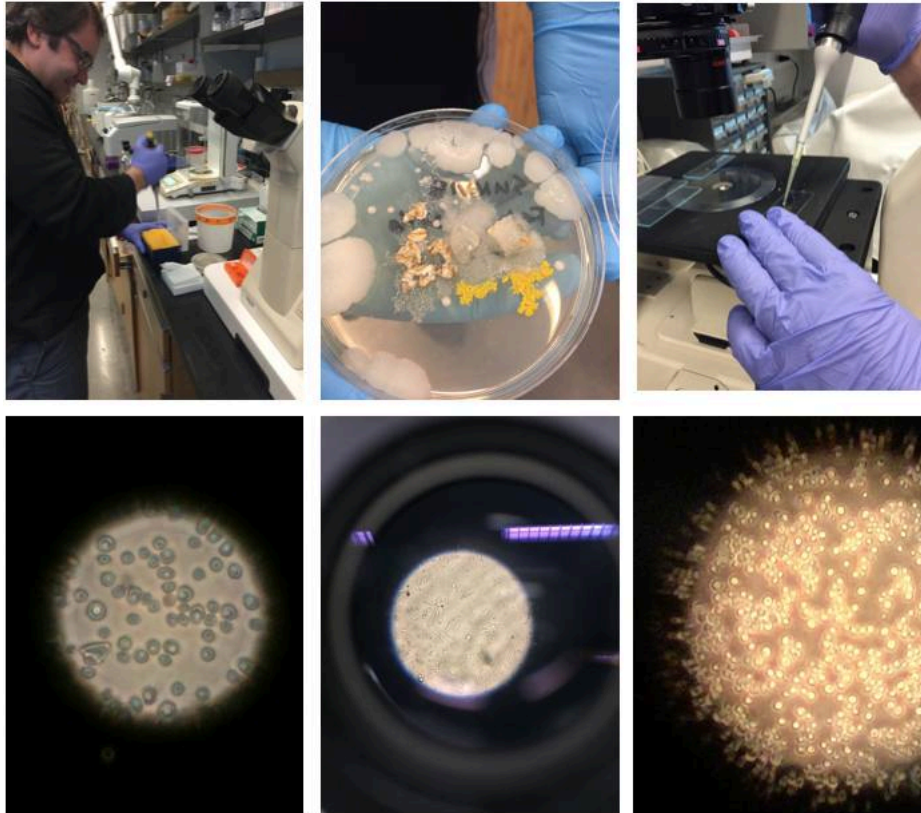
Cooper Union Lab

My next task was to use a much more powerful microscope to actually see microbes. One of my collaborators facilitated a visit to the Cooper Union Lab of Dr. Oliver Medvedik (co-founder of the community bio lab, GenSpace). Using the Nikon Diaphot 200²⁶, Dr. Medvedik used cultures in his lab to give me a “tour” of the microbial world. Even though the

²⁶ “Nikon Diaphot200 Inverted Fluorescence & Phase Contrast Tissue Culture Microscope w/Camera Port.” Spach Optics. http://www.spachoptics.com/DIAPHOT_200_p/nikon-diaphot-200.htm (April, 2016).

microscope was much more powerful, it was very challenging to document properly as there was not a mechanism to take photos or video.

FIGURE 23 | IN THE LAB AT COOPER UNION WITH DR. OLIVER MEDVEDIK.



Nanotronics Microscope

Finally, I visited Nanotronics, a company that “[c]ombine[es] optical microscopy, computational super-resolution, artificial intelligence, and robotics ...[to] bring the world’s most advanced microscope to every manufacturing sector.”²⁷ Nanotronics offered the most insightful look into the microscopic world. Being able to use the most cutting edge and state of the art equipment produced stunning images that are truly evocative of a world that is, as Kevin Slavin says, “more alive than anything we could ever build.”

²⁷ “To Build the Future, You Need to See It.” Nanotronics. <http://www.nanotronics.com> (April, 2016).

To get a sense of where these microbes live, we used various elements from the built environment as samples to look at under the microscope. We used wood and metal scraps, pieces of broken asphalt, flowers, bee-hive debris, and a local swab of the office. The Nanotronics equipment we used was:

>> Nanotronics nSpec Microscope with Nikon Objectives

>> Allied Vision Technologies GT 2750 Camera

FIGURE 24 | (TOP) A FEW OF THE SAMPLES USED FOR OBSERVATION IN THE MICROSCOPE. (BOTTOM) KEVIN SLAVIN AT NANOTRONICS.

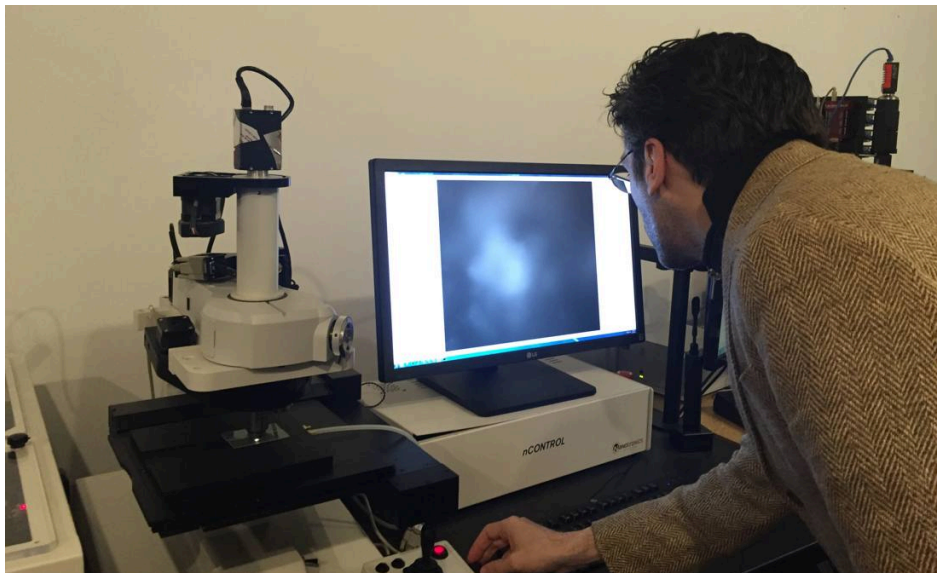
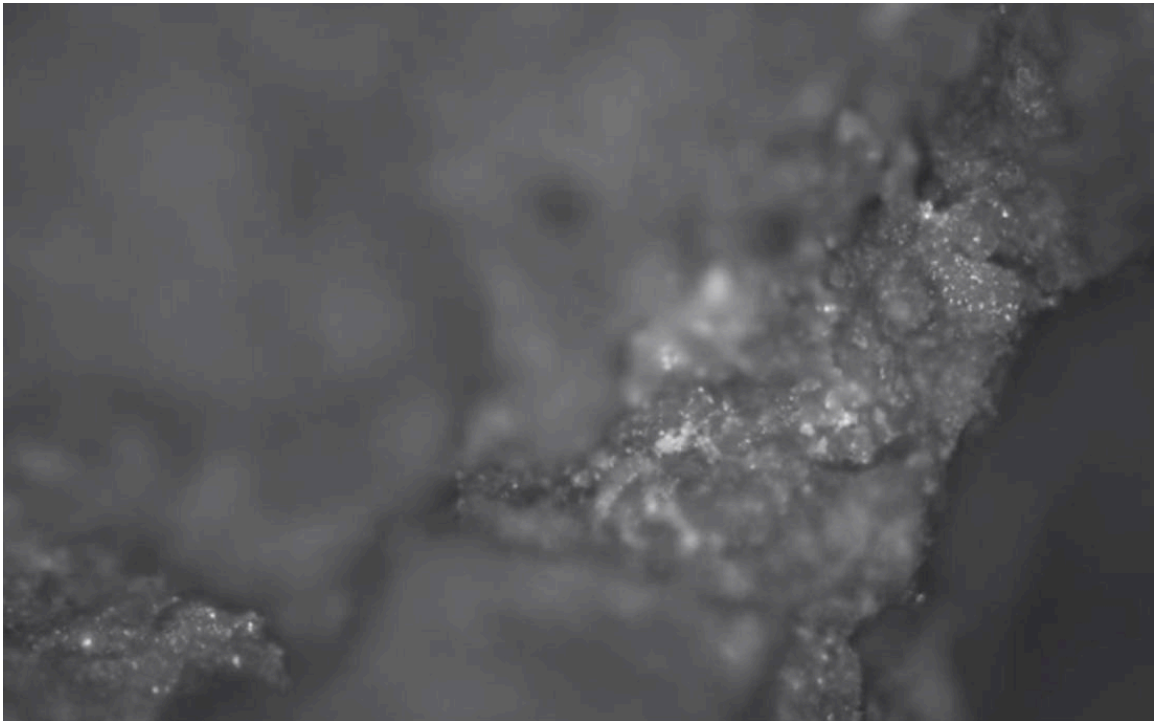
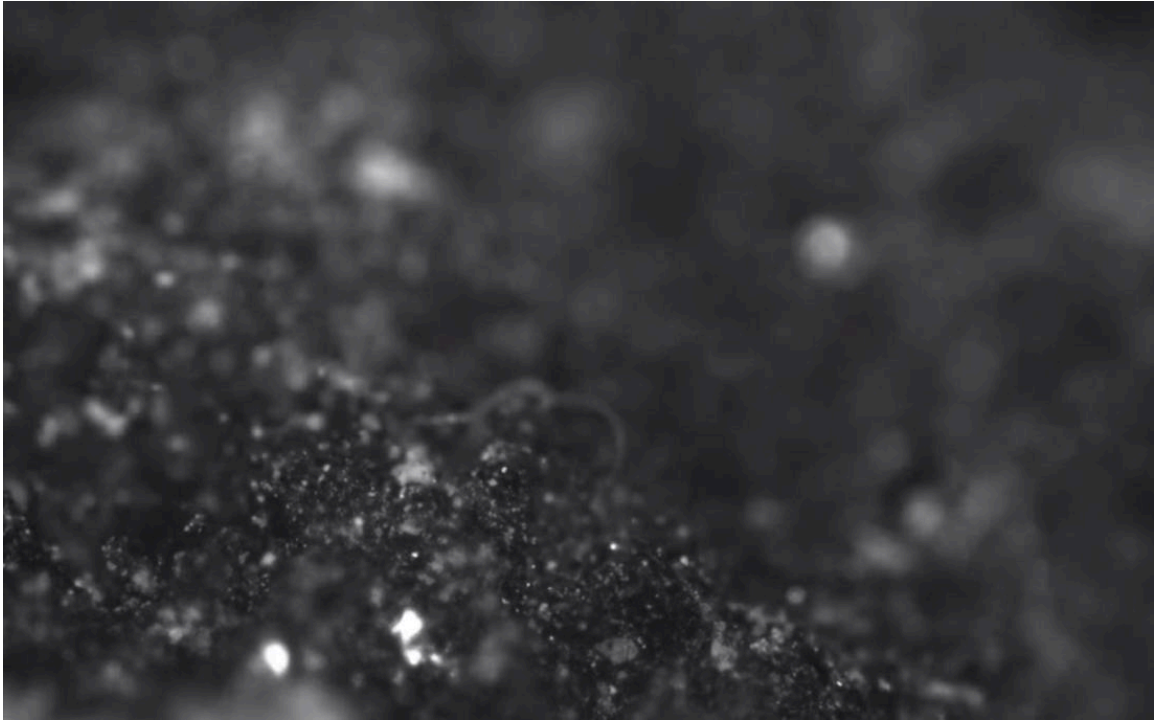
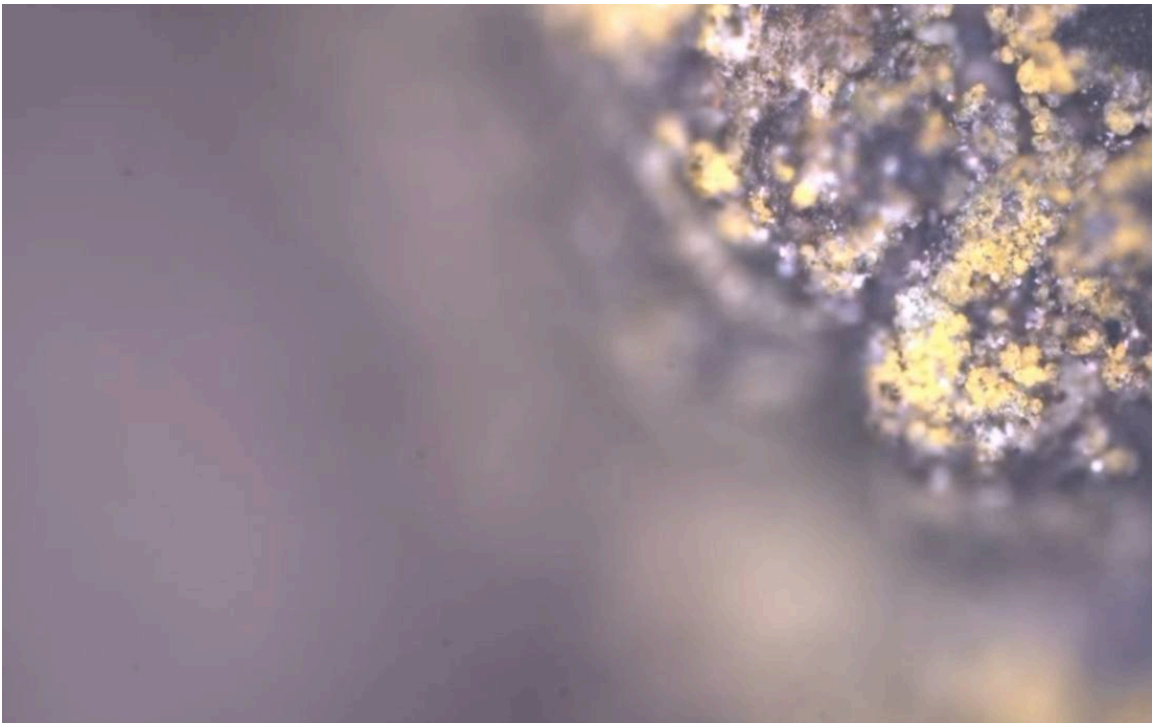
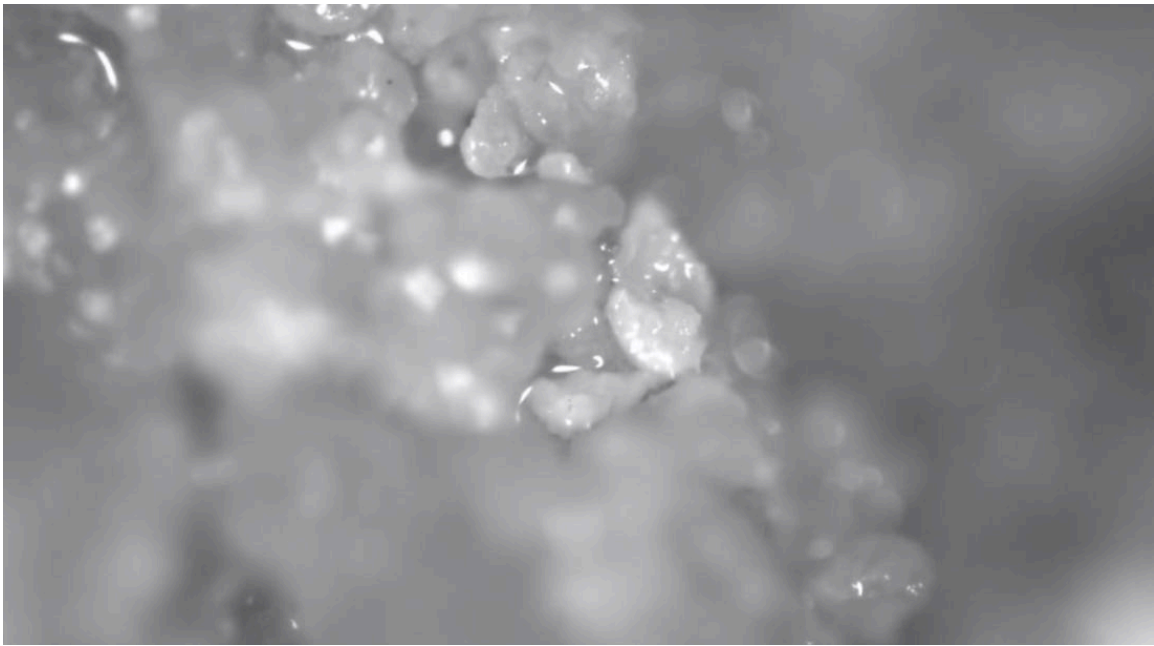


FIGURE 25 | NANOTRONICS MICROSCOPE IMAGES - IN ORDER: (1) ASPHALT, (2) BRICK, (3) MAGNOLIA PLANT, (4) BARK WITH MOSS, (5) BEE DEBRIS.







The images that were produced with the Nanotronics microscope are breathtaking. Some appear to be, not images from a microscope, but images from a telescope peering out into the cosmos. This isomorphism across scales points to a poetic beauty that is woven into the story of this work, the deep connection between the microcosm and the macrocosm: the microbe and the human, the human and the city, the city and the planet, the planet and the galaxy, and beyond.

PROTOTYPING: THERMAL IMAGING, FLOW FIELDS AND OPTICAL FLOW

In early March 2016, the team held a design workshop, or a meeting of minds, to hone in on the design aesthetic (as described in the *Design Aesthetic* section) using the early prototypes as sign posts found along a pre-established path: video, 3D environment, and post-processing effects. The decisions we made about design aesthetics are rooted deeply in a scientific framework. The major design decisions that came out of this meeting were 1) to use thermal imaging as the video source, flow fields to model microbial movement and 3) to use optical effects to mirror microscopy. For me, as the key World Development Designer, these

were pivotal decisions affecting my prototyping process, and they shaped the future direction of the work.

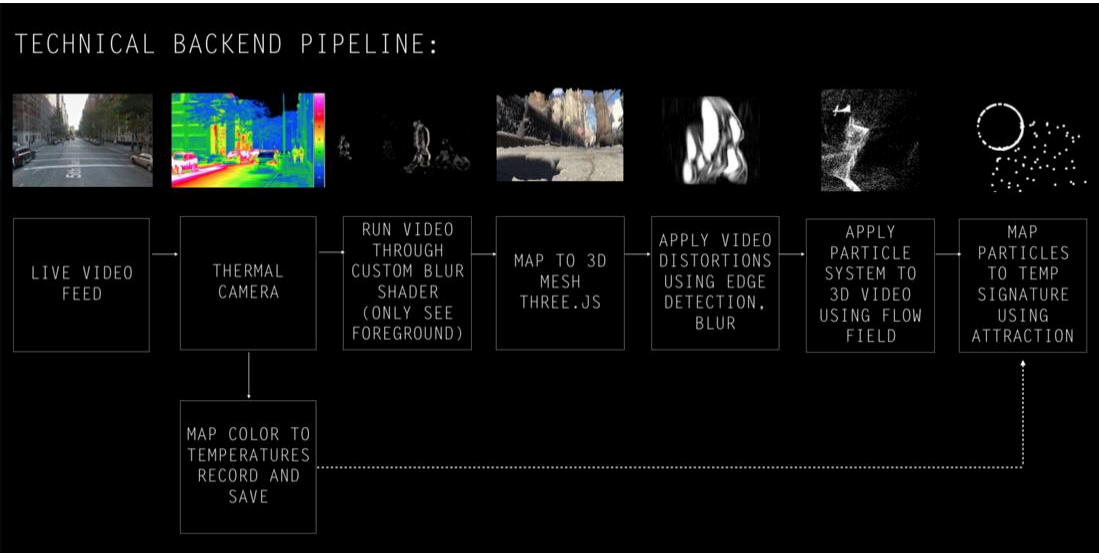
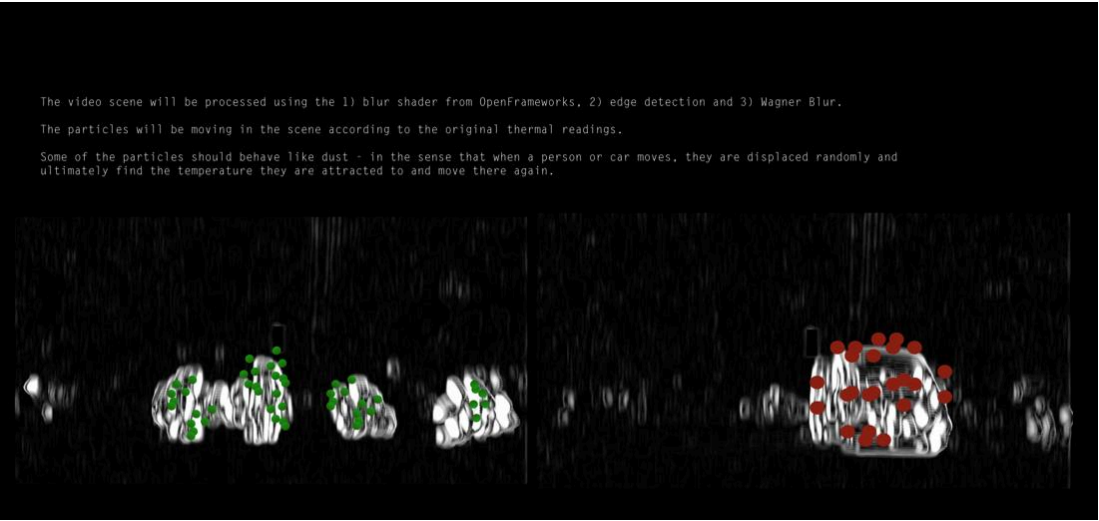
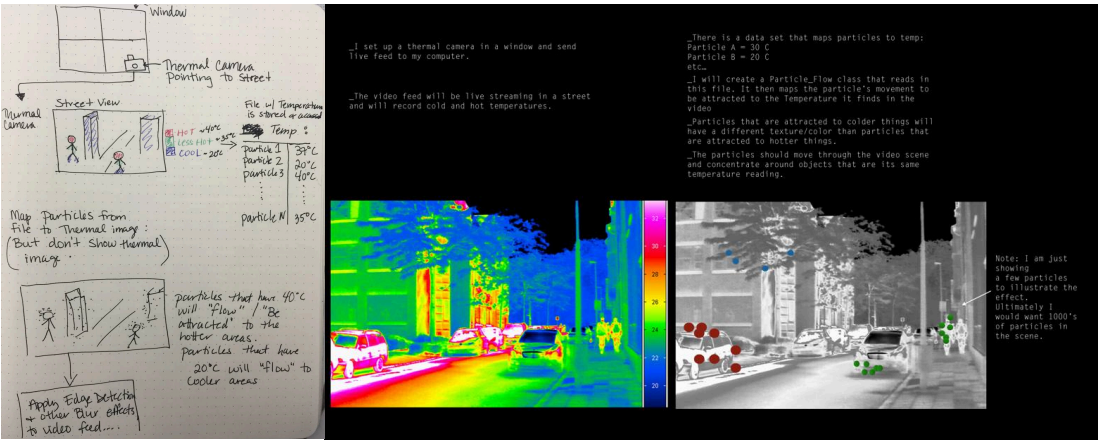
FIGURE 26 | MARCH 2016 DESIGN WORKSHOP - FROM LEFT: REGINA FLORES MIR, MIGUEL PEREZ, CHRIS WOEBKEN, ELIZABETH HÉNAFF, AND DEVORA NAJJAR.



Technical Backend Sketches

As the technology designer, one of my roles was to transform the team's vision of the look of the piece into code and a technical map. My first task was to sketch and make schematic diagrams of how the backend pipeline would work technically. Using these sketches, I was able to begin my next phase of prototyping.

FIGURE 27 | (TOP) INITIAL DEVELOPMENT SKETCHES. (MIDDLE) SCENE SKETCHES. (BOTTOM) INITIAL SCHEMATIC OF THE TECHNICAL BACKEND PIPELINE.



The initial plan was to use both a video feed with a normal camera and a thermal camera. The normal video would be piped through the software I had made (to a custom shader to remove the background, to a three.js mesh, to a post-processing effects processor). The thermal feed would be used to map the particles from the flow field into the scene.

Thermal Imaging Experiments

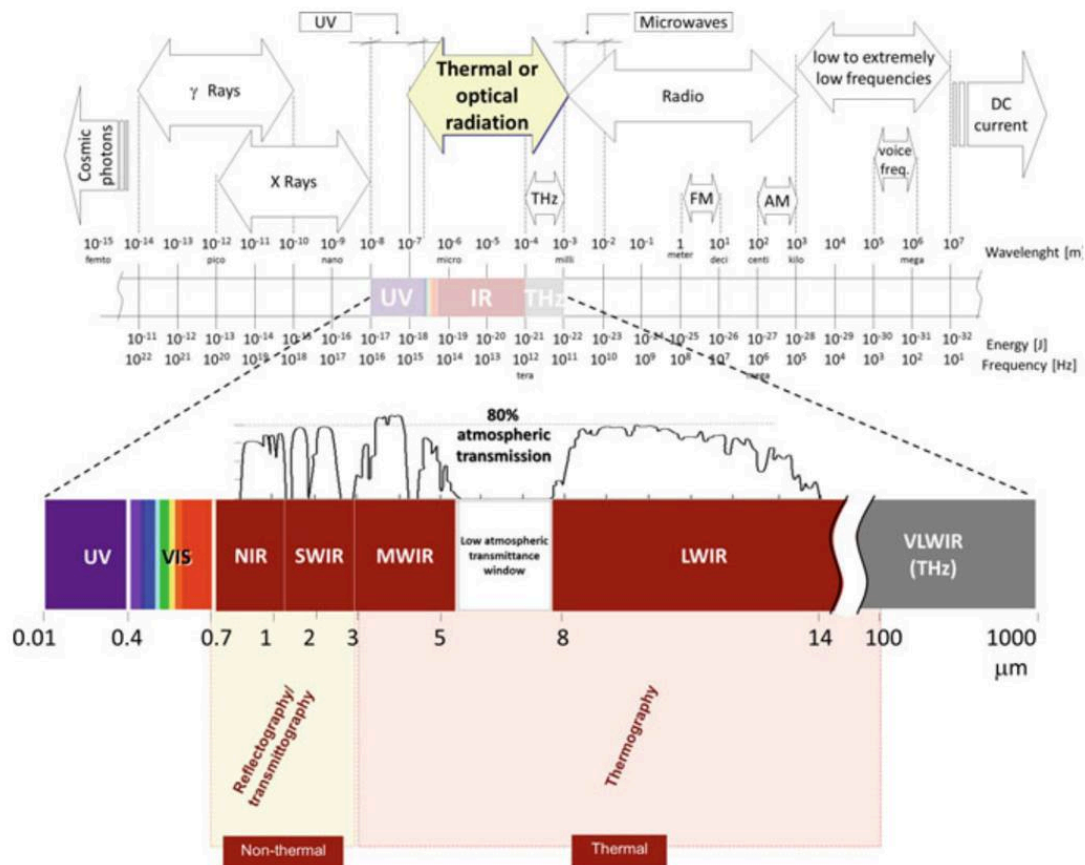


FIGURE 28 | ELECTROMAGNETIC SPECTRUM HIGHLIGHTING THERMAL RADIATION REGION. SOURCE: CLEMENTE IBARRA-CASTANEDO

Perhaps because I am a former astrophysicist who specialized in radio astronomy, when I received a design direction to “use thermal video,” my first reaction was to question what “thermal” video is. Having an intimate relationship with the electromagnetic spectrum, I was unclear about the technical specifications of *thermal*, i.e., which light

wavelengths would suffice for the project. This launched me into a two-day ethnographic study of the *creative technology community* and the *outdoor gaming community* (people who kill animals for sport tend to be a large market for thermal cameras) to understand the equipment typically used in these two disparate areas. I was trying to reconcile use of the word “thermal” as it relates to the “infrared spectrum” to the typical creative technological usage of these cameras. After extensive research into thermal cameras and the best option, given needs and costs, I determined the iPhone extension cameras would be preferable.

FIGURE 29 | EARLY EXPERIMENTS USING AN INFRARED CAMERA MADE BY HACKING A WEB-CAMERA - QUICKLY PROVED TO BE THE WRONG PATH GIVEN CERTAIN DESIGN CONSTRAINTS FROM MIT.



I used a Seek Thermal Compact XR camera (with a rig designed by Chris Woebken) to take test footage on a NYC street.

FIGURE 30 | SEEK THERMAL CAMERA EXPERIMENTS.



FIGURE 31 | EARLY THERMAL PROTOTYPES USING THE SEEK IOS CAMERA.



The Seek XR Extended Range camera ultimately was abandoned in favor of the FLIR One iPhone camera due to the FLIR's 1) wider depth of field and 2) more sharp image.

Temperature Tracking & Flow Field Early Experiments

How do microbes perceive their environment? In what ways do they *see*? *How do they move in their environments?* These are the design questions that led us to use thermal imaging (microbes can perceive temperature, and for each species there is an optimal temperature at which it thrives) and flow fields (based on computational fluid dynamics and the movement of microbes through the different microclimates that exist on an organism). Thus, the idea was to map particles via flow fields to the video using thermal tracking.

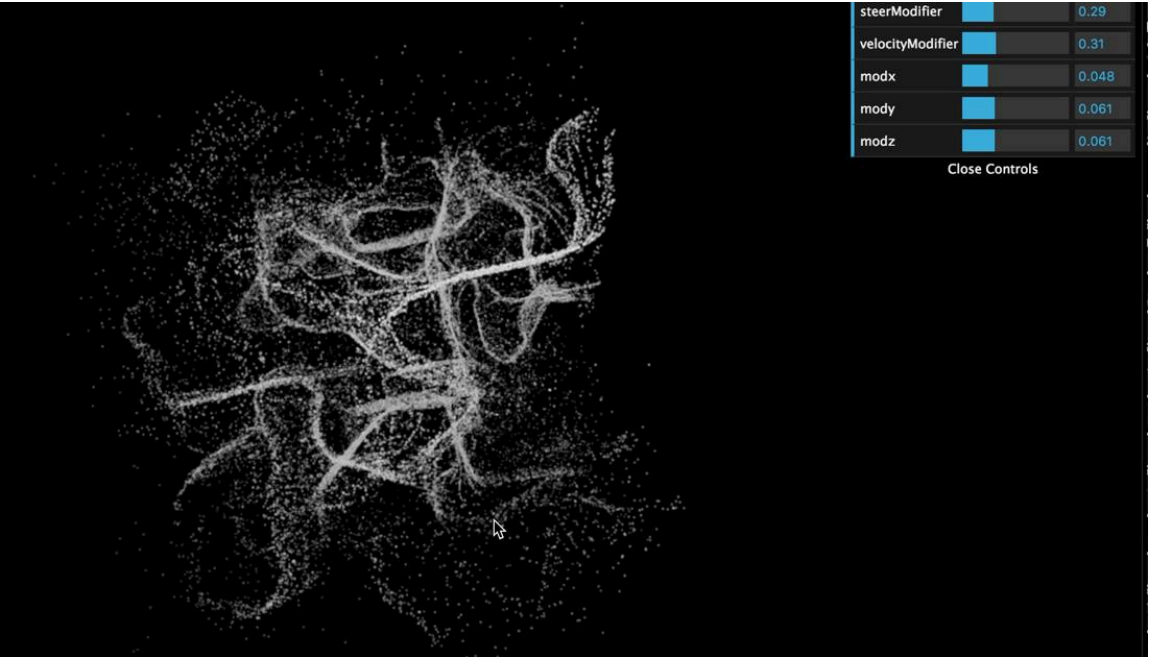
To incorporate these new design ideas, I started experimenting by building upon the previous software I had built in three.js. To model the movement of microbes in the urban environment, I experimented making 3D flow fields in three.js.

Creating the flow field was a simple enough task, but the challenge came when trying to make the particles in the flow field interact with the video. Through what mechanism will the video mesh communicate with the particles? This simple question launched an investigation into the world of computer vision.

FIGURE 32 | EARLY SOFTWARE PROTOTYPE SHOWING THE VARIOUS CANVAS RENDERINGS OF DIFFERENT FEEDS. TOP ROW: (LEFT) THERMAL VIDEO FEED. (RIGHT) NORMAL VIDEO FEED, (MIDDLE) CONVERTING NORMAL VIDEO FEED TO BLACK AND WHITE -. BOTTOM ROW: USING POST PROCESSING EFFECTS (EDGE DETECTION) ON THE NORMAL VIDEO FEED.



FIGURE 33 | BASIC FLOW FIELD USING THREE.JS.



Computer Vision Optical Flow: Farnebeck & Lukas Kanade Algorithms

Computer vision is a field of computer science that deals with the complex nature of instructing machines to *understand* (i.e., process data) images and video as the human eye does. As my thesis only deals with this topic in an ancillary way, I will not write in depth about the algorithms, their complication and various technologies, being used in the field. However, I will discuss briefly optical flow—the algorithms used to track moving objects in a video scene or “... the pattern of apparent motion of objects, surfaces, and edges in a visual scene caused by the relative motion between an observer (an eye or a camera) and the scene.”²⁸ I initially began experimenting with built-in JavaScript libraries, using first the oflow.js²⁹ library and then the tracking.js³⁰ library. However, there were limitations in the returned data. There were not enough data points to meaningfully track the scene. The program oflow.js returned less than 10 points, and tracking.js returned only one, the corner of a box that encapsulates a moving object. Therefore, neither was optimal for my needs. I also tried using the jsfeat.js³¹ library’s “Lukas Kanade optical flow” package, which became a problem to implement due to a lack of on-line documentation. Finally, I spent time looking at OpenCV³² (an open source computer vision software) and thinking about how to port Python into JavaScript. At this point in the research process, there simply was not time to dive deep into optical flow algorithms; there was a need to produce results, not more tests. Thus, I was never able to fully research this avenue. For future iterations of this work, more investigation should be made using OpenCV and specifically the Lukas Kanade algorithm.

²⁸ “Optical Flow” Wikipedia. https://en.wikipedia.org/wiki/Optical_flow (March, 2016).

²⁹ “oflow.js optical flow detection.” GitHub. <https://github.com/anvaka/oflow> (March, 2016).

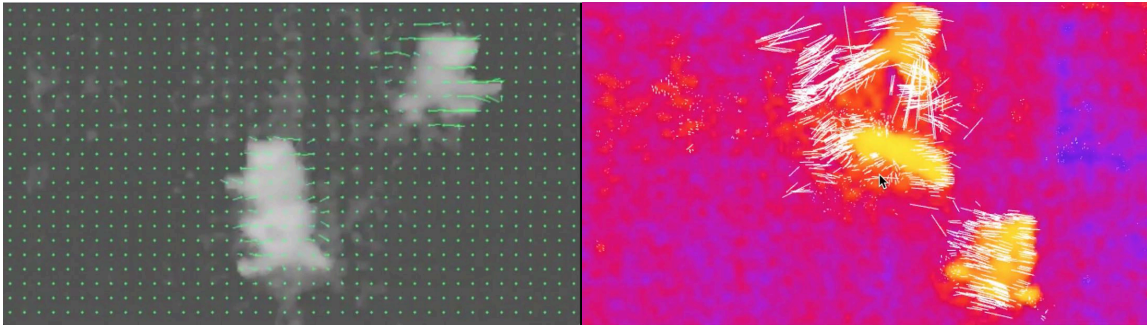
³⁰ “tracking.js A modern approach for Computer Vision on the web.” <https://trackingjs.com/> (March, 2016).

³¹ “Data Structures.” Jsfeat. <https://inspirit.github.io/jsfeat/> (March, 2016).

³² “Motion Analysis and Object Tracking.” OpenCV.

http://docs.opencv.org/2.4/modules/video/doc/motion_analysis_and_object_tracking.html (March, 2016).

FIGURE 34 | (LEFT) OPENCV IN PYTHON. (RIGHT) OPENCV IN OPENFRAMEWORKS KEY POINTS EXAMPLE.

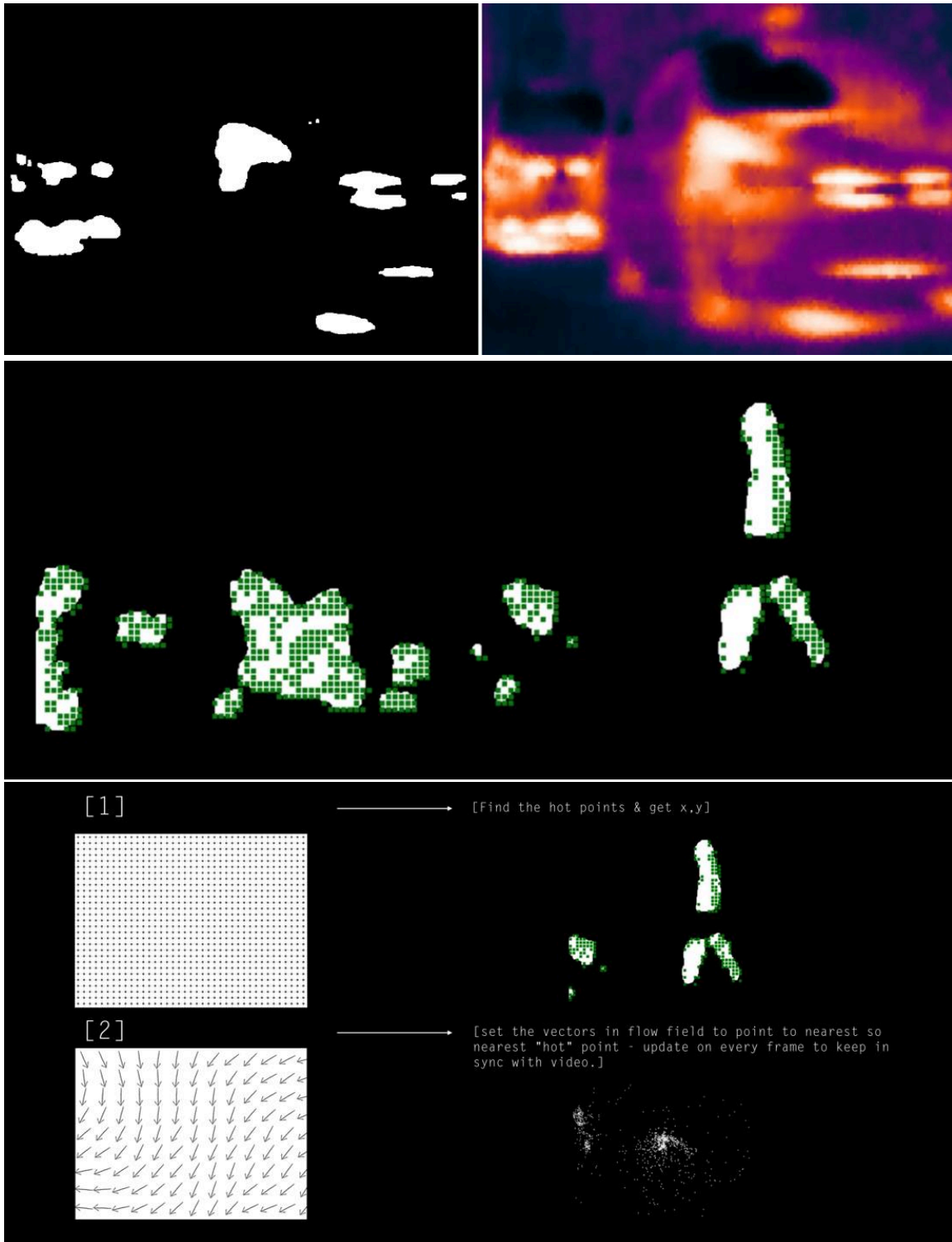


Early Optical Flow Algorithm

After struggling with the existing JavaScript libraries, I ended up writing my own optical flow algorithm specifically designed to use temperature tracking. Taking the thermal video as input, I transformed the temperature image from one with the full range of colors to one with only black and white, where white represents "hot" and black represents "cold." I then ran my optical flow algorithm on that processed video to track the movement and get data for the "hot" readings from the temperature video. The output was a set of points, called *flowData*, which gave x/y coordinates (normalized between 0-1) that denoted the positions where the movement was occurring. All of that processing happened off-screen on an invisible canvas; however, for illustration I have provided images of it below.

The flow field is a grid of vectors that tell the particles which direction to move in, and it is calculated by using the points from the optical flow data. The vectors in the flow field point to the nearest "hot" point from *flowData* (which is updated every frame of the video). The particles, which have an internal velocity, are then set to flow corresponding to the vector field. The velocity is updated every frame using readings from the flow field *steer modifier* (a parameter being set in the GUI which can be manipulated on the fly).

FIGURE 35 | (TOP) TRACKING HOT POINTS IN THE THERMAL SCENE. (MIDDLE) RUNNING OPTICAL FLOW ON THE HOT POINTS AND GETTING THE X/Y POSITIONS AS DATA TO PASS TO THE FLOW FIELD. (BOTTOM) SCHEMATIC EXPLAINING FLOW FIELD.



Refining the Algorithm: Temperature Mapping

The optical flow algorithm I wrote myself never really produced great results and was the source of some major pain points in the design process. As a workaround (a hack to be honest) to the optical flow algorithm, I created a refined algorithm and dubbed it “optical flow,” but it was not using true flow detection.

I added in temperature regimes, or bins, to be able to distinguish between hot and cold points. Bin 1 was from 40°C to 30°C and Bin 2 was from 30°C to 10°C. The refined algorithm made the thermal video very small (80x60 pixels), to ease up on processing speed which was starting to become an issue, and then stepped through each pixel, converting it from RGB to luminance. It then checked if the luminance fell within a given temperature range as defined by the bin (bins were on GUI sliders allowing the user to determine a range) and this turned it into “flowData.” Of course “flowData” was not really doing any flow detection, rather it just checked which pixels in the small video were bright enough to be considered “hot” (or “cold”).

FIGURE 36 | EARLY ITERATION WITH JUST ONE TEMPERATURE BIN. (LEFT) AERIAL SHOT OF SIDEWALK. THE HOTTEST AND BRIGHTEST SPOTS IN THE SCENE ARE PEOPLE AND A DOG. THE PINK DOTS OVERLAID ARE ILLUSTRATIVE OF THE PIXELS BEING TRACKED THESE POINTS ARE WHAT THE PARTICLE SYSTEM (MICROBIAL CLOUD) WILL BE MOVING TOWARDS. (RIGHT) THE SAME SCENE WITH THE PINK TRACKING DOTS TURNED OFF.

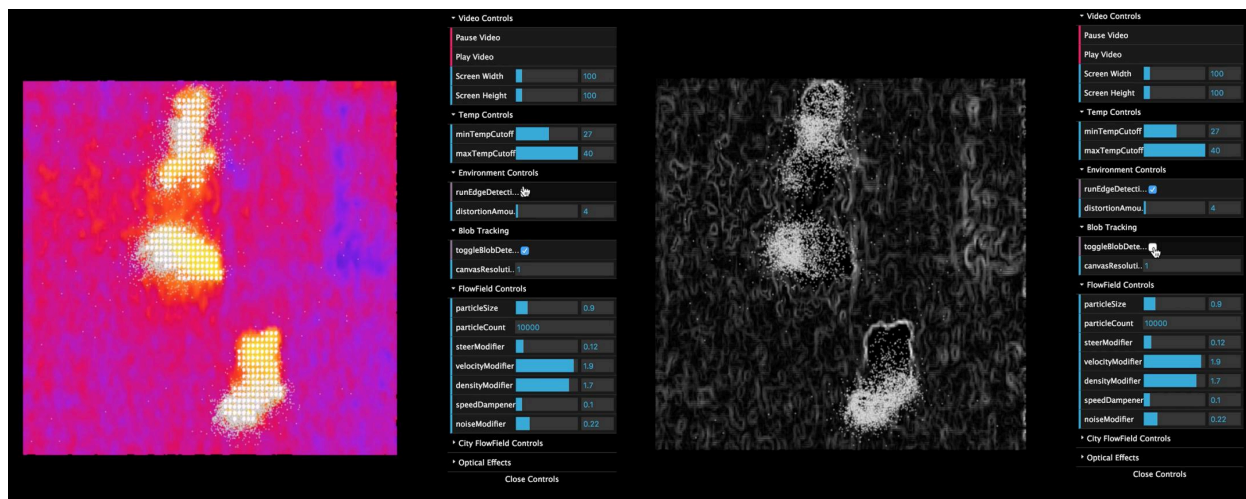
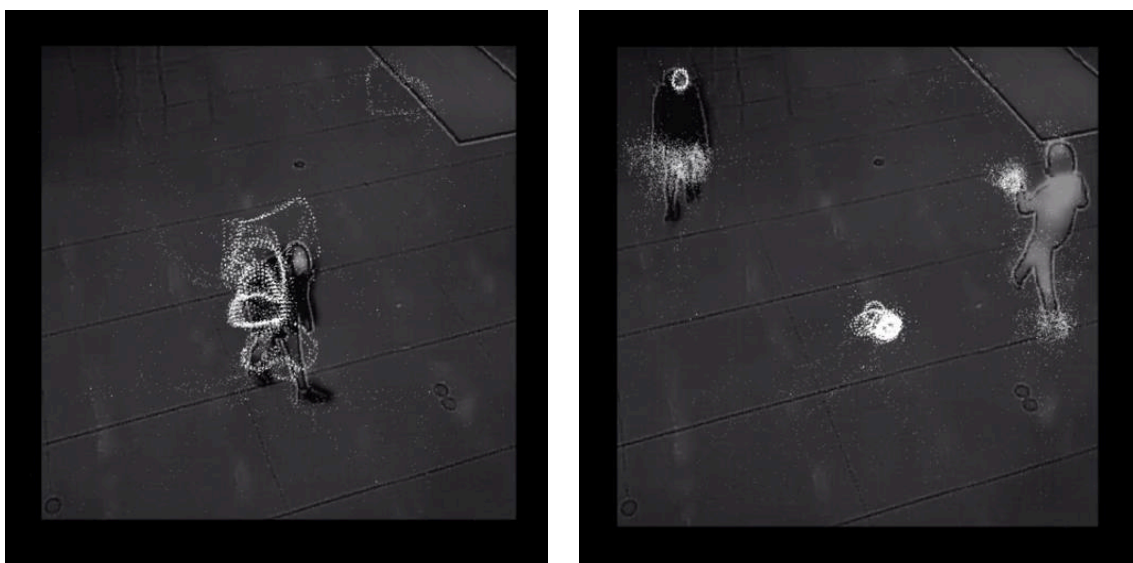


FIGURE 37 | STILLS FROM THE APP USING BLACK AND WHITE THERMAL IMAGING.



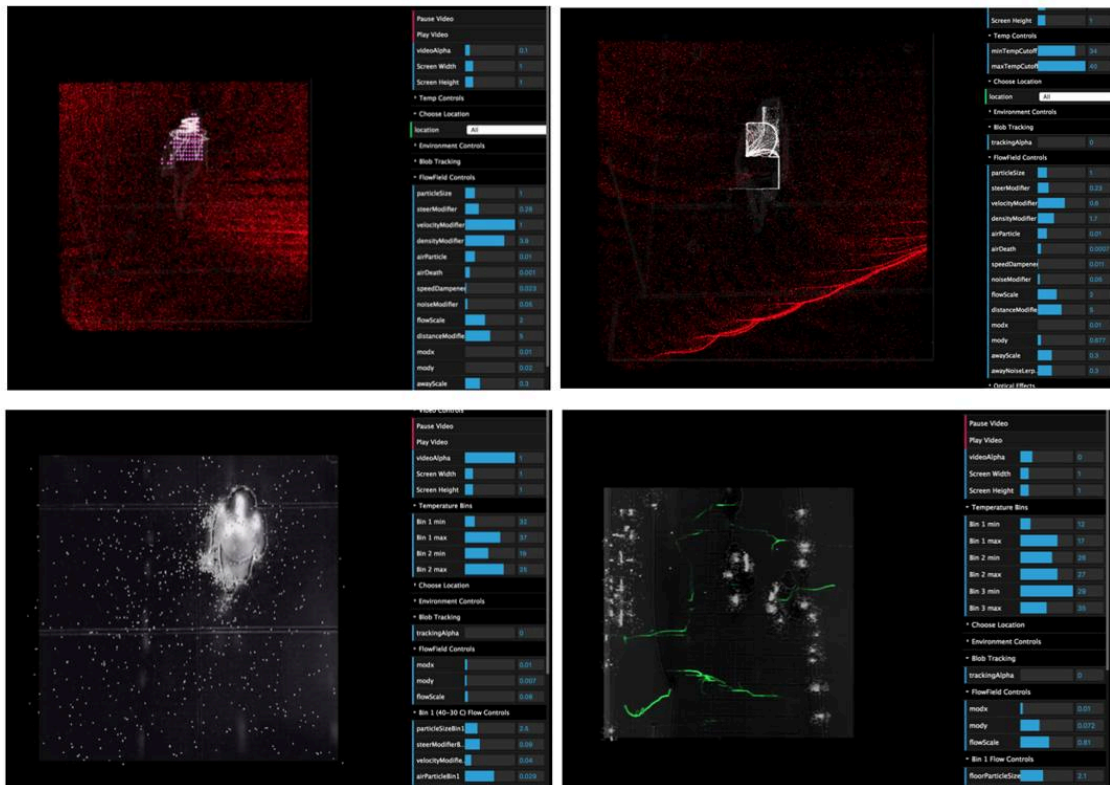
Adding Multiple Systems

The next phase of work came in mapping microbes (particles) to the environment, i.e., the areas in the scene that are not moving, like the floor, walls, etc. My first approach was to add in an additional temperature bin to account for a wider range of microbial activity. I experimented with assigning these “floor particles” to different flow fields.

Adding in multiple systems began to introduce performance issues. That is to say, even using a high-end machine with state of the art processing power, rendering the project in three.js (with video, thermal tracking, and multiple particle systems) was very processor-intensive and became a hindrance to aesthetic quality. It was around this time in the design prototyping process that Kevin Slavin decided to rethink the design work-flow in order to be able to reproduce the work at scale (i.e. adding in more and more complexity). Thus, he brought an After Effects technician onto the team to work closely with the design leads to recreate the aesthetic effects in After Effects, which is a more

stable environment. (See the *Venice Biennale* section for more details). Given time constraints and the pressure of completing the project, Kevin did not want to gamble with performance issues in a creative technology platform.

FIGURE 38 | VARIOUS ITERATIONS OF PROTOTYPES MAPPING ENVIRONMENTAL PARTICLES.



I think with more time, these performance issues could have been resolved, and in fact, I did tweak the code in a later iteration to be a bit faster.

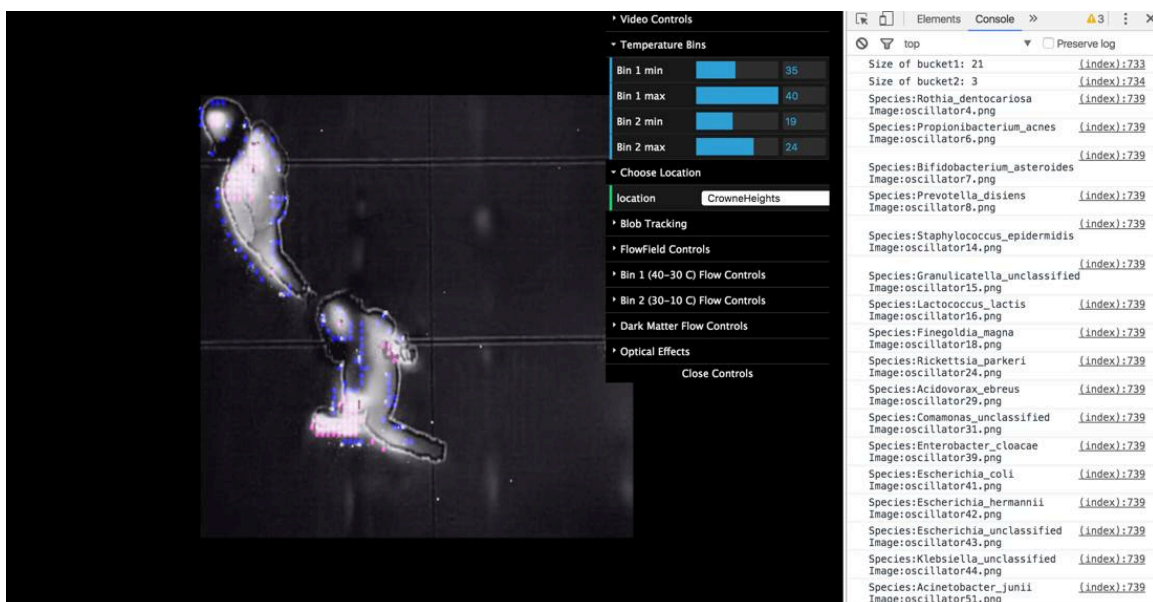
Mapping Microbes to Particles by Temperature, Abundance, and Location

A unique set of microbial species exists in each location (Astoria, Fort Greene, and Crown Heights). Each species thrives within a particular optimal temperature and is associated with a set percent abundance (how

much of that species is in a location). The software assigns 10k particles for each temperature bin, i.e., 10k hot particles and 10k cold particles. I added in a feature that allows the user to select a neighborhood from a drop-down menu. Once the neighborhood is selected, the software assigns each microbe to a unique image (created by Chris Woebken) according to its percent abundance. (So, if there is 90% of species A and 10% of species B, then there would be 9,000 particles with image-A and 1,000 particles with image-B.) The image is mapped onto the particle and flows in the scene according to its individual temperature.

FIGURE 39 | (TOP) SAMPLE JSON FILE SHOWING TEMPERATURE FOR EACH SPECIES. (BOTTOM) BROWSER SHOWING CONSOLE TO ILLUSTRATE THE PROCESSING THAT OCCURS ON THE BACKEND WHEN A USER SELECTS A NEIGHBORHOOD.

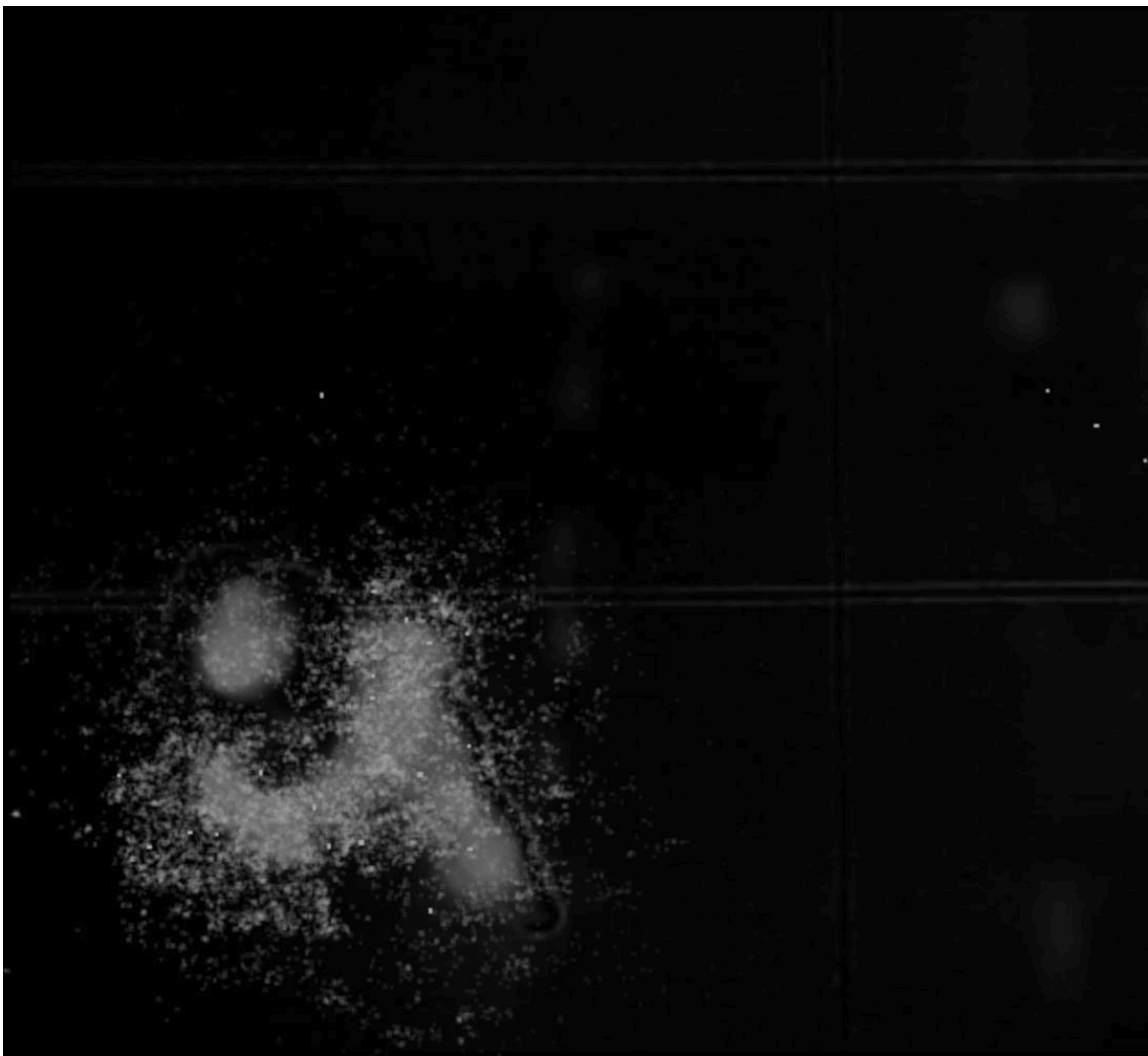
```
[
  {
    "location": "Astoria",
    "percentage": 0.59015,
    "species": "Granulicella_unclassified",
    "temp": 20.0
  },
  {
    "location": "Astoria",
    "percentage": 0.49463,
    "species": "Agromyces_unclassified",
    "temp": 30.0
  },
  {
    "location": "Astoria",
    "percentage": 6.591689999999999,
    "species": "Curtobacterium_unclassified",
    "temp": 30.0
  },
  {
    "location": "Astoria",
    "percentage": 0.50272,
    "species": "Rothia_dentocariosa",
    "temp": 37.0
  },
  {
    "location": "Astoria",
    "percentage": 0.53873,
    "species": "Rothia_unclassified",
    "temp": 30.0
  },
  {
    "location": "ForteGreen",
    "percentage": 0.06866,
    "species": "Acinetobacter_schindleri",
    "temp": 30.0
  },
  {
    "location": "ForteGreen",
    "percentage": 62.20783,
    "species": "Acinetobacter_unclassified",
    "temp": 25.0
  },
  {
    "location": "ForteGreen",
    "percentage": 30.585089999999997,
    "species": "Pseudomonas_putida",
    "temp": 27.5
  },
  {
    "location": "ForteGreen",
    "percentage": 35.03863,
    "species": "Pseudomonas_unclassified",
    "temp": 37.0
  },
  {
    "location": "ForteGreen",
    "percentage": 0.6207600000000001,
    "species": "C2likevirus_unclassified",
    "temp": 30.0
  },
  {
    "location": "CrownHeights",
    "percentage": 0.9617600000000001,
    "species": "Pseudomonas_fragi",
    "temp": 30.0
  },
  {
    "location": "CrownHeights",
    "percentage": 0.44893,
    "species": "Pseudomonas_mandelii",
    "temp": 30.0
  },
  {
    "location": "CrownHeights",
    "percentage": 14.31231,
    "species": "Pseudomonas_putida",
    "temp": 27.5
  },
  {
    "location": "CrownHeights",
    "percentage": 50.38316,
    "species": "Pseudomonas_unclassified",
    "temp": 37.0
  },
  {
    "location": "CrownHeights",
    "percentage": 0.24475,
    "species": "Stenotrophomonas_maltophilia",
    "temp": 30.0
  }
]
```



MIT Media Lab Members Week

MIT Media Lab Members Week is a week-long exhibition of work by all the groups at the Media Lab. The members and financial sponsors of the groups are given an opportunity to see the progress made in the various projects. My piece was shown at the Playful Systems Lab and during Kevin Slavin's talk. Mori Building Corp. representatives were in attendance. By all accounts, the work shown at Member's Week was very well received.

FIGURE 40 | STILL FROM THE PIECE SHOWN AT MIT MEDIA LAB MEMBER'S WEEK EXHIBITION.

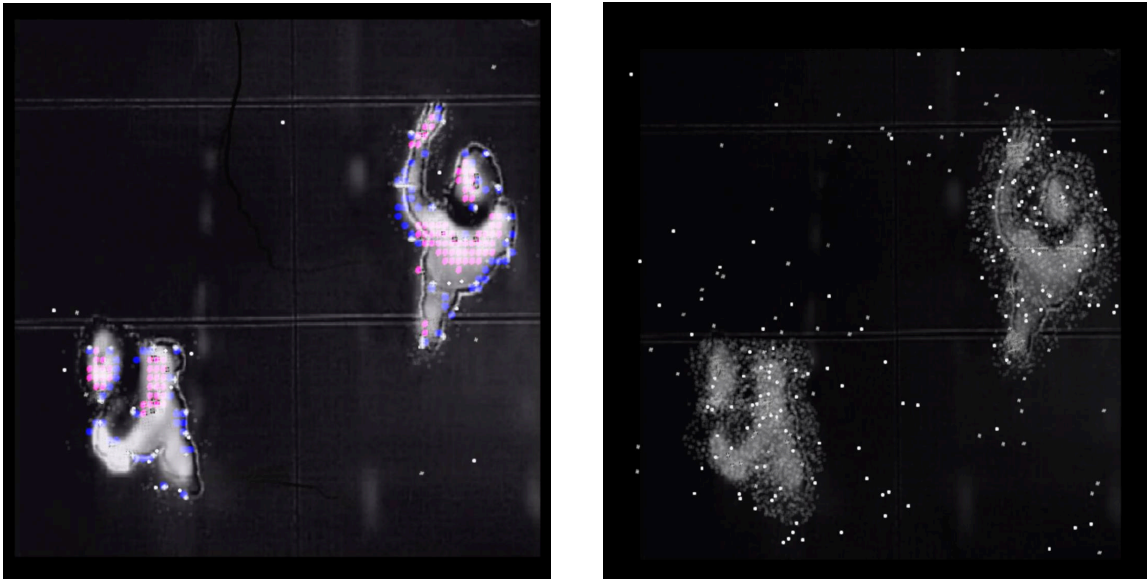


FINAL DESIGN

FINAL WEB-BASED EXPERIENCE DESIGN

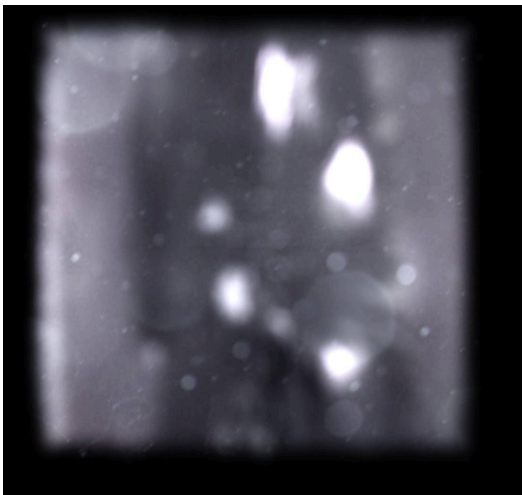
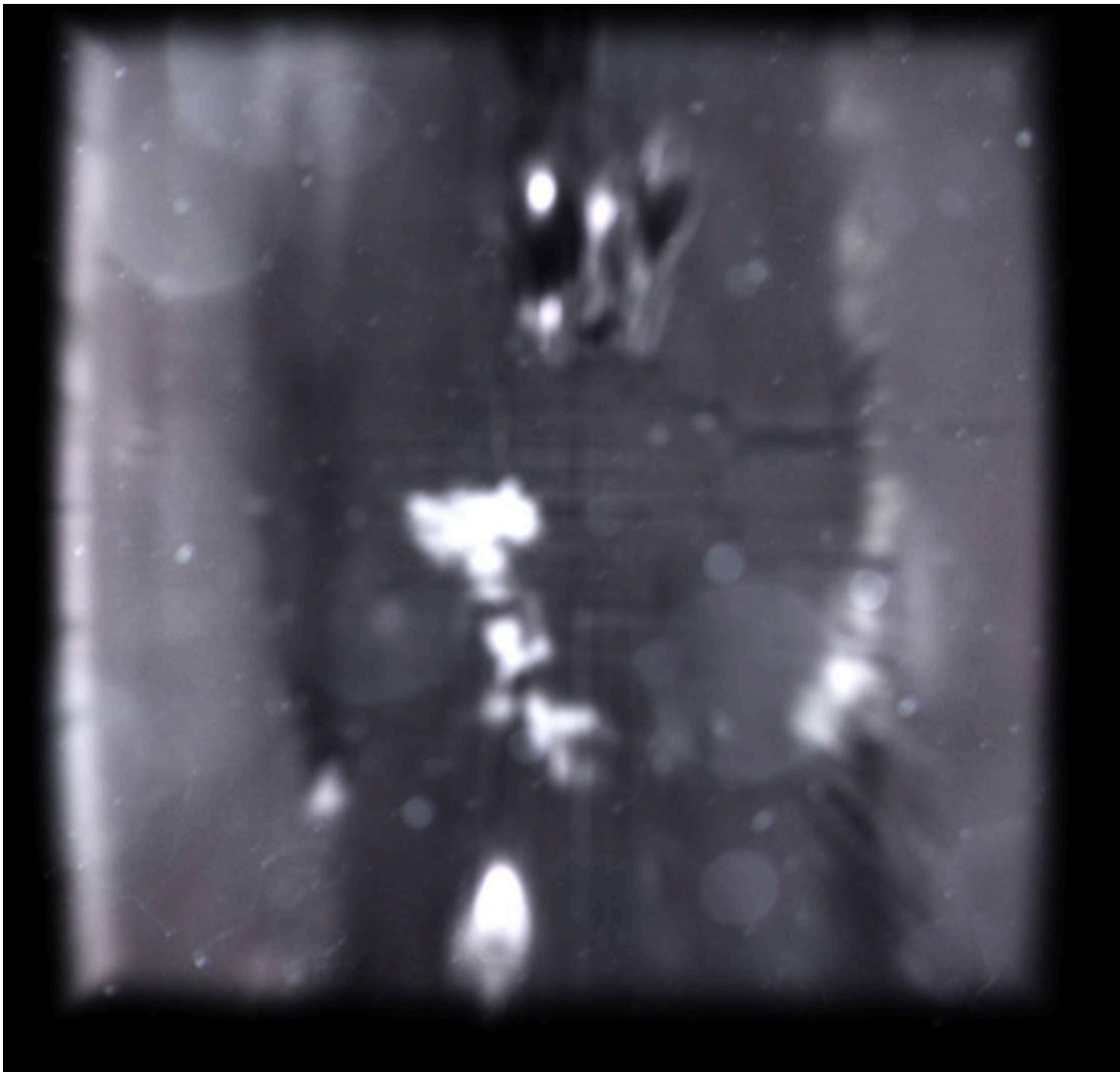
For the final design, I added in a post-processing library called *Wagner*³³ made by Jaume Sanchez, whose web-pseudonym is *clickToRelease* and *Spite* on GitHub. The library allowed me to recreate some of the microscopic effects seen using the Nanotronics microscope, such as blurring, dirty lens, zooming, dark field³⁴, color camera, and brightening effects. My hope is that these images are evocative of turning the microscope onto the city so that the user feels as though the metropolis and, perhaps, himself/herself are inside the petri dish.

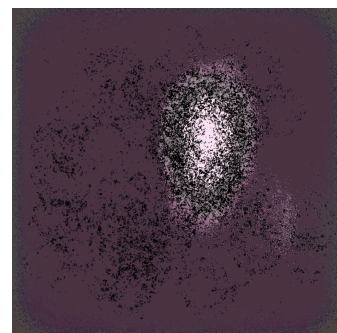
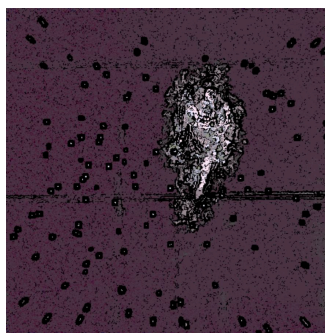
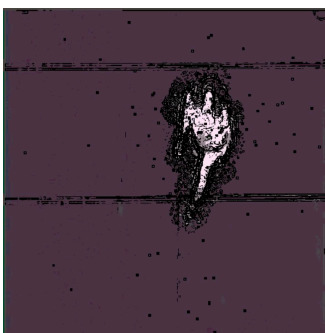
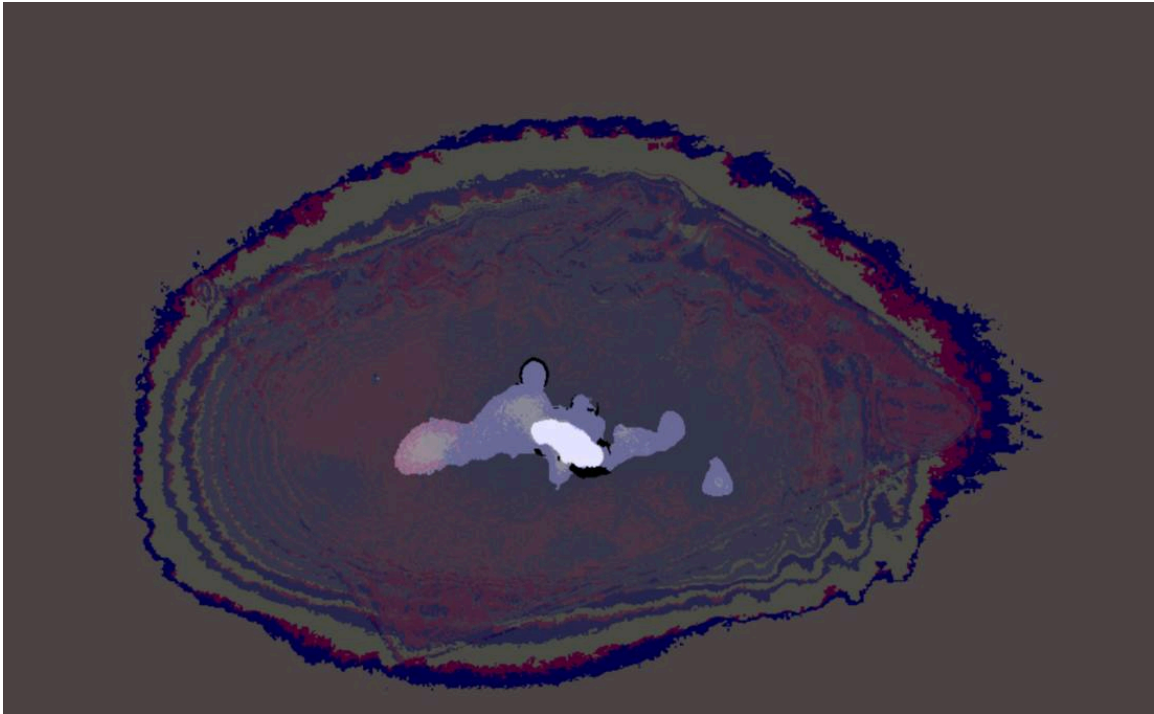
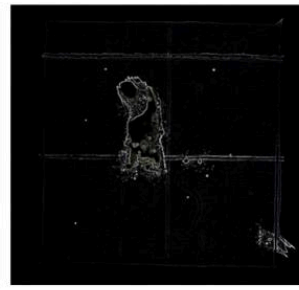
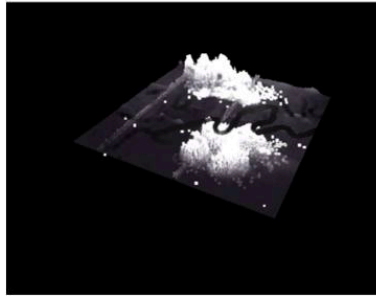
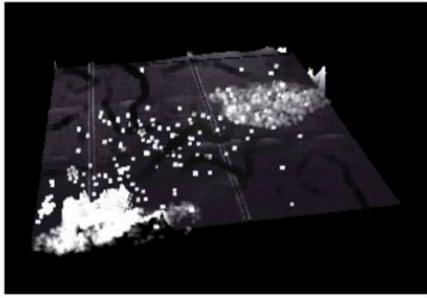
FIGURE 41 | THE FOLLOWING SERIES OF IMAGES ARE STILLS FROM THE FINAL PIECE PRESENTED WITHOUT EXPLANATION TO ALLOW THE READER TO TAKE AWAY HIS OR HER OWN EXPERIENCE.



³³ "Wagner" GitHub. <https://github.com/spite/Wagner> (March, 2016).

³⁴ "Dark field microscopy (dark ground microscopy) describes microscopy methods, in both light and electron microscopy, which exclude the unscattered beam from the image. As a result, the field around the specimen (i.e., where there is no specimen to scatter the beam) is generally dark." "Dark Field Microscopy." Wikipedia. https://en.wikipedia.org/wiki/Dark_field_microscopy (April, 2016).





WEB SITE DESIGN

The project will live on in the URL *microbiome.nyc*. The website front-end was designed with Tangram as the main page (using some of the work from the failed experiments). The process page was made using FullPage.js. The Taxo Maps page was made using d3.js. The Microbiome and Simulation pages were made in three.js.

The site serves as a presentation of this body of work and attempts to weave a narrative about the entire research project.

It has also become, however unintentionally, the place for advertising the work in the scientific community. I received this email from a former member of the Patho Map project who used to work in the Mason Lab:

FROM: XXXX@XXXX.COM³⁵

SUBJECT: FWD: WOA—YOU GUYS ARE REDEFINING AWESOME

DATE: May 6, 2016 4:47 PM

TO: REGINA FLORES MIR

... CHRIS JUST SHARED THE MICROBIOME.NYC PAGE WITH ME AND I'M COMPLETELY AMAZED. THE VISUALS AND DYNAMICS OF THE WEBSITE ARE SUPERB. SUCH A NOVEL AND AMAZING WAY TO VISUALIZE AND PRESENT THE DYNAMICS OF THE MICROBIOME. I WOULD LOVE TO CHAT WITH YOU GUYS MORE ABOUT THE METASUB PROJECT BECAUSE I'VE BEEN THINKING OF HOW CAN WE VISUALIZE 50 CITIES OF DATA AND I THINK THE ONLY WAY TO DO IT IS TO CREATE A NOVEL APPROACH, BECAUSE THE ONES MOST PEOPLE USE IN MICROBIOME STUDIES JUST ISN'T GOING TO CUT IT.

REALLY THOUGH I'VE JUST BEEN PLAYING WITH THE WEBSITE FOR THE LAST HOUR
... PHENOMENAL JOB!

³⁵ Name hidden for privacy.

FIGURE 42 | LANDING PAGE FOR THE WEBSITE MICROBIOME.NYC.



EXHIBITION

EXHIBITION DESIGN

This body of work will featured in the Parsons School of Design MFA Design and Technology 2016 thesis exhibition, *Between Spaces*. The exhibition will highlight a series of pieces that create a narrative for the participant to see the body of research and prototyping work that went into the project.

Unfortunately, due to the limited space available in the thesis show (with over 75 projects featured), my work will be bound to a small area. With more space and more time, I would have liked to experiment with the exhibition design in order to create a more complex experience for participants. For example, I would have created an entire room that a user could enter to be completely engulfed in a world of microbes and to use a live thermal feed instead of pre-recorded video so that the users see themselves projection-mapped to all six walls of an enclosed room.

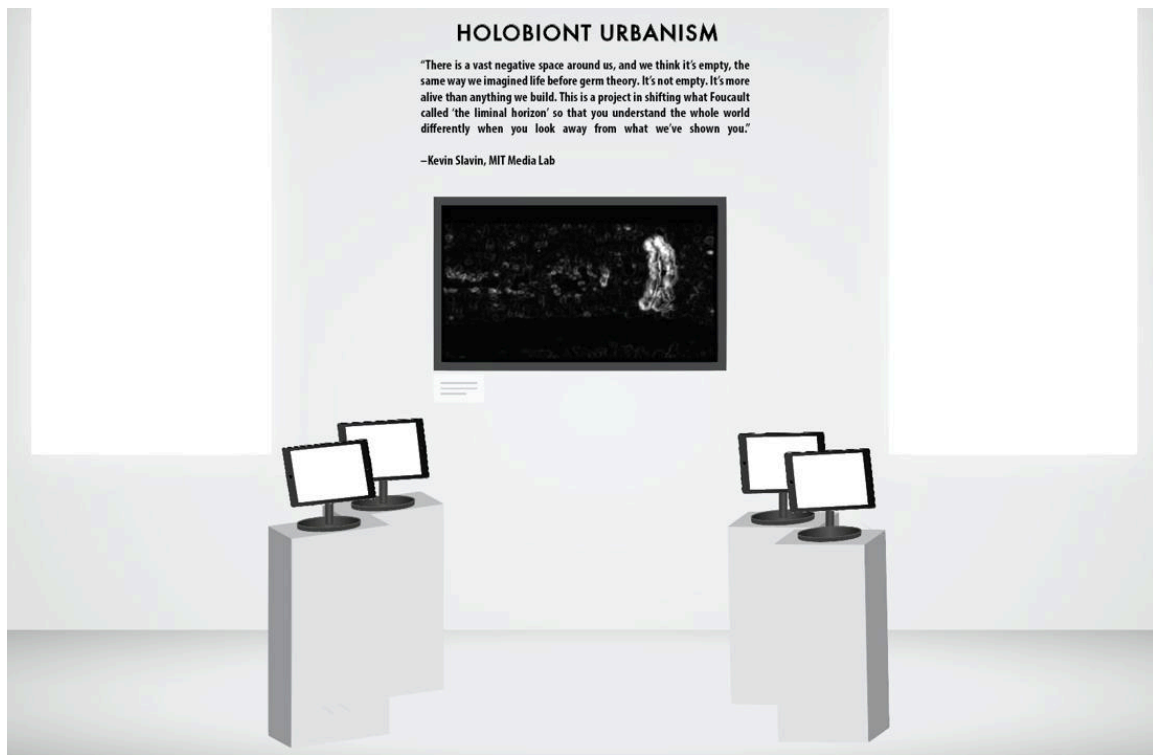
In the thesis exhibition I will show a large main screen mounted on a wall in front of four iPad Pros resting on rectangular gallery stands, the whole creating a rectangular shaped space. The pieces that will be featured on the iPads are:

- >> Process reel on loop highlighting the project background
- >> Nanotronics microscope reel highlighting the microscope footage
- >> Data Visualization piece showing the hive data mappings in three.js with auto rotating—UI disabled
- >> Taxonomical maps of neighborhoods by sample

The main screen will feature:

- >> Web-based experience on loop on the main screen—UI disabled

FIGURE 43 | MOCK-UP OF EXHIBITION DESIGN FOR THE PARSONS MFA THESIS SHOW .



VENICE BIENNALE

The project will also be featured at the 2016 Venice Architecture Biennale under the name *Holobiont Venice*. Ultimately, my work was used as a prototype map for a technician to recreate the piece in After Effects. My actual piece (the web-based simulation that I am showing in the thesis exhibition) will not be presented at the Venice Biennale exhibit. Rather, it was used as a design framework and prototyping tool to arrive at the final design concept. As Kevin Slavin put it in a memo to the team:

The overall goal in this final push... is to regain some of the depth and mystery you see in here [referring to my work]. This obviously isn't something to replicate with a different process, but the question is how to understand what is happening in the original image and how to find a

congruent sense of mystery in this new process. To say it again: not to do exactly ^ this, but to do something that does what ^ this does, at scale.

Nevertheless, images of my work were included with the project's catalog submission and will be featured in the official Venice Biennale 2016 catalog which I have referenced in the *Unpublished Source Material* section. In addition, the phylogenetic maps (shown in the *Data Art and Visualization* section) will be featured in the videos being shown in Venice. For Venice, additional data sets from Sydney, Melbourne, and Venice will be used and for which I have also created the phylogenetic maps that will be featured.

FUTURE RESEARCH

MICROBE TAXONOMY DESIGN

One question that I never fully answered in this process was: what do the individual microbe species look like? What are the distinguishing characteristics between them? And what are they based on? Ultimately, a microbe character taxonomy should be made. The taxonomy design is an aspect of the work that I prototyped and sketched but ultimately did not complete within the context of my thesis work. For the Venice Biennale, that aspect of the work was delegated to another designer, Chris Woebken (from the design studio, The Extrapolation Factory), to complete, while I focused on the World Development.

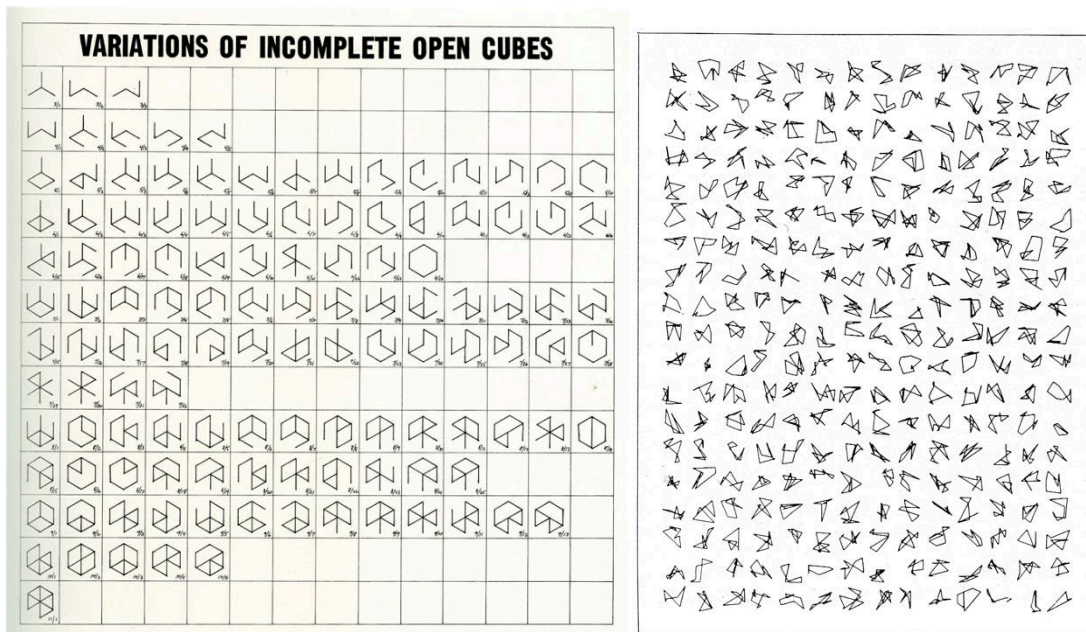


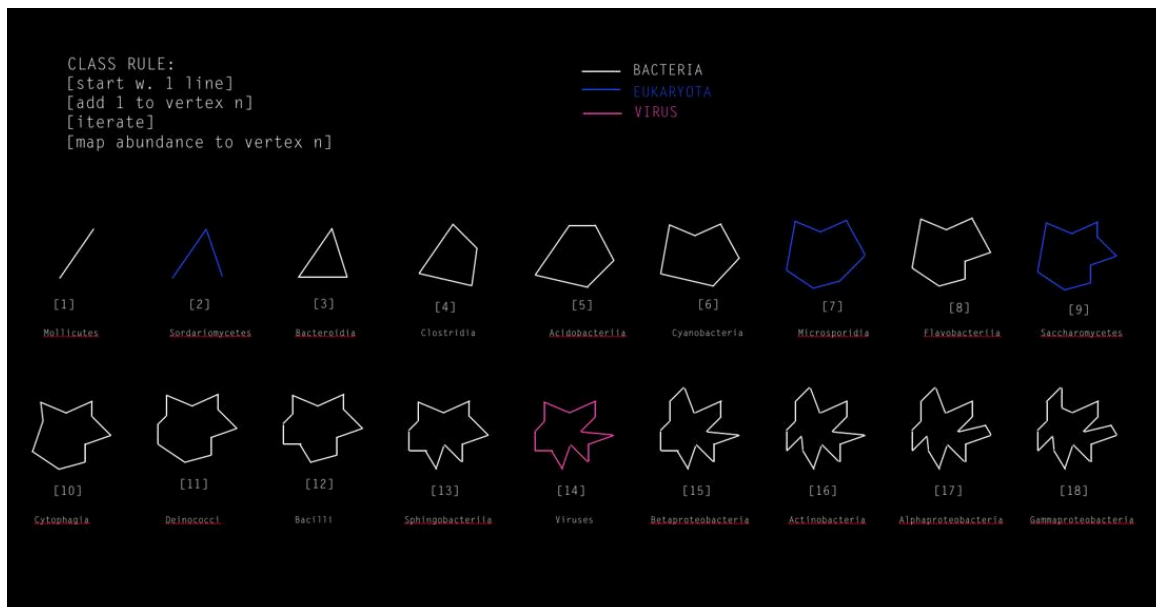
FIGURE 44 | (LEFT) SOL LEWITT 'VARIATIONS OF INCOMPLETE OPEN CUBES.' (RIGHT) GEORG NEES, '8-ECKE.'

My initial idea for creating a *species taxonomy* was based on Sol Lewitt's *Variations of Incomplete Open Cubes*. Sol Lewitt is a designer whose work I find incredibly inspirational, and I try to reference it as much as possible. Thus, my idea was to create a rule-based system and

use code to come up with a generative set of shapes. The recursive algorithm would be based on the percentage abundance of the species in the data. For the least abundant species the program would generate a line. For each succeeding species abundance, the program would generate a connecting line. This pattern would continue creating generative shapes.

During my data analysis research, experimentation led me to simply represent each species as a black dot in order of appearance in the data. I created an infographic with this data, and inspired by the 1960's minimalist art movement, I added a Courier font to hint at the typeface from a typewriter so as to harken back to a 1960's aesthetic.

FIGURE 45 | ROUGH SKETCH OF A MICROBE TAXONOMY USING GENERATIVE ALGORITHM TO CREATE SHAPES.



New York City Microbiome

Astoria | Crown Heights (L) | Crown Heights (TB) | Fort Greene

All Samples

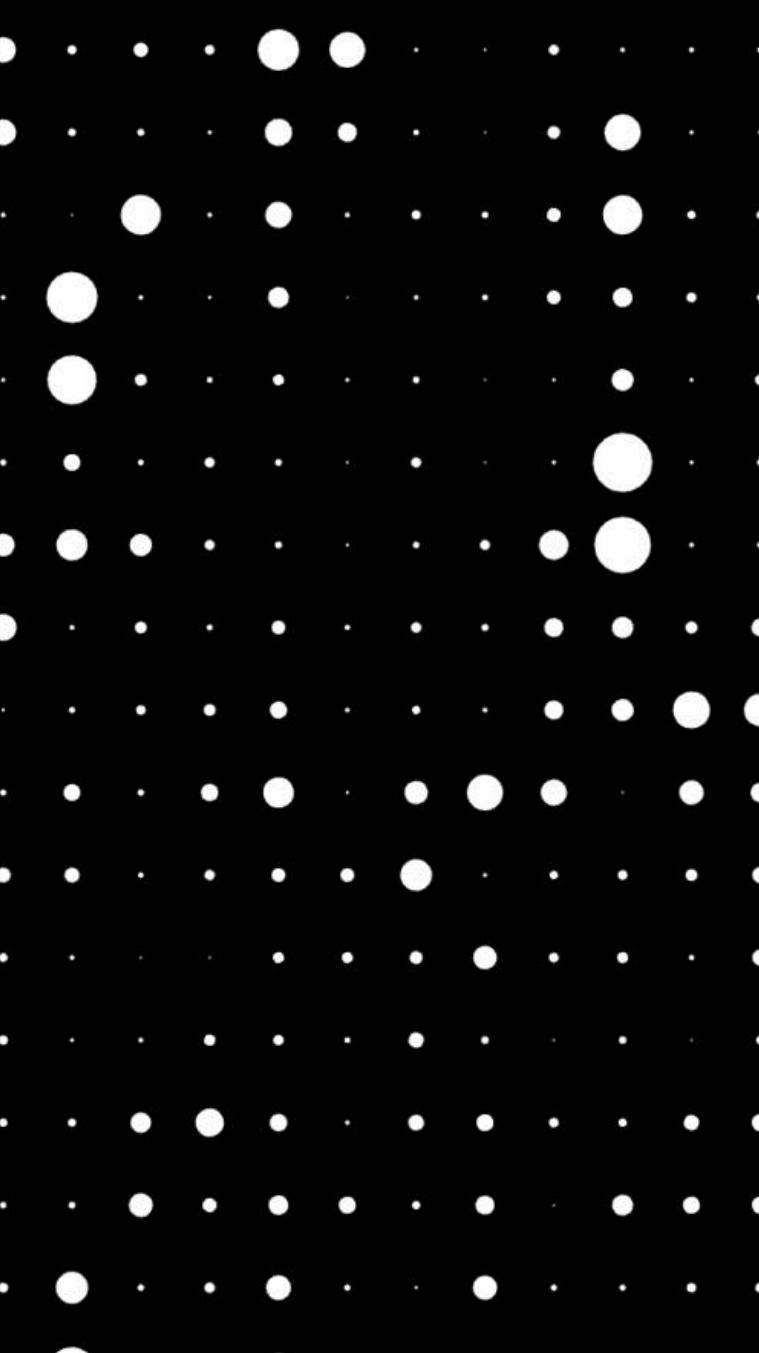


FIGURE 47 | MICROBIOME TAXONOMY NOTES SHOWING TOP LEVEL STATISTICS ON THE MICROBES AND THINKING ABOUT HOW TO REPRESENT THE INDIVIDUALS.



Microbe Dark Matter:

As I merged my previous work in astrophysics with this project, I believed I had coined the term “Microbial Dark Matter.” I learned about a week before publishing this document that this term had been in the public sphere since about 2005. Wikipedia has a very short entry on the topic quoted here:

Microbial dark matter comprises the vast majority of microbial organisms (usually bacteria and archaea) that biologists are unable to culture in lab due to lack of knowledge or ability to supply the required growth conditions. It is hard to estimate the relative magnitude of the dark matter, but the accepted gross estimate is that less than one percent of microbial species in a given ecological niche is culturable. In recent years effort is being put to decipher more of the microbial dark matter by means of learning their genome DNA sequence from environmental samples and then by gaining insights to their metabolism from their sequenced genome, promoting the knowledge required for their cultivation.

DNA DARK MATTER

Dark matter makes up about 85% of the total mass in the universe. Astrophysicists do not know *what* the dark matter is, but they know that *it is there* and that it plays a fundamental role in the cosmic structure as well as in “the anisotropies observed in the cosmic microwave background,”³⁶ which was my former field of study.

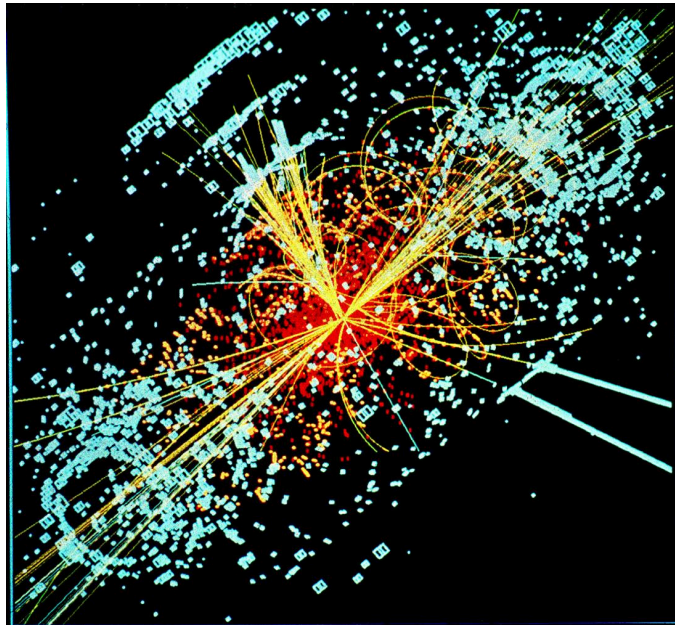


FIGURE 48 | SIMULATED LARGE HADRON COLLIDER CMS PARTICLE DETECTOR DATA, IMAGE SOURCE CERN.

In metagenomics there exists a similar mystery, which I am affectionately naming *DNA dark matter*. When scientists sequence DNA samples, they compare their sample to the millions of known DNA sequences in massive databases. However, since the development of DNA research began as a way to understand human beings, the databases are inherently biased towards organisms that affect humans. Thus, in an environmental study there will be a vast amount of data returned that is simply unknown, without a match, a mystery, essentially dark matter. Researchers know that it is there, but they do not know what it is and

³⁶ https://en.wikipedia.org/wiki/Dark_matter

so discard it. In other words, all the data analysis I have done and reported on in this paper is only a fraction of the complete data set. Most of the data was thrown away. Genetics researchers have developed a technique called K-mer for analyzing the complete data set. Essentially, K-mer analysis counts the number of unknown DNA strings to create frequency distributions. The researchers cannot know what is there, but they can count *it* up to see how much of *it* is there.

For a creative technologist, this opens up a world of possibilities for translating data into compelling visualizations and experiences. For example, could we translate k-mer frequencies into sound frequencies? What would DNA data sound like? As my MFA thesis writing professor, Edward Jefferson, said (I am paraphrasing): “Data visualization is the release of understanding to the mind. Not an illustration (an illustration means you already understand), but it is about visualizing data in new ways in order to understand.”

What might we see if we could see DNA dark matter? What might we hear if we could walk through a soundscape created from data? These are areas worth exploring, and I look forward to expanding my own creative abilities in potential future iterations of this work.

THE INTERNET OF BEES

As noted earlier, urban metagenomic research is currently in its infancy and we humans are only beginning to understand the vast transversal worlds that live around us. The research undertaken as part of my MFA thesis is also just the beginning of the research being done at the MIT Media Lab. For the Venice Biennale, the team collected bee debris data from Venice, Italy, Sydney, and Melbourne, Australia. Ultimately, the plan is to begin collecting data in Tokyo, Japan, where the Mori Building Corporation will feature this work in their art gallery in Roppongi Hills Tower. Perhaps even more cities will come online to

create a global network of urban bee metagenomic research centers. Imagine a future where there is an Internet of Bees (IoB)—a massive network of distributed databases of urban metagenomic data collected from major cities all over the world that could be publically accessed with an API. The future is bright and covered in bee shit!

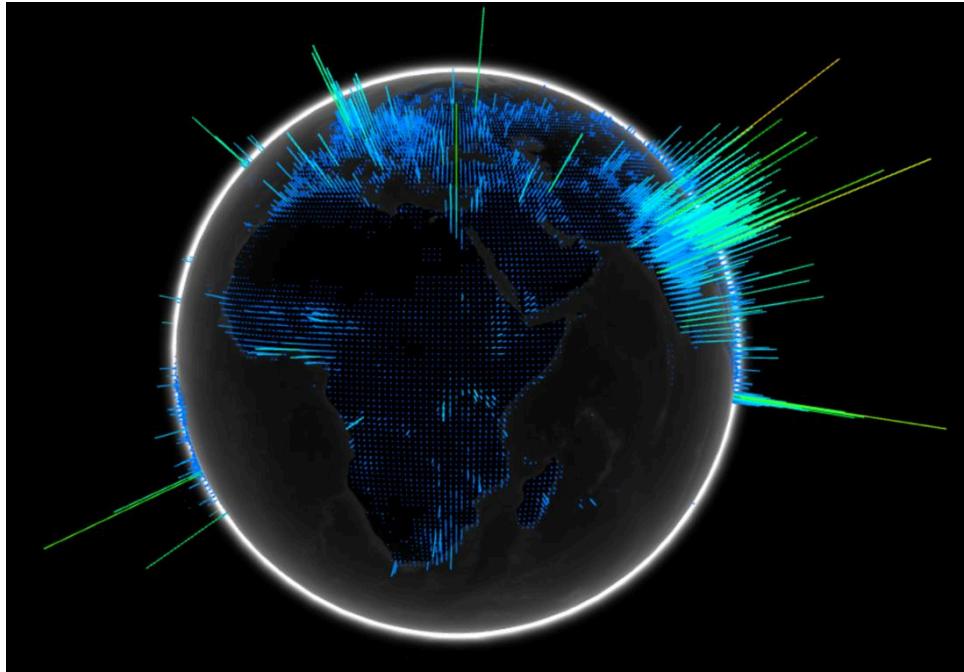


FIGURE 49 | GOOGLE CHROME EXPERIMNET, IMAGE SOURCE: GLOBE.CHROMEEXPERIMENTS.COM.

Building Community of Practice:

Early on in the research phase of this work, I set out to form a *community of practice*. (In fact, this is how I met Kevin Slavin.) I interviewed a wide range of luminaries in the fields of Bio Design, Bio Fabrication, Bio Computation, and Bio Engineering. While the design work for this thesis, in the end, was data visualization and creative coding, the primary impetus for this work was influenced by my interest in Bio Design. The culmination of the *community of practice* research can be found on my blog (reginafloresmir.com) under the tag *Bio Design*.

DESIGN PRECEDENTS

While it is impossible to write in detail about every work, art show, installation, and book that has helped inform this thesis project, the precedents collected below are the key pieces that have been the most inspirational and have been seminal in the development of this project.

DESIGN & SCIENCE

Geodesic Dome / Buckminster Fuller

Key Elements of Interest: geometry, architecture, design and science.

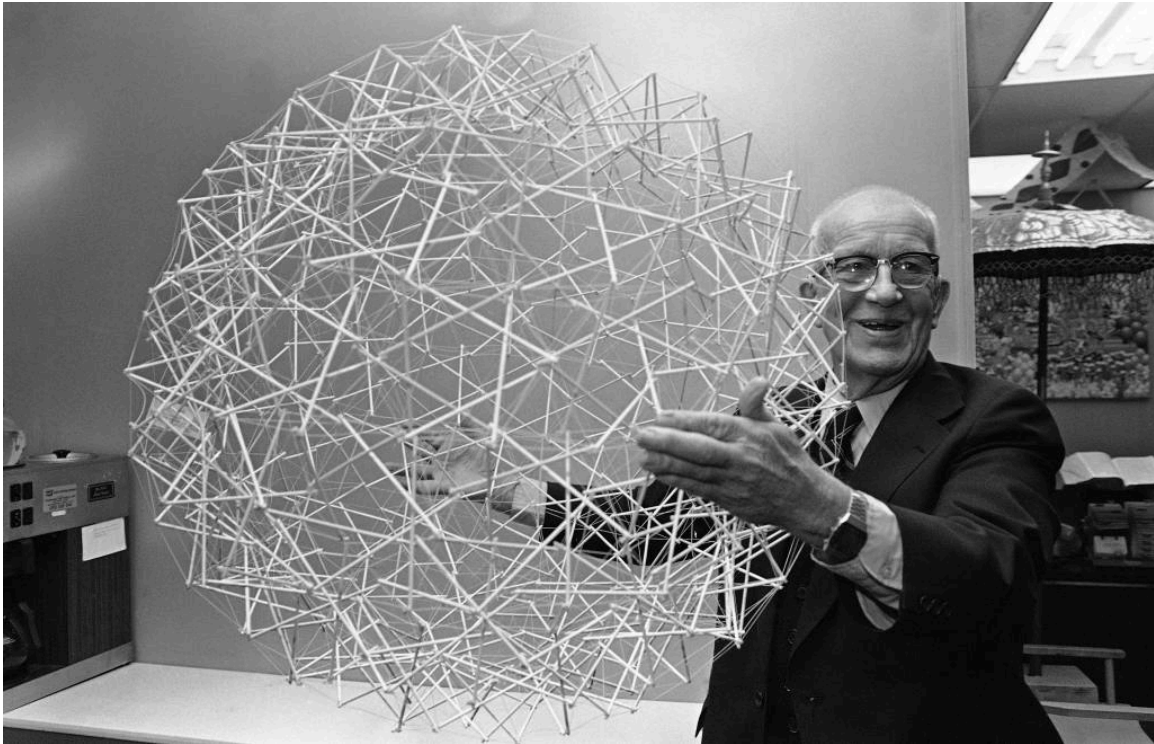


FIGURE 50 | THE LEGENDARY R. BUCKMINSTER FULLER, IMAGE SOURCE: BFI.ORG.

The life, work, and ideology of Buckminster Fuller have been key inspirations in almost all my work. His ideas of design science and designing across scales have been very influential. The *Geodesic Dome*—the architectural structure—is the living icon of his life's work. This

same structure later became the capstone of the discovery of the Buckminster Fullerene, for which Kroto, Curl, and Smalley won a Nobel Prize in Chemistry in 1985. Fuller's ideas about geometry and patterns in nature found in his book, *Synergetics: Explorations in the Geometry of Thinking*, have been significant, not just in this project, but in my life in general. This connection between designer and scientist, both conceptually and literally, is one of the main themes in the research and, ultimately, the body of work of *Holobiont Urbanism*.

Powers of Ten | Charles and Ray Eames

Key Elements of Interest: micro/macro duality, design and science, biology and astronomy, designing across scale.

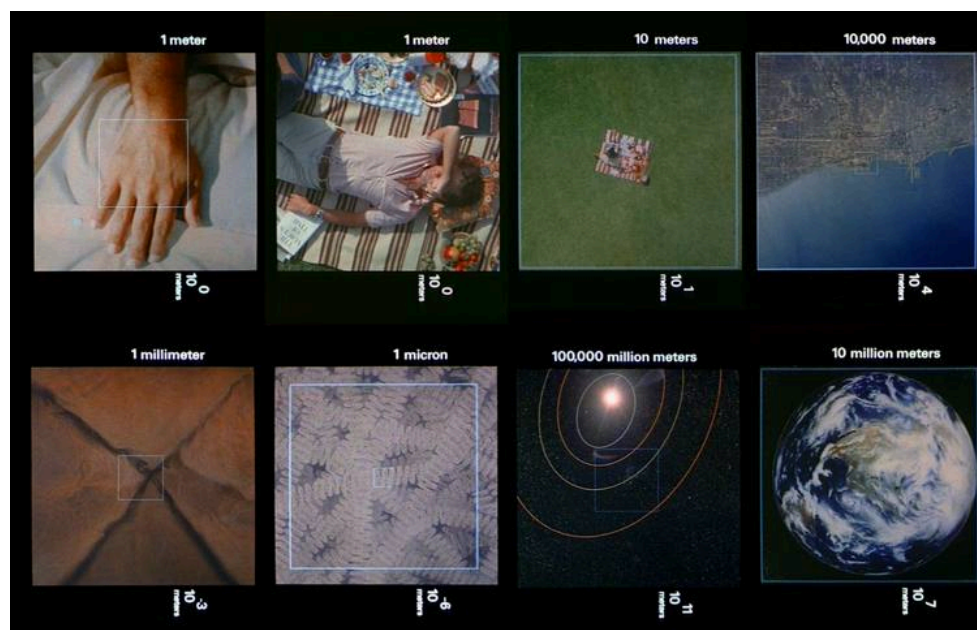


FIGURE 51 | CHARLES AND RAY EAMES, POWERS OF TEN. IMAGE SOURCE: EAMESOFFICE.COM.

If I had to select one seminal work that embodied the philosophy of this thesis work, it would be *Powers of Ten*. Written by renowned designers Charles and Ray Eames, this work uses a design framework that mixes scientific understanding (of 1977) to give participants a unique look into the known world. The idea of zooming through space and time to

encounter new worlds and discover new perceptions of reality is quite literally what the original goals of my work had tried to accomplish.

Powers of Ten illustrates the universe as an arena of both continuity and change, of everyday picnics and cosmic mystery. It begins with a close-up shot of a man sleeping near the lakeside in Chicago, viewed from one meter away. The landscape steadily moves out until it reveals the edge of the known universe. Then, at a rate of 10-to-the-tenth meters per second, the film takes us towards Earth again, continuing back to the sleeping man's hand and eventually down to the level of a carbon atom.³⁷

DATA ART

Immaterials/ *Timo Arnal, et al.*

Key Elements of Interest: conceptual ties to visualizing the invisible parallel worlds.

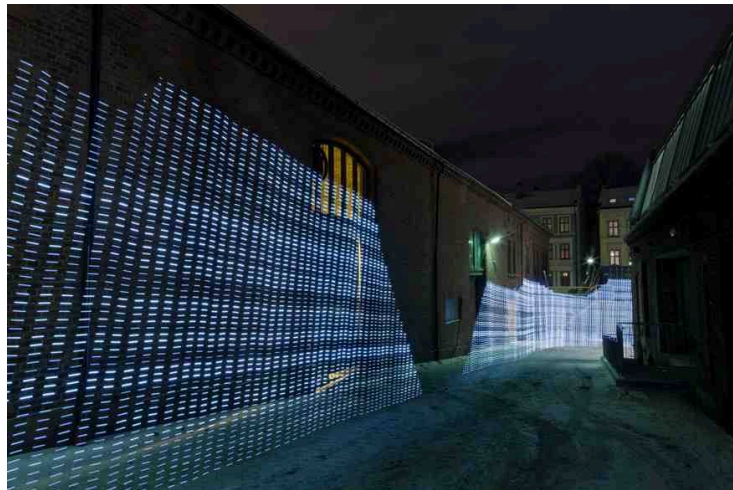


FIGURE 52 | TIMO ARNAL, IMMATERIALS.

³⁷ "Powers of Ten and the Relative Size of Things in the Universe." Eames Official Site. <http://www.eamesoffice.com/the-work/powers-of-ten/> (March, 2016).

Immaterials had been the key precedent driving the work of this project before I came on board the team. (In fact, its original title was *Biological Immaterial*, an ode to this work. Timo Arnall worked on some of the early prototypes, and his name is featured in the Venice Biennale catalog.) The work seeks to visualize the electromagnetic signals that increasingly surround us in our modern world.

Our environment is comprised not only of the physical, visible architecture and infrastructure that we can see and touch, but also of immaterial and invisible technological infrastructures that have an increasingly profound impact on how we experience the world. WiFi, GPS, RFID and mobile networks are the invisible materials, mechanisms, and infrastructures which enable contemporary digital culture. But our inability to see these systems hinders our capacity to understand their importance.³⁸

Ghost Cell | Antoine Delach

Key Elements of Interest: microscopic, city as medium.



FIGURE 53 | IMAGE SOURCE: ANTOINEDELACH.COM.

³⁸ Arnall, Timo. "Immaterials Artist Timo Arnall on Seeing the Invisible." Lighthouse. <http://www.lighthouse.org.uk/news/immaterials-artist-timo-arnall-on-seeing-the-invisible> (April, 2016).

This project, a visual interpretation of a city, attempts to render the world through a microscopic lens. “Scientific and dreamlike documentary at once, *Ghost Cell* is a stereoscopic plunge into the guts of an organic Paris seen as a cell through a virtual microscope.”³⁹ Similar to the prior work, *Immaterials*, this work seeks to visualize the city in a new way—to skew the perception of what the city is. This project is evocative and, through its form of an organic reinterpretation of Paris, forces the viewer to reexamine the city. *Ghost Cell* is both creative and thought-provoking and very quickly captures the viewer’s attention.

Sight Lines / ScanLab

Key Elements of Interest: data art, city as medium.

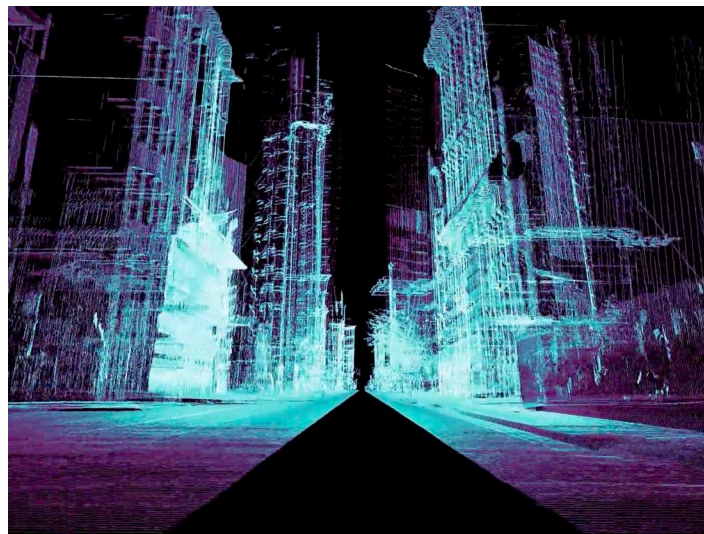


FIGURE 54 | SCAN LABS.

ScanLab used 3D laser scanning at the architectural scale to produce a short film “[t]o see how driverless cars might perceive—and misperceive—the world.”⁴⁰ Using the streets of London as the canvas, this project, like *Immaterials* and *Ghost Cell*, tries to reinterpret the “known world.” While *Immaterials* makes visible that which is invisible,

³⁹ *Ghost Cell* Trailer <https://vimeo.com/139651679>

⁴⁰ *Sight Lines* Trailer <https://vimeo.com/145248208>

Ghost Cell and *Sight Lines* try to reimagine the known through new eyes. While the aesthetic is distinctly digital, the re-rendered city scape has a hollowed feeling that is evocative of a new form of perception. The viewer has an understanding that the human eye is just one of many ways of perceiving the world.

BIO DESIGN

Bio City Map of 11 Billion: World Population in 2110 | Terreform One

Key Elements of Interest: Buckminster Fuller, micro/macro duality, biology as an alternative narrative, large-scale installation, parametric design, computational modeling, and addressing global issues from the age of the anthropocene.

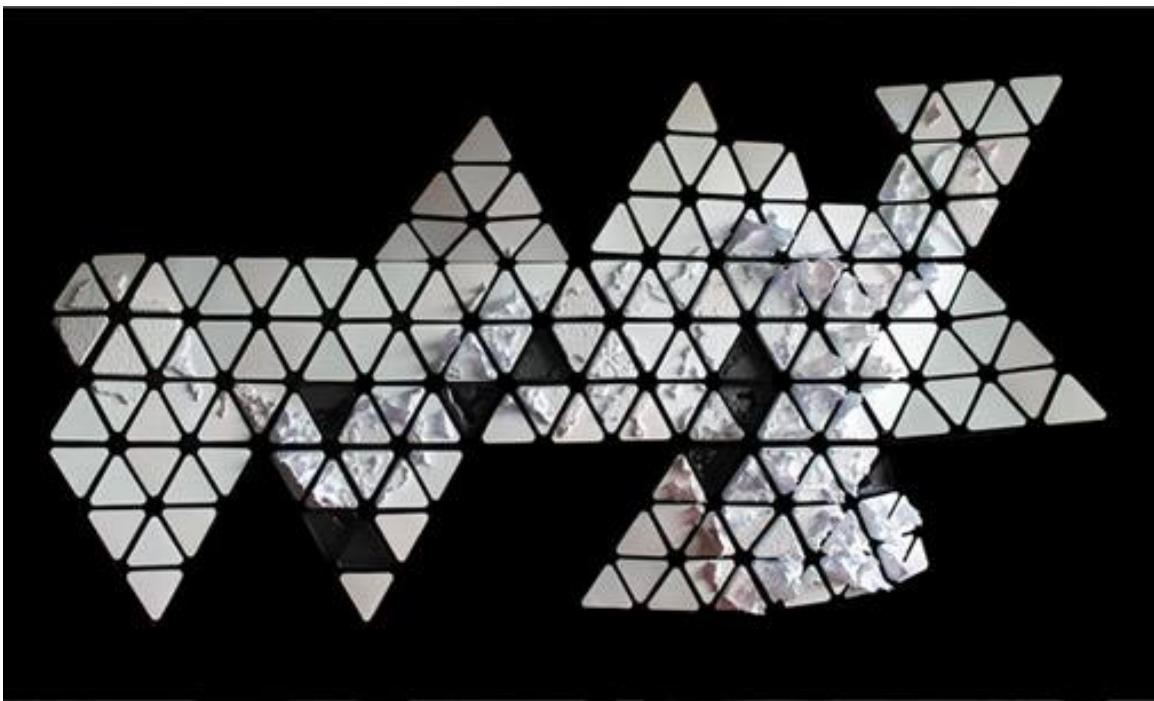


FIGURE 55 | TERREFORM ONE, BIO CITY MAP OF 11 BILLION.

The project, an official selection of the Venice Biennale in 2014, is perhaps one of the most quintessential biodesign projects in my eyes. It examines how the Earth may look in the year 2110 when the human

population swells to 11 billion, but it does this in a totally innovative way by using E. coli bacteria as a model of rapid population generation. The “Bio City Map displays population density as a parametric graph on the front and the back is made with living biosynthetic transgenic matter” (Terreform). The incorporation of biology into this project not only enhanced the user’s understanding of the problem but also served as a feedback loop to show how the integration of biology with artifice must be a solution to combat the impending problems that will arise with “water scarcity, food shortages, overcrowding, air quality depletion, and traffic congestion” (Terreform).

Silk Pavilion / MIT Mediated Matter Group

Key Elements of Interest: digital and biological fabrication, architectural scale, geometric design, computational modeling, and aesthetic appeal.



FIGURE 56 | SILK PAVILION, MATTER.MEDIA.MIT.EDU.

Neri Oxman, head of the MIT Media Lab's Mediated Matter Group, developed the concept of *material ecology*, which mixes the built environment with environmental consciousness. Oxman defines *material ecology* as

the study and design of products and processes integrating environmentally aware computational form generation and digital fabrication... [T]he field operates at the intersection of Biology, Material Science, Engineering and Computer Science with emphasis on environmentally informed digital design and fabrication.⁴¹

This project is very much a scientific endeavor, using components such as digital design, biological computation, and biological fabrication that are meant to help us understand how we could use biological materials for designing at scale.

The Silk Pavilion explores the relationship between digital and biological fabrication on product and architectural scales. The primary structure was created of 26 polygonal panels made of silk threads laid down by a CNC (Computer-Numerically Controlled) machine. Inspired by the silkworm's ability to generate a 3D cocoon out of a single multi-property silk thread (1km in length), the overall geometry of the pavilion was created using an algorithm that assigns a single continuous thread across patches providing various degrees of density. (Silk Pavilion)

ALGORITHMIC ART

Growing Objects / Nervous System

Key Elements of Interest: generative mathematical models, software as medium, nature inspired design, and 3D printing.

⁴¹ Oxman, Neri. *Material Ecology*.

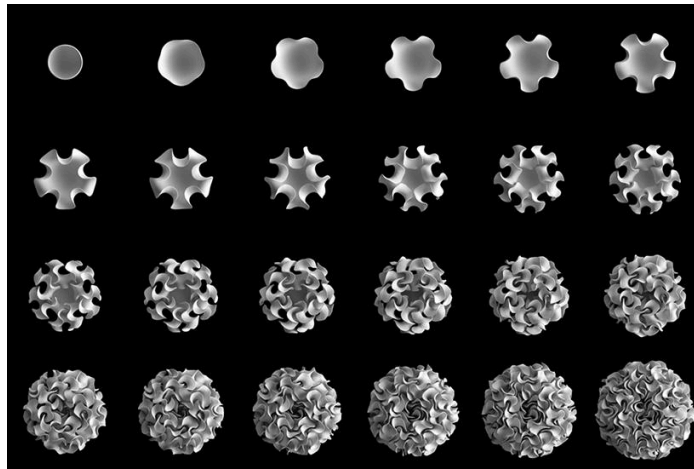


FIGURE 57 | NERVOUS SYSTEM.

Nervous System defines their work as that which

... explores processes which cause structure and pattern to emerge in nature... [and] adapt[s] the logic of these processes into computational tools; translating scientific theories and models of pattern formation into algorithms for design... These algorithmic investigations of nature were each documented by digitally fabricated sculptures... (Growing Objects)

The *Growing Object* exhibition is exemplary of how the design studio uses natural systems as inspiration to help dictate the form of their art objects. They

do not directly mimic specific phenomena, [rather] are instead open-ended explorations of the mathematics and logic behind them. The generated forms propose a new way of thinking about how we can design or 'grow' our environment. (Growing Objects)

UNPUBLISHED SOURCE MATERIAL

IN CONVERSATION WITH KEVIN SLAVIN

Written by Kevin Slavin to me as a response to a question I asked about the project.

“Yes, this comes back to the core goals of the project from a cultural point of view, which is understanding that the autonomous-decision-making-self is a fiction that we deploy for ourselves, mostly in order to get through the day without panicking. In reality, we are assemblies of many independent selves, moments, motives and ultimately — entirely foreign organisms — that collectively find their way towards meaning same way the market does, or cities do, both without a primary neo-cortex like “author.”

To put it another way: the goal of the project is to make you feel as tiny as you are. That 5-minute talk I gave about a pathogen that leads to the prevalence of cat videos is a joke, but it’s also real, and it’s to beg the question: how much of human behavior actually lies outside human agency as we understand it? Could ISIS be explained by a dietary anomaly? By an airborne toxin? Is it so crazy to imagine it as such?

[.....]

So to return to the project, the original brief is: to get the people of Tokyo to understand that they are living in a _complex_ adaptive biological system called Tokyo_ ... not just some fancy built environment. The city isn’t _like_ a biological superstructure, it _is_ a biological superstructure. In the way that no architect tells termites to build their incredibly complex structures who tells humans to build these things called cities?

Is it really the neo-cortex, or is it—like risky sexual behavior for those infected with Toxoplasmosis—to favor the rest of the world inside us? Are we totally different from the ants with Ophiocordyceps Unilateralis, when we learn that depression is actually a bacteria in the human gut? How does it benefit that organism to shut humans down? How does it change the sense of what it means to be alive when you realize that it’s not your brain that’s in charge of you?



FIGURE 58 | NATIONAL GEOGRAPHIC⁴²

To return again to the project, the goal here is to make visible the broad system that surrounds you and make you feel tiny, no more or less alien to the city than a polar bear.

Since you'll never see this world around you, how can we make you feel it, bring you into a world such that when you look up you understand — subtly, suddenly, or both — that this invisible world around you, it's also inside you, that you are just a ship carrying around this active freight.

You need to be able to see the city and understand who the real citizens are, the ones who do not fool themselves like we do.

*There is a vast negative space around us, and we think it's empty, same way we imagined life before germ theory. It's not empty. It's more alive than anything we build. This is a project in shifting what Foucault called the liminal horizon so that you understand the whole world differently when you look away from what we've shown you. *That is the ambitious sole criterion for the success of this project. **

... Start with 1) building a volumetric representation of the negative space of the city, for this is where the microbiome lives, and 2) start building a visual microbiological language to help us imagine what we can't see."

⁴² "Collective Mind in the Mound: How Do Termites Build Their Huge Structures?" National Geographic. <http://news.nationalgeographic.com/news/2014/07/140731-termites-mounds-insects-entomology-science/> (April, 2016).

INTERVIEW TRANSCRIPTS WITH DR. ROXANA HICKEY

Interview Date: February 17, 2016

Regina: So, you were saying that we don't actually know that much about microbes..

Roxi: Well, we don't know much about how they're interacting with each other in terms of cell to cell interaction. Most of what we know is about the types of microbes, just like sequencing, sort all of these names, we can identify those based on their DNA information, but in terms of visualizing microbes in the roles that, as far as I know, ...it's still pretty much in its infancy, I can explain about different types of how microbes look and some of them can move on surfaces or in liquid, and I can explain some ideas that we have about these structures in a three dimensional way like in clumps and dusts and on the surface...

Regina: What I want to know is if you're standing on a street, and I know you're in Washington or Oregon, but let's pretend you're standing on a busy New York City street, and what do, as a human being, what do you see? Do you see lots of buildings, do you see cars, do you see people, do you see, like maybe a rat, like a random tree, in thinking about that as the kind of "environment" where am I going to find microbes? Are they floating in the air? Are they all over humans? Are they on the sides of buildings? Describe to me...if I could put on special goggles that could see microbes, what would I see?

Roxi: I guess I'll start with the air. We know that humans shed about a million skin particles per hour into the air, and on the skin particles, imagine like little flakes floating around, you would have microbes attached to those flakes, just like hanging on. You would probably also have other stuff floating through the air like polyp grains or plant material or insect material that would also have microbes attached to them. You look at a person and you get the skin, you have the bacteria, they're kind of spread over the surface of the skin, very sparsely...you have microbes inside the body, in the mouth, in the gut, and in those places that are more protected you might have things like bio-zones which is where you have kind of sticky substance where microbes are secreting different types of materials that allow them to adhere together.

In buildings most of what we sample is dust, and dust is composed of a lot of different things. When people are present obviously there's human skin cells and hair. You might also have insect parts and fibers from clothing and other materials and within all of that you're going to find microbes just attached to different kinds of things.

Regina: What about adhering to the walls of buildings?

Roxi: Well, we know that there's microbes there, I'm not sure how--if they're growing on surfaces or if they're just stuck there somehow with parts of...no one's actually looked at it, but in my imagination, what I see on a surface like a wall that has a little bit of texture and you probably have microbes sitting on that surface. You have things like...think about, the drains in your kitchen sink and your bathroom, you know your shower, you're more likely to find bio-zones there so anything that's like sticky or slimy will definitely have tons of microbes growing there. What else? Surface of the t-bowl...We leave our skin oil and products that we use like soaps and lotions, all of those things. And anything that's like sticky or oily is going to grab microbes. Is this helping?

Regina: Yes, this is totally helping! So when we say in "in the air" what would, how would you characterize the density? Would you say it's like overwhelmingly dense or is it sort of like, you know, sparsely dense, hahaha I know these are not very scientific terms but hahaha...

Roxi: It's probably sparse. I personally don't know too much about the estimates of particles moving through the air, cause that's going to differ a lot, depending on ventilation, how big the room is...

Regina: Again, imagine you're outside.

Roxi: OK, then I would imagine it to be, it's really going to depend...if you have pollen floating around, in spring time, you know, you can see when those particles are floating around, and there are microbes attached to each particle. I mean I don't think you could ever feel it super dense. It's always going to be less dense flying through the air than you would find on a human body.

Regina: So on the surface of a human body it's much more dense? Than the air?

Roxi: Yes, I think the estimate for the skin microbiome is something like 10 to the fifth cells per square meters of skin I think.

Regina: So far we have humans, more dense, air not so dense, and building surfaces kind of like...

Roxi: Definitely less dense than skin, but more dense than air...

Regina: Would you say porous buildings (concrete) are more potential for microbes?

Roxi: I wouldn't go that far.

Regina: ...Buildings are more porous than cars, so how would you characterize cars as compared to air, humans, and buildings?

Roxi: Cars are moving around; any time there's moisture or something sticky there's more. They're getting splashed, and anytime there's moisture, there's more attraction for things to stick to it like microbes...but again a building is constantly exposed. I think it would be really interesting if someone would go out and try to estimate the concentration of microbes everywhere, but I don't think it's been done.

Regina: I totally understand that you're speculating!.. In New York, every so often you'll have a little park area where there's trees and some bushes, little tiny green, so how would you characterize those green spaces as opposed to air, persons, building, and car.

Roxi: The thing about green spaces, any time you have plants, they're going to have their own. They'll have microbes attached to the surfaces and circulating through, I think, like a root system, I'm not sure about inside the plant, but those are going to have bacterial or microbial communities that are distinct from what humans carry and what's just floating around in the air, and some of the things on plants are going to get attached to what's floating around in the air, but you'll certainly have greater density ...I recall a colleague saying that she read that plants are thought to have the same concentrations of bacteria on their leaves that people have on their skin....Plants are more dense than the air...and concrete surfaces in those areas, you'll have dust moving around and particles collecting. People can be depositing microbes on surfaces they touch, and other people are picking those up later.

Regina: Again, speculating, how would you characterize the speed...if you look at the Chris Mason lab...particles flying around...Is that kind of the way microbes fly?

Roxi: I think for the most part microbes are going to be traveling attached to particles...I don't know for sure but I doubt that you'll have any microbes travelling around by themselves like free cells because they're so small. Like dust particles in a sunlit room...you should be able to see individual bacterial cells or fungal cells riding on those particles.

Like in *Never Ending Story* with the kids riding around on that dragon thing. For indoors, dust microbiomes... Every time you walk through a carpeted area you're stirring those settled particles, and every time you do that the microbes can be distributed throughout the air on those particles. They can be on particles or droplets of moisture, millions of small droplets of moisture. Bacteria are attracted to one of those droplets.

Regina: Would you say then, it sounds like, a built environment, a city--a mixture of people, buildings, cars, some animals and green

space, the vast majority of microbes are coming from humans and animals, followed by air?

Roxi: Well actually outdoors...most of the microbes that are encountered outdoors are associated with either soil or plants and these are things that are probably being transmitted by dirt and other plants and associated particles...if you go INDOORS, that's where you start to see more human contribution because they're more concentrated. But outdoors most of the stuff that you collect is going to be soil and plant associated.

Regina: So to rephrase, talking about a city, the vast majority of microbes present are going to be from green spaces which comes from plants and soil and I guess maybe animals to some extent; like in New York we have squirrels and birds. Do they have microbes?

Roxi: Oh yes, all animals and plants have microbes on them.

Regina: So, from a city's perspective, you're standing on Broadway, and if you have on your goggles, and all you can see is people, cars, and buildings, so your microbe detector goggles are lighting up a little bit, but as you walk away from Broadway and get into Central Park, you microbe detector goggles are really going off because Central Park is really full of microbes.

Roxi: Yes, the buildings and cars, I guess, those are probably coated with like a SHEEN of microbes, then you go to Central Park with all the trees, polyps coming out of that, and you've got all the...flying around, and all of those things are going to have microbes. People are probably going to be the most factored of the group because their hands are constantly touching things, so yes with these goggles I imagine everything is lighting up...bionic things.

Regina: If you have on the goggles, are the microbes on people's skins evenly distributed around their skin or would you say it's more concentrated on their hands and their mouths.

Roxi: I don't know for sure, but I think that oily, like your hands and your face and your chest, I think that they call it incubation sites; oily sites are going to have higher concentrations whereas your elbows are probably not going to have much. They're dry and rough. I don't think much is going on there. So definitely, the hands would be more densely coated...Ashley Bateman,

Regina: What is the kind of work you do in the lab?

Roxi: I just started about six months ago, I've been getting involved with various sites of built environments, are you familiar with the microbial cloud? The microbial cloud research?

Regina: No, is there a link to that?

Roxi:So one of the things I'm doing is particles emitted from people in a room and their DNA. If you're a person sitting in a room, you're emitting particles with microbes on them, and we can actually detect different people's microbes by having, you have a distinct microbiome from the next person, and we can tell that just from sampling the air around the room.

Regina: Wait, wait, wait! Say that again! So that if I'm sitting in a room with some other dude, you have a machine or some kind of a detector that can say that this is your microbiome and that's his microbiome?

Roxi: We're not to the point where we can say that this is this person, but we've done experiments where one person sits in the room, and then another person sits in the room, and when you look at those two air samples that were collected from the air around them and the bacterial DNA, they look different from each other, so we can tell that there were two different people.

Regina: Wow, that's fascinating!

Roxi: And there's other weird stuff, so like women and men look different because actually a lot of the microbes that we find when a woman is in the room are things that are associated with a vagina, and when men are present, men don't have vaginas, so they don't have vaginal microbes floating around.

Regina: Wait, wait, But I have clothes on and my vagina is like shut off from the room! It can still be emitting?

Roxi: Unhuh. Well it doesn't come straight out of it, but you're probably trans-moving microbes around...We also find fecal associated organisms in air or on surfaces that we interact with, so it...

Regina: So there's fecal matter everywhere?

Roxi: Yeah, yes. James Meadow. There was a lot of popular press about this; he did another study...and he found that on chairs you found a lot of fecal associated materials, material that you would normally find in a human gut, so the media went crazy with that.

Now, of course, most of the microbes found in the gut are not harmful.

Regina: I'm not personally alarmed. I just want to tell you that I did some user testing for my project, and this is a direct quote from a friend: "Microbes, I always thought they were germs." I think the vast majority of people, when I say "microbes," they think germs, and I need some anti-microbial hand wash, I need to sanitize. People don't understand that probably the vast majority of microbes are good for you and the environment. We want to be...raising awareness is not quite it,

but like getting people to be in awe of the vastness, the vast universe that exists at our level. We're all aware of the stars...we're aware because of the Hubble Telescope, but people tend not to understand that and don't think about that there's a universe at our level. What are your thoughts about how we might inspire people? We're looking for stories in our data set.

Roxi: I think it's really great what you're doing. The human microbiome community has been trying to convince people of, that there's a lot of microbes and most of them are good and we need them for our survival... You can talk about how plant systems microbes do essential functions like fixing nitrogen and performing nutrient recycling that matters for the health of the plant, and we need the oxygen from plants to survive. Microbes are used a lot in bio-remediation. We can use microbes to break down toxic chemicals, to treat waste, conversion of waste products. In our own bodies we use microbes to metabolize nutrients that we can't metabolize. They compete with pathogens for colonization on our skin or in our gut and actually prevent things from making us sick. In the vagina we need them because they cause the acidic environment that prevents other things from coming in and invading. You need bacteria in the vagina to maintain health. In every environment that you can imagine there's microbes there doing something that makes us healthier or makes us safer.

Regina: I wonder if we've always been curious about finding microbial communities on Mars for example as a sign of life.

Roxi: I don't think so, but I know that there's microbes at the depths of the ocean which is kind of astonishing....is doing something about microbes on the space station. I'm sure that if any microbiologist got their hands on a sample [of Mars], microbes is the first thing they would have looked for.

Regina: I don't think they brought back samples.

Roxi: There was a lot of sterilization 'cause they're afraid of contamination, bringing something to Earth.

.....

INTERVIEW WITH DR. ELIZABETH HÉNAFF

Interview Date: April 21, 2016

Regina: I wanted to talk to a little bit about metagenomic sequencing and MetaPhlAn. So first of all I want to understand, before the Human

Microbiome Project, which was around 2008, were people doing metagenomic sequencing? Was that a *thing* before the Human Microbiome Project?

Elizabeth: It was kind of a *thing*, but it was done in a different way, so the timeline of how people study microbes was first when people realized that they could get bacteria to grow on plates. Like the first person who discovered that you could put nutritious agar on a plate and you would swab stuff on it, and things grew. That's how bacteria was identified and discovered, that something that is very microscopic and that you can't see, forms a colony, and so that colony is visible to the naked eye. And so the first kind of, like, identification of bacteria, and the field of microbiology, evolved from studying bacteria that are able to grow on a plate and then form a colony. So when you see a colony, you know that there's bacteria that are living there, and different kinds of bacteria grow in different ways. And so, you know when you leave a Tupperware in the fridge for a very, very long time, you see like white fuzzies, green fuzzies, and things with little stems and spores and other things that are gooey. Those are all different types of bacteria that each form a particular type of colony, and then a trained microbiologist can recognize those. It'd be like "Oh yeah, that kind of white fuzzy is bla, and that kind of green fuzzy is like whatever," and so that's kind of how it all started. And then people started in using DNA sequencing to identify, so that's a very useful technique. It's limited to bacteria that form colonies; some don't, and when you're using it as a way to study what kind of bacteria are present in an environment, it's limited to bacteria that are willing to grow on a plate, and then you can get fancy and make different kinds of media or different kinds of conditions, to get different kinds of bacteria to grow. But not all bacteria will grow in the lab.

So that's kind of the limitations of culture-based methods, and so then people started using DNA sequencing to identify bacteria in different ways than what we're doing now. So what they would do, they would use PCR. Are you familiar with PCR? PCR stands for polymerase chain reaction, and what it allows you to do is to amplify a particular sequence of DNA from a sample. So if you know the sequence that you're looking for, you can build primers that are complementary to that sequence and say if you have a mix of DNA--Take your cheek swabbing for example and you want to know whether you or me have a particular gene and if we know what sequence we're looking for--we can do a PCR, and if that sequence is present in the sample, it will amplify many, many copies, and if it's not present, it won't.

So that is kind of a precursor in a certain way to the whole genome sequencing methods that are used now. And so, what people would use is a particular region of the ribosomal RNA **gene** which is called 16S, so when you read metagenomic papers, and when they refer to using 16S sequencing to do species classifications, that's what they're talking about. When they pull out the sequence of this particular gene, and so

this gene is useful because it is not very variable across species so it's similar enough across species so that you can actually compare it from one species to another, but it's different enough between species that it gives you a differentiation.

So for example the extreme of that would be if we were to choose a gene that was present in some species and absent in others, you wouldn't even be able to compare these two because the genome isn't even in that other species, and if you choose a gene that is too conserved, so it's identical across all kingdoms of life, then it doesn't give you enough differences to differentiate species. But this particular gene is similar enough and different enough, that it allows you to differentiate species...

Regina: So going back to the timeline. When did metagenomic sequencing become a *thing*?

Elizabeth: I don't know what the first seminal paper is using whole genome sequencing, but I think that the first really big effort for that was the human microbiome project.

Regina: But I thought that metagenomic sequencing referred to environmental samples?

Elizabeth: Not necessarily. Meta means above, or encompassing the genome, so it refers to sequencing the entire set of DNA of the whole population of bacteria.

Regina: Got it. So the Human Microbiome Project was the first big effort to do it on the human scale, but what was the first big effort to do it on the environmental scale? PathoMap?

Elizabeth: PathoMap, yes.

Regina: So from the Human Microbiome Project, it went to PathoMap. Well really, the Human Microbiome Project was mapping the human being, kind of like the Human Genome Project. Then came PathoMap, which is looking at the environment, or I should say the subway, in particular.

Elizabeth: Yes, yes. Public transportation.

Regina: Then came MetaSub which is sort of like an extension of that, and then it seems like there was a paper that Devora sent to Facebook. Seems like there are other people now, like I know Jessica Green, and there's other people doing like office spaces? Internal inside spaces?

Elizabeth: Yes, yes. There's also a water sampling effort, also the hospital microbiome project.

Regina: And NASA has that twin study...you know I think part of the twin study was studying the microbiome of the space station. But in terms of mass scale, in terms of using bees as collectors, this project will be the first one of its kind..

Elizabeth: Yes, yes, yes. And I think an interesting point to make about that is that otherwise when you sample the built environment, you sample hyper localized...Your samples are hyper localized. Like getting things from a two foot by two foot square, and so you're extrapolating when you consider that that's representative of a whole environment.

Regina: Well, we're not really sure where the bees are sampling from though.

Elizabeth: We're not really sure, no. But the fact that we're finding bacteria that are associated with a bunch of different stuff, like plants and animals and water. They are somehow getting a pretty comprehensive cross-section.

Regina: So that's helpful. So metagenomic sequencing, so basically you take these samples from the beehive, you purify them in the lab and then you take...

Elizabeth: I think the appropriate expression for that is 'extracting' DNA. I know the Devora uses the term 'purifying' but the appropriate term is actually 'extracting. '

Regina: OK. So they extract the DNA from the sample, and then you send that DNA to a lab to be sequenced, and it compares it to sort of like a big data base of other DNA that they know...

Elizabeth: Yes. There's two steps of that. The first step is sequencing it, so that's the chemical reaction that gives you a digital readout for the actual molecular data, so you go from that reaction, input are molecules and output is digital readout. so that's like one step. And the other step is taking that digital readout and then analyzing it to answer different questions, like what kind of species are there and what are they doing.

Regina: And asking the question 'what kind of species there are' is really inherently a problematic question because you can only compare to things you already know. You know a priori that there's a whole bunch of stuff that you don't know what it is and you have to just toss it out.

Elizabeth: Yes.

Regina: And I would like to refer to that as the 'metagenomic dark matter.'

Elizabeth: OK

Regina: What do you think of that?

Elizabeth: OK. I think it's totally legit. I think the idea that the problem that the databases are inherently biased is a true problem. We can only identify things that we know already exist.

Regina: And the "metagenomic dark matter", that's the stuff that you wanted to do the K-mer analysis on, right?

Elizabeth: I wanted to do the K-mer analysis on the whole data set.

Regina: Which is inclusive of the dark matter?

Elizabeth: Yes, but I think that there's a subtle difference, I don't know if it's necessary for you to make it, but there's two different kinds of unusable data for the classification problem. So assume we have bacteria that's known into the database, and we know its entire genome and in our bags of sequences we have sequences that correspond to that whole genome. Only some of those sequences are going to be informative for classifying that bacteria because a lot of that bacteria's genome is going to be similar to other bacteria's genomes, so whenever we sequenced those ambiguous parts of the genome are not useful for differentiating one bacteria to another, but they are part of a bacteria that we know and so the only sequences that are useful to differentiate or to classify that bacteria are the parts of that particular bacteria's genome that are unique to it. So even for a known bacteria there is a small fraction of its genome sequences that are useful for classification and then a lot of it is redundant between it and many other bacterium, and so it's not useful for classification. It is useful, however, for functional analysis for example.

Regina: Oh shit, I forgot about the functional analysis... We didn't even use it really.

Elizabeth: Yes, we didn't even use it, really. It was part of the exploration.

The other kind of dark matter, or unusable sequence data, is sequences that come from bacteria that are not in the databases, and so they don't match anything. So you have two kinds of useless data: sequences that match any bacteria in the database--which is not useful for

discriminating between one and the other--and sequences that match no bacteria.

Regina: And that's what I think is the most interesting stuff.

Elizabeth: Why?

Regina: Because it's not that it ...it's almost like we know that something exists, we don't know what it is, but we can't...it's almost like the astronomy analogy. There's a signal, we can't, but we don't have a telescope powerful enough to know the signal. Like we know in math that theoretically there should be a signal out there, but we just don't have a telescope powerful enough to reach it.

Elizabeth: Or the instrument we have to register it is not registering it in a meaningful way.

Regina: Exactly... So tell me, talk to me about MetaPlAn. I read through the paper, but maybe it would be helpful just to hear you...

Elizabeth: Yes, so MetaPlAn is a tool that implements an algorithm that will perform the search of... or the given set of reads from a metagenomic sample, will take all of those reads and compare them to a set of sequences that they have compiled of those unique identifier sequences for all of the bacteria that are present in the public databases.

Regina: So it's like the most state of the art algorithm.

Elizabeth: It's one of the better ones, yes, definitely.

Regina: And that's the one that we use, the MetaPlAn?

Elizabeth: Yes.

VENICE BIENNALE CATALOG

Written by Kevin Slavin to the Global Art Affairs (GAA)- Foundation for submission to the 2016 catalog. GAA organizes as an official part of the Venice Architecture Biennale 2016.

*Kevin Slavin, Playful Systems / MIT Media Lab
w. Miguel Perez, Jun Fujiwara, Devora Najjar, Chris Woebken, Regina Flores Mir, Elizabeth Hennaf, Chris Mason, contributions from Timo Arnall, Jack Schulze*

In cities, it's hard to shake the sense that homo sapiens is the apex participant; we rarely build anything with any other species in mind. And for homo sapiens, it's also difficult to imagine "being" as anything outside the three pounds of brain matter, the core of consciousness.

But we are learning that our sense of the world -- and who we are -- has to accommodate another three pounds, deep in the gut. This is the "gut biome," referring to the roughly 10,000 different microbial species living inside you. Some of these species are not yet identified, but by 2016 some have revealed that they may account for who we are just as much as our environment, or our genes. By count, we may have more of their DNA in our bodies than our "own."

These microbes may account for why we are fat, or depressed, or more anxious, or less anxious, or even more risk/accident prone. Who we are, then, is not a person, but a superorganism, in which our "human" parts of us are in dialogue with quiet migrants who may well run the show.

To find out what's in the gut biome, we can genetically sequence an individual's poop. These are as individual as our genes, or our fingerprints. Much of it comes from what we breathe and touch. So what is the gut biome of Brooklyn, or Tokyo, or Venice? Are they as individual as the people within them? How would we discover what they are? How would we represent them?

Our work -- done with the generous support of the Mori Building Company of Japan -- sets out to answer these questions. First, to detect the invisible world around us and second, to bring that world to life. The videos we are generating are a landscape of these cities; the microbiological cities that don't build images of their own.

Our first obstacle was to reliably gather urban material to sequence. It's ambitious to pull microbes in from the open air. We had to find a way to get swabs from specific neighborhoods without depending on hundreds of volunteers with nylon swabs gathering microbes from sidewalks, gardens, windowpanes.

We found, finally, extraordinary collaborators: urban honeybees. As citizen scientists, they gather microbial material within 1.5 miles of their hive, and always bring it back to the same place. We don't ask them to do anything different: we just ask to see what they've brought home. We ask this with genomics -- some advanced computation -- which breaks down the "bee debris" and allows us to see what the bees have gathered.

We're still learning. Along the way to learning what we're looking at, we are learning how to see from the microbes POV. It's a world almost parallel to our own, we move through it every day. If you look

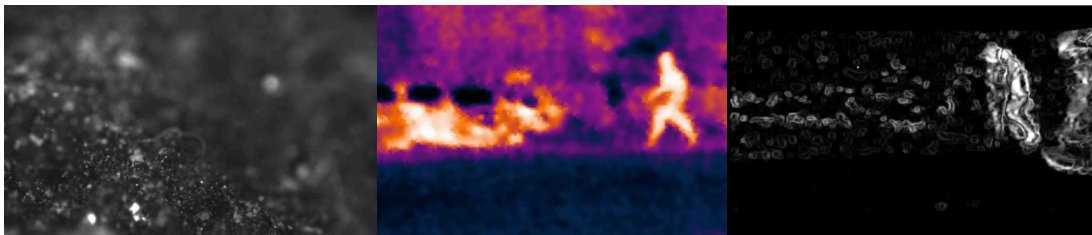
carefully, you'll be able to make out human forms moving in the videos. From a microbial point of view, those humans are just another way to get to work.

As we get the lab data, we see that cities are different from one another, in microbiology as surely as in culture, planning, and architecture. This may be why cities "feel" different, or why they thrive or die. We're only beginning to discover this new world, the one that's been at our fingertips all this time.

Shown here for the first time, our videos that are sketches from this new world. Whether they serve as postcards or maps, we hope they remind you of home.



Caption: Metagenomic Beehives, Kevin Slavin and Miguel Perez, MIT Media Lab. Photos: Miguel Perez.



Caption: Process sketches, Kevin Slavin and Regina Flores Mir, MIT Media Lab. Images: Regina Flores Mir.

FIGURE 59 | PHOTOGRAPHS SUBMITTED BY KEVIN SLAVIN TO THE GAA. CAPTIONS WRITTEN BY KEVIN SLAVIN.

APPENDIX

ADDITIONAL DOCUMENTATION

1. Final project website: <http://microbiome.nyc/>
2. A compilation of the microscopic videos from Nanotronics:
<http://microbiome.nyc/nanotronics/>
3. Complete code from live website:
<https://github.com/reginaflores/microbiomeNYC>
4. Week-by-week documentation:
 >> Thesis 1 blog:
 <http://www.reginafloresmir.com/blog?category=Thesis+1>
 >> Thesis 2 blog:
 <http://www.reginafloresmir.com/blog?category=Thesis+2>

CODE BASIS

The complete code is referenced above on GitHub. Below is a snippet of code that is meant to be illustrative of the medium and type of work used for this project. Shown is the algorithm used to create the “optical flow” hack.

FIGURE 59 | SNIPPET OF CODE ILLUSTRATIVE OF PROJECT EFFORTS.

```
function opticalFlow() {
  offscreenContext.drawImage(videoThermal, 0, 0, canvasVideoWidth, canvasVideoHeight);
  var imageData = offscreenContext.getImageData(0, 0, canvasVideoWidth, canvasVideoHeight);
  var d = imageData.data;

  flowData.forEach(function(flowArray) { flowArray.length = 0; }); // reset the arrays of points
  var tempsInBucket1Count = 0;
  var tempsInBucket2Count = 0;

  for (var i = 0; i < d.length; i += 4) {
    if (tempsInBucket1Count >= maxPointsNum && tempsInBucket2Count >= maxPointsNum) {
      break;
    }

    var color = [
      d[i],
      d[i + 1],
      d[i + 2]
    ];

    var x = (i / 4) % canvasVideoWidth;
    var y = Math.ceil((i / 4) / canvasVideoWidth);

    var temperatureInBucket1 = isInBucket(color, derivedParams.actualMinTempCutoff1, derivedParams.actualMaxTempCutoff1);
    var temperatureInBucket2 = isInBucket(color, derivedParams.actualMinTempCutoff2, derivedParams.actualMaxTempCutoff2);

    if (temperatureInBucket1 && tempsInBucket1Count < maxPointsNum) {
      tempsInBucket1Count++;
      flowData[i].push(
        new THREE.Vector2(
          x / (canvasVideoWidth),
          (1 - y / (canvasVideoHeight))
        )
      );
    }
    else if (temperatureInBucket2 && tempsInBucket2Count < maxPointsNum) {
      tempsInBucket2Count++;
      flowData[i].push(
        new THREE.Vector2(
          x / (canvasVideoWidth),
          (1 - y / (canvasVideoHeight))
        )
      );
    }
  }
}
```

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